OpenSourceRail

Open Engineering Platform

An illustrated guide to the railway, its people and its open engineering work



Wheel-supported reference and candidate-bound research. Manufacturing, physical and operating acceptance remain open.

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I. An open railway for people

Purpose, passenger experience and the design reference.

1.1 A note to the reader

This book is for people who want to understand the railway as a whole: the passenger experience, the city it serves, the train and guideway, the workshops that could make and maintain it, and the decisions still required before anyone could build or operate it. It is also an account of the open engineering platform that connects those subjects.

OpenSourceRail begins with a practical question. Can a comfortable, accessible urban railway be designed so that cities can understand its complete cost and develop lasting local capability to make, repair and improve it? The project explores a relatively simple wheel-supported train and station-powered propulsion, while allowing other arrangements to compete when they serve passengers better. It treats design, construction, operation and renewal as one problem. A cheaper train has little value if it needs an unaffordable station, an unrepairable part or an unworkable service pattern.

The present reference is a six-car train of 17-metre cars, with passive steel wheels on conventional rails carried by concrete. Each car is planned for 44 fixed seats and two wheelchair places, with no planned standing. These are development choices, not promises of an approved train. Propulsion, braking, headway and the final route must be established for each complete candidate. Safety, passenger protection and accessibility are requirements throughout.

What the reader will find

Part I explains the passenger purpose and the three connected products. Part II guides a reader through the published v0.6.1 software: its city atlas, Workbench journeys, candidate research, engineering views, delivery drafts and visible holds. Part III follows a city proposal into a worked Samawah corridor revision. Part IV presents ten illustrated city route examples. Parts V through VII explore the physical railway, its product and assembly views, and the work of making, costing, controlling and maintaining it. Parts VIII and IX show how digital twins are tested and how their evidence should be read. Part X follows the v0.7 effort to connect one corridor, vehicle and module into a continuous project. Part XI tours the screens people use. Part XII describes adoption, contributions, terms and further reading. The PDF outline provides chapter navigation.

The city maps in Part IV are excerpts of calculated planning outlines; their routes and access are unsurveyed. Product images from separate research candidates are labelled separately, and screenshots identify the software view from which they were captured. Illustrations are labelled as concepts or engineering views. A concept image can help explain a proposal without proving dimensions, performance or qualification. Calculated results are presented with their assumptions and limits. The published v0.6.1 release is a software release: it makes planning, research and review tools available, while physical manufacture and passenger operation remain subject to separate engineering and approval work.

Reading the evidence

Four descriptions recur in this book. A proposal is a design intention. A calculated result follows from a stated model and its inputs. A verified software result shows that a workflow ran and its outputs can be checked. A physically accepted result would require appropriate measurements, approved limits and independent review. The first three can help design a railway; none automatically becomes the fourth.

The current railway is not a single finished object. There is a selected development reference and there are research alternatives. When a vehicle, station or propulsion arrangement changes, its new geometry, mass, forces, power, heat, construction and cost consequences must travel together. The book calls this a candidate. Figures from a different candidate are useful for comparison, but cannot be inserted into the selected train as though they had been measured there.

The machine-readable source receipt beside the PDF records the exact inputs, illustrations and book file. It is for verification, so long file identities and execution logs are kept out of these chapters. Readers who want the detailed methods can follow the named project guides in the final chapter.

Edition: 13 November 2026. The v0.6.1 Open Engineering Platform software release is published; v0.7 engineering and integration continue.

1.2 Why build an open railway

Urban transport shapes where people can live, study and work. A new railway can shorten journeys and make a city more accessible, but it also commits the city to decades of structures, vehicles, energy use, staffing, renewal and repairs. OpenSourceRail starts from that lifetime commitment. Its aim is an affordable railway that carries useful numbers of passengers in ordinary service and leaves the owner able to understand and improve what it has built.

The project joins three products. The first is an open set of design and delivery tools: city planning, engineering models, quantities, prices, manufacturing plans and maintenance records. The second is the railway and its industrial system: trains, stations, guideways, workshops, tools and repair methods. The third is open control software, researched for train protection, supervision and faults. The software can be developed and tested before there is a railway; it cannot authorize movement of an unqualified train.

Local capability is part of the design

Local manufacture is more than a final assembly address. A workshop needs drawings, materials, process knowledge, inspection methods, fixtures, replacement parts and the authority to revise a design. Concrete precasting, structural fabrication, interiors, wiring, winding, assembly and routine maintenance may offer different opportunities in different places. Specialist bearings, electronic devices, cells or safety equipment may still come from outside the country. The origin and value added of each item should be recorded honestly.

Repairability changes engineering choices from the start. A removable panel needs accessible fasteners and space for a tool. A door mechanism needs a route out of the car. A bogie needs lifting points, service clearances and a realistic replacement sequence. A beam or bearing must be inspectable after installation. These details may add a little material or a manufacturing step, yet reduce years of downtime and dependence on one supplier.



Concept of a local vehicle assembly workshop

Concept illustration. The body, bogies, doors, glazing and interior appear as separable assemblies; their actual workshop methods and tolerances require controlled design work.

Compare complete systems

Transport concepts are often compared using one attractive number: a vehicle weight, a top speed or the price of a beam. A useful decision needs a common passenger service and a complete system boundary. A light vehicle may require more expensive propulsion equipment. A shorter station may need more trains. A low initial price may increase energy, maintenance or renewal costs. The design tools therefore carry the same candidate through vehicle, civil, electrical, service and economic accounts before ranking alternatives.

Some inputs can be calculated early, while others require a survey, a supplier or a physical test. The early estimate should show that difference. Unknown ground conditions do not become free foundations; unlocated equipment does not become a zero-mass vehicle. This makes the design more useful to a city even when it is incomplete, because the next decision and the cost of uncertainty are visible.

Open design knowledge also makes independent challenge possible. Another engineer can inspect an assumption, repeat a calculation, propose a different material or test a different train. A failed option remains valuable if it shows which constraint matters. OpenSourceRail's long-term value is therefore not one claimed winning configuration. It is a method for creating, checking and improving practical railway choices while retaining public knowledge of why each choice was made.

1.3 The passenger journey

A railway starts as a promise to people: reach the station, enter it, board safely, travel comfortably, make a connection and leave without unreasonable delay. The physical design should be judged against that journey. A drawing of a fast train says little about walking distance, crowding, step-free access or what happens when the service is disrupted.

Access before speed

Station spacing changes the balance between access and travel time. More stops can shorten the walk but add dwell, braking and acceleration. Fewer stops can speed a train while moving the benefit away from people who need it. A city proposal must therefore consider housing, work, schools, public space, transfers, barriers and the safe routes between them. The exact station point also affects utilities, property, foundations, entrances and rescue access.

The six-car reference places two doors on one side of each 17-metre car. It plans 44 fixed seats and two wheelchair places per car, with no planned standing. Across the train that is 264 fixed seats and 12 wheelchair places. The seating policy is a comfort decision, not permission to ignore crowd, evacuation or accidental-load cases. One retained structural control considers 160 people per car; it remains a separate load case even though routine service is planned for a smaller occupancy.



Passenger cabin concept with seating and a wheelchair space

Concept illustration. Door positions, boarding gaps, wheelchair circulation, restraints, fire performance and all installed clearances still need measured and controlled engineering.

A service is more than a timetable

The familiar two-minute interval is a named comparison case. With 276 planned places per train it yields 8,280 places per hour in one direction by arithmetic. It does not establish that train protection, dwell, terminal turnback, power or fleet availability can deliver such a service. A different formation or headway could be better for a specific city. An earlier 120-person train and 14,400-place comparison remains available for research, but its occupancy and service assumptions cannot be substituted for those of the current reference.

A credible timetable needs the route geometry, gradients and curves; the available traction force rather than only the required force; independent braking under adverse adhesion; station dwell and platform exchange; protected

separation; terminal operation; and recovery from a failed train. Energy and crew requirements follow from the same timetable. OpenSourceRail retains a 72 km/h launch cap in one stopped-network proposal, but the final speed is a candidate and corridor decision.

Ordinary days and difficult days

Passenger protection is most visible when a system fails. The railway must consider a train stopped between stations, a door fault, loss of wayside power, a hot day, a wet rail, a closed platform or a blocked evacuation route. Service batteries support lights, ventilation, communication, doors and protection where required; they are not traction batteries in the reference train. Independent brakes and rescue arrangements remain necessary even if a station propulsion system can recover energy in normal operation.

Accessibility reaches beyond the train floor. It includes station entrance, lift or ramp reliability, the platform and train interface, wayfinding, wheelchair and mobility-device space, sensory access and assisted evacuation. Each proposed station and vehicle change should preserve that complete path. The images in this chapter show an intention to make travel comfortable and inclusive. The actual result will depend on tested geometry, operations and feedback from the people who use it.

1.4 A reference, with room to explore

The development reference gives the project a common starting point: six 17-metre cars on passive two-axle steel-wheel bogies, running on conventional rails carried by concrete. The vehicle has no normal-service traction motors, traction batteries or levitation magnets. A passive aluminium secondary is retained as an interface for comparing station linear-motor propulsion. It is an interface choice, not a decision that linear motors have been installed.

This starting point simplifies some questions. A wheel-supported train can use familiar rail and running-gear principles. Removing normal-service traction equipment from the vehicle may make a car simpler to maintain, but puts more work into station equipment and its interfaces. Those system effects must be counted. Service batteries, independent brakes, protection, HVAC and rescue equipment still belong on or around the train.



Illustrated wayside linear-motor comparison

Concept cutaway. Wheels carry the vehicle; a passive fin sits between station-side motor primaries. Motor geometry, powered length and repeated-duty performance are not selected by this image.

Change the whole candidate

Research is permitted to challenge the reference. Powered wheels, onboard recovery, different station drives, alternate bogies, shorter cars, different formations and civil sections can all be explored. A change is meaningful only if its consequences are followed through the complete railway. A powered bogie, for example, adds motors, gearing, converters, cooling, wiring, protection, mass, maintenance tasks and perhaps different axle loads. It may remove or reduce wayside propulsion equipment. A fair comparison accounts for both changes and provides the same passenger service.

The project distinguishes the physical candidate from an evaluation of that candidate. If the geometry is unchanged but a new material property or magnetic-field hypothesis is tested, the physical train has not changed; the calculation has. A selection then records why one candidate is preferred for a particular purpose. These distinctions keep a favourable result from one model from being attached to another train without an explicit new evaluation.

Design question	What must travel with an alternative
Vehicle formation	Platform length, seats, fleet, mass, coupling, rescue and construction effects

Design question	What must travel with an alternative
Propulsion	Installed force, power, thermal duty, equipment, protection and maintenance
Running gear	Axle loads, wear, noise, bearings, brakes, access and inspection
Civil form	Load paths, bearings, foundations, erection, access and renewal
Service	Demand, dwell, frequency, energy, operating staff and spare fleet

There is no mandatory 900-metre or 100-metre powered installation. Those lengths appear in retained comparisons, and future designs can choose other arrangements. Nor does a short journey calculated from a prescribed acceleration prove that a motor can supply it. The railway must close force, current, heat, deformation and clearance together over actual operating conditions.

The older permanent-magnet maglev work remains an important historical research family, with its own source records and lessons. It is not the wheel-supported development reference. Preserving alternatives makes the project more open; keeping their evidence separate makes that openness technically credible.

1.5 Capacity, frequency and useful service

A railway can carry many people on paper and still offer an awkward journey. The passenger does not experience a capacity figure; they experience the walk to the station, the wait, the ability to board, the time on board and the certainty of reaching their destination. Capacity has to be tested against those experiences, and against what the railway can deliver reliably on an ordinary hot afternoon, not only under a perfect timetable.

Places and trains are different measures

The six-car reference has 44 fixed seats and two wheelchair places in each car. Across the formation that is 264 fixed seats and twelve wheelchair places, or 276 named passenger places. At one train every two minutes, simple arithmetic gives 30 trains per hour and 8,280 places per hour per direction. This is a useful reference control for comparing concepts. It is not an operating capacity until station dwell, safe train separation, terminal turnback, boarding patterns, availability and passenger demand have been shown to support it together.

The project retains a 120-person-per-train comparison that yields 14,400 places per hour at the same two-minute interval. That is an alternative formation and accommodation question, not a property of the seated reference train. The 160-person-per-car crowd load is a structural control. It asks whether the body can withstand a demanding occupancy case; it does not tell the operator to plan standing passengers in everyday service. Keeping these three quantities distinct prevents a strong beam calculation from becoming an unintended passenger policy.

The bottleneck moves

A shorter headway can increase offered places, but it also makes dwell variation more important. A wheelchair boarding event, a busy interchange or a door fault can hold the following train unless the stations, control system and recovery plan have enough margin. More trains require more vehicles, more power, more inspections and often more depot capacity. A longer train may relieve crowding but lengthens platforms, increases station and structure loads, and can complicate access on a tight site.

The retained-stop reference uses a 72 km/h launch cap. It is a design control for the journey calculation, not a promise that every alignment can reach that speed. Acceleration, braking, gradients and closely spaced stops can dominate travel time. So can a station placed on the wrong side of a road. A useful service comparison therefore holds the actual corridor and passenger objective constant while varying train length, frequency or propulsion. Otherwise a cheap vehicle can appear attractive only because it offers fewer journeys or serves fewer people.

Decision	Passenger effect	Engineering consequence
Add a station	Shorter access for some riders; longer ride for others	Platform, civil work, dwell, power and fleet may change
Run more often	Shorter average wait and more offered places	More trains, energy, staff and protected control capacity
Use a longer train	More places per departure	Longer platforms and different structural loads
Change stopping pattern	Different travel and transfer times	New timetable, capacity and recovery analysis

Reliability is part of capacity

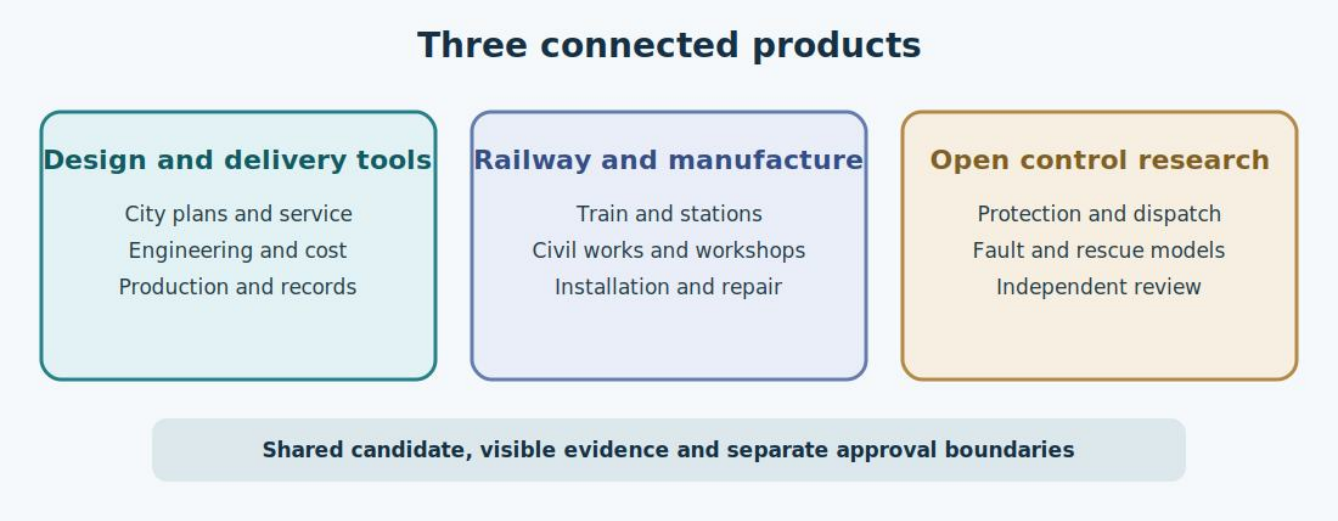
An offered timetable is only useful when the fleet and infrastructure can maintain it. The model should include spare trains, inspection time, maintenance access, station equipment faults, power outages and the time needed to rescue a stopped train. A failure that blocks a single guideway can turn a high theoretical capacity into a long queue. The cost of that queue matters to a city's decision even if it does not appear on the construction bill.

OpenSourceRail's direction is to compare complete service packages. A candidate needs a route, station pattern, train and propulsion arrangement, protected separation concept, maintenance plan and cost range. The current tools can connect many of these assumptions and expose what changes after an edit. The passenger demand, safety margins and failure rates needed for an operating timetable still require local evidence and independent

review.

1.6 Three connected products

OpenSourceRail is easier to understand when its three products are seen together. One is a set of open tools for planning and delivering a railway. The second is a proposed train, civil structure and local manufacturing system. The third is research into open train control. Each has a different level of maturity. The published software can support a useful planning conversation today, while a passenger railway still needs site evidence, hardware, tests and the approvals of the people responsible for operating it.



Three connected OpenSourceRail products

The products exchange design and evidence. Their software and physical approval boundaries remain separate.

Design and delivery tools

The software begins with a city and a transport question. A planner can inspect mapped corridors, stops, street context and assumptions; compare service patterns; and calculate outline fleet, energy, civil quantities and costs. The Workbench brings these records into one view and shows the source of a result and any hold that limits it. An engineering team can move from that proposal into a vehicle, propulsion and structure candidate, then develop parts, factory work and delivery drafts against the same design.

The important feature is continuity. A route edit should change the service and cost accounts that depend on it. It should not cause the body of an unchanged train to become a different vehicle, or let a cost from one drive architecture appear under another. In the released v0.6.1 software, the Samawah city journey demonstrates a fresh calculation, a repeated unchanged request and a changed-demand request. In v0.7 development, the Samawah station comparison carries a specific stop edit through service, costs and a revised dossier. Both demonstrate software behavior within their own declared limits, not a surveyed construction design.

Railway and manufacturing system

The physical reference is a six-car, wheel-supported train on conventional rails carried by concrete. The preference for relatively simple cars, repeatable civil parts and local repair is a design hypothesis to be tested against cost and passenger service. The project retains competing research architectures, including powered wheels and station energy recovery. Every physical option needs its own installed mass, force, braking, civil, factory and whole-life comparison. No attractive component study by itself selects the whole railway.

The manufacturing product extends beyond a train drawing. It includes material and supplier choices, tooling, inspection, assembly sequence, transport, construction staging, access for replacement and records of what was actually built. Local content is genuine domestic work and value added, with imported materials and specialist components named honestly. A local workshop can make a useful part only when the drawings, process, tests and quality responsibilities match its equipment and skills. The current catalogues and draft packages show much of this structure; controlled production and physical qualification remain open.

Open controls and supervision

The control research studies how a train receives movement authority, how routes are protected, and how dispatch, doors, braking and faults interact. Rust software makes some state and refusal behavior testable and reviewable. These are research and simulation results. They are not a commissioned automatic train-control system or permission to run a passenger service.

Equipment observation is a separate function. The optional FUXA and gateway integrations can represent alarms, trends and maintenance events for a configured deployment. Observing a cabinet temperature or logging a door fault must not be confused with commanding a train. A real installation would need authenticated equipment bindings, tested communications, operating procedures and independent safety review.

Why the products meet

A city's choice of stations and headway changes fleet and power demand. The chosen train changes platform dimensions, structure loads, parts and repair work. A component's inspection result can alter the maintenance plan and possibly the service. The three products are connected so those consequences can be seen together, yet each approval boundary remains explicit. A good software result makes the next physical question clearer. It never makes that question disappear.

II. The published v0.6.1 platform

A guided tour of the released software and its limits.

2.1 The published v0.6.1 platform

The first packaged OpenSourceRail platform release gives a reader several ways to explore the project. One can browse a city, inspect a proposed service, follow an engineering candidate, examine a drawing, and ask how a design might be made and maintained. The release is a working software and research environment. Its output is most useful when a reader keeps the difference between a calculated outline and an approved railway clearly in view.

Three products in one project

The first product is the design and delivery platform. It brings city geography, transport proposals, vehicle and civil designs, calculations, quantities, costs and proposed delivery records into a traceable workflow. A change to a station or train should be visible in the results it affects. The platform is already useful for exploring and reviewing design-stage questions, though not every intended handoff is complete.

The second product is a railway and manufacturing system. Its development reference is a six-car, wheel-supported train on conventional rails supported by concrete, with a strong preference for local manufacture and repair. Vehicles, guideway, stations, propulsion, braking and workshops must form one buildable candidate. The release includes drawings, inventories and numerical studies; it does not contain a physically qualified train or construction package.

The third product is open train-control research. Rust software explores train protection, interlocking, station dispatch, supervision and fault response. Its simulations and checks can reveal inconsistent commands or unsafe state transitions. It has no live railway movement authority. The business and equipment-observation integrations discussed later are also software components; they do not turn the research system into an operating railway.

Start with what is available

The published city catalogue contains outlines for 266 cities and 1,565 lines. These are mapped planning cases with route and station information, service assumptions, energy and cost scenarios and local-evidence records. They provide breadth for comparison, not 266 surveyed construction designs. The Workbench can display the geography, sources and calculated results and can run a bounded Samawah study. That study is the clearest fresh example of the v0.6.1 software journey.

The release also includes fourteen-scope cost and delivery drafts. A reader can inspect what is included in a proposed railway estimate, where a price comes from and which quantities remain provisional. Engineering galleries offer rotatable and editable geometry for selected vehicle and civil examples. These are useful views of design thinking, not approved production drawings.

Some capabilities are previews. Baghdad has a retained route, service and cost comparison that can be read without recalculating the city. Candidate research has source-pinned alternatives, but the current Workbench portfolio selector is held. ERPNext, Frappe HR, FUXA and gateway adapters can prepare reviewable drafts and examples, while actual purchasing and railway control remain outside the release. Full native CAD and mechanics workflows vary by model and tool availability.

Why the visible holds matter

A calculated result is trustworthy only for the inputs and method that made it. The older integrated engineering dossier has a broken source binding at the published release. The Workbench therefore keeps global portfolio and cost cards held instead of presenting them as current. The Samawah calculation uses its own checked inputs and remains available. Likewise, the native RustGIS project export from the published atlas is held because its GIS and outline publications do not match; the read-only Baghdad atlas still works.

This is how an open platform should behave when evidence is incomplete: show the useful work, name the limit and refuse to borrow a result from a different source. Later development may close or change a hold, but the v0.6.1 release remains a fixed record of what users could actually inspect and run. The next chapters take a

reader through that release, beginning with one ordinary Workbench session.

A modest adoption path

A city need not begin by choosing a train. It can start by agreeing on a passenger problem, asking residents how they reach existing services and using the atlas to compare several corridor ideas. The Samawah journey shows how to turn one scenario into a traceable calculation. The cost scopes identify which local prices, surveys and supplier conversations would have the most value. Only after those questions are clearer should the city ask whether a vehicle, propulsion family and civil arrangement can meet the service at an acceptable whole-life cost. In that sequence, the platform supports an informed public and engineering discussion without implying that a proposal is ready for tender or passenger service.

2.2 A reader's first session

The simplest route into OpenSourceRail is to open the published Workbench and follow the Samawah example. The release package includes an installation guide and a recorded clean-install demonstration. For this software journey, the optional native CAD, mechanics, GIS and business services are unnecessary. That separation helps a new reader understand the platform before assembling a full engineering workstation.

Orient yourself before changing a value

Begin with the city view. Choose Samawah and inspect the line and station map, the displayed source information, demand assumptions, planned headway and access notes. The map is a proposal over available geography. It does not certify the walking route to a station, the land needed for a pier or the accuracy of an unsurveyed alignment. A useful first question is therefore: which facts are local observations, and which are shared model assumptions?

The Workbench separates source context from an evaluation. Source context includes the geography and other information selected for the study. An evaluation applies an explicit service request to that context and produces a calculated answer. The same request should lead back to the same verified answer; a changed request deserves its own result. The interface exposes this relationship so a reader can compare the two cases without guessing which map, train or cost basis was used.

Read a result in layers

After the baseline Samawah request completes, start with the service summary. Which lines are included? How many trains are implied? What energy and civil quantities are counted? Then open the cost view. Its fourteen scopes include more than the train or the concrete: construction, stations, power, labour, maintenance, replacement and end-of-life work all have to be considered. Low, base and high values describe model uncertainty and assumptions; they are not three supplier offers.

Next inspect the detailed result and its delivery draft. The result should show its inputs and its unresolved questions. The delivery draft translates model quantities into proposed tasks and resource needs. It is useful for asking whether a local workshop or construction sequence has been considered, but it does not authorize procurement or fabrication. A business-system draft is deliberately disabled until reviewed conversion and real transactions are appropriate.

Repeat, then change one thing

Repeat the identical request. The published demonstration confirms that the Workbench reuses the sealed result rather than performing unnecessary work. For the reader, the important point is consistency: the same inputs should not quietly produce a different city price or vehicle candidate. Reuse also matters on a computer with limited disk and memory, because a design team may review the same scenario many times.

Then change the example peak demand from 4,000 to 9,000 passengers per hour per direction. The Workbench creates a distinct evaluation and shows its service and cost effects. This does not assert that Samawah has observed either demand level. It demonstrates how an owner can ask, “What if our peak demand is higher?” while keeping the previous case available for comparison. The same discipline should apply to a future route or vehicle revision: change the relevant input, recalculate its dependants and retain the parent.

Take away the right result

The Workbench offers a summary, detailed result, source record, performance record and delivery plan for the study. These help another person review the calculation and repeat it under the same software. The exported package is not an engineering approval or a surveyed city plan. A new user should leave this first session knowing what was assumed, what changed, what was reused and what field or supplier evidence is still missing. That is a stronger outcome than merely finding an attractive total on a dashboard.

Discuss the result with another person

A useful follow-up is to ask a planner and a vehicle engineer to describe the same study in their own words. The planner might notice a weak station catchment; the engineer might notice that a proposed frequency relies on unverified stopping. Both can refer to the same source and evaluation without mistaking their different questions for conflicting data. A cost specialist can then identify the largest transferred rates. This shared review exposes the most important uncertainty before anyone invests in a more detailed model. It also shows why the Workbench presents links and holds alongside the attractive numbers: the next decision depends on understanding the conditions of the answer.

2.3 Exploring the city atlas

The city atlas is an invitation to ask better questions about a place. It joins a map to route proposals, station markers, service assumptions, energy, fleet, civil quantities and costs. It also carries the source and uncertainty behind those views. This is more informative than either a line drawn on a street map or a single headline estimate, because the reader can examine what the line is meant to do.

Begin with the route and its neighbourhood

Choose a city, then follow one proposed line from end to end. Look for the origins and destinations it connects, interchange opportunities and places where the route crosses water, major roads or dense urban fabric. An elevated guideway may reduce occupation at street level, but it needs foundations, stations, access and construction space. The atlas can show available road, building, water, land-use and place context where those sources exist. It cannot replace a site survey or a conversation with the people whose streets would change.

The catalogue includes original network proposals and alternatives such as straighter routes or ring forms. Those variants invite comparison of reach, length and service, not a beauty contest between drawings. A straighter line may be faster but miss a useful destination. A ring can improve transfers while requiring more guideway and more complicated operating patterns. Every alternative needs a stated passenger purpose and a realistic access route to its stations.

Move from map to service

Open the service view after understanding the line. Station spacing, dwell, headway and speed assumptions influence journey time, fleet and energy. A train that carries 276 named places at a two-minute interval offers 8,280 places per hour by arithmetic, but an actual timetable also requires protected separation, terminals, boarding time, maintenance and spare trains. The atlas can calculate planning relationships. It cannot measure demand that has not been observed or prove a safe operating headway.

One useful exercise is to compare a city-centre segment with a longer outer segment. Short walks near the centre may justify more stops; an outer line may place greater value on travel time. A single vehicle type is not automatically the lowest-cost service on both. The city result should make such trade-offs visible in offered places, fleet, energy and station scope.

Understand the two kinds of edit

The planning workspace can prepare route, stop, service and origin–destination proposals. A planning handoff preserves those proposals and their source context; it is not itself a calculated railway study. A calculated atlas study, by contrast, has evaluated service, fleet, civil and cost under a declared candidate and assumptions. This difference matters when a reader sees an edited map: moving a marker is not the same event as recalculating a complete project.

The published v0.6.1 release demonstrates a bounded fresh Samawah calculation and includes a retained read-only Baghdad comparison. It does not include a native RustGIS export-and-return journey from the atlas. That route is held at the release because two source publications do not match. Later v0.7 development has worked on the return path; it should not be retroactively attributed to the fixed v0.6.1 package.

Read the cost view as a question list

Fourteen cost scopes help a reader ask where a proposed railway's resources would go. They connect vehicle, structures, stations, power, construction, fleet, staffing, energy, maintenance, renewal and eventual disposal. The atlas's low, base and high cases are conditional estimates. A city should identify transferred rates, missing utility diversions, uncertain ground and unpriced assets before treating any value as a budget.

The most productive atlas session ends with a shortlist of decisions and evidence requests: which corridor deserves a survey, which station needs an access study, which service assumption dominates fleet cost, and which local price or ground condition could change the preferred option. The catalogue covers 266 cities, but that

breadth is a library of planning prompts, not 266 independent construction packages.

Invite local correction

An open atlas should be easy to challenge. A resident may point out a barrier that a map misses, an interchange that works poorly at night or an entrance location that is unsafe in summer heat. A local engineer may know where utilities, flooding or construction access constrain a neat line. Such information belongs as a named revision or source, so the original proposal remains understandable and the new one can be tested. The project gains more from a corrected station and an honest uncertainty range than from a catalogue that appears complete but cannot explain how its choices were made.

2.4 A Samawah Workbench journey

Samawah is the published release's most complete fresh software example. The Workbench loads a mapped city outline, accepts a bounded service request, calculates a result, reuses an unchanged result and produces a separate answer when demand changes. This lets a reader observe the platform's central promise: a design question, its assumptions and its answer remain connected.

The first calculation

Begin with the displayed city and line assumptions. The source record tells the Workbench which network and local information to use. The service request records values such as demand and headway. Before submitting it, note the station and access assumptions. A railway map may show a plausible line while leaving the actual platform, walking route or land rights open; the comparison is still useful if those unknowns remain visible.

Submit the baseline request and follow its status until the result is ready. The summary reports service and fleet, energy assumptions, civil quantity classes and the full cost account. A low, base and high estimate shows how rates and model assumptions affect the total. A reader should resist the temptation to call the base figure “the price.” It is a conditional estimate for this source, train and service, with local prices and site quantities still to be improved.

The unchanged request

Run the same request a second time. The release demonstration shows verified reuse of the sealed calculation. The same result is not a claim that the railway has been physically tested twice. It is a check that the software recognizes identical inputs and avoids a needless recalculation. For a small city edit this is good engineering practice: a repeated screen view should not regenerate a structural model, nor should a changed rate silently reuse an obsolete cost.

The visible result record helps distinguish source, evaluation and selection. The source is what the model used. The evaluation is the calculation under a particular request. A selection is a decision to use a candidate for a stated purpose. The first two can be checked by software; neither is permission to build or operate the line. These distinctions may seem abstract until two similar maps produce different prices. Then they explain whether the difference came from geography, demand or a design choice.

The changed-demand request

In the published demonstration, the reader changes peak demand from 4,000 to 9,000 passengers per hour per direction. The Workbench gives the revised request its own evaluation. The two figures are scenario values, not a measurement of Samawah's peak. Their value lies in showing how a service and cost model responds to a passenger question. The higher case may alter fleet and operating requirements even though the drawn line remains the same. The old result is retained, so the comparison remains inspectable.

This is a different exercise from the v0.7 station-merge comparison described later in the book. Demand changes the requested service. The station merge changes a planning feature while keeping the route and selected reference explicit. Both must preserve their own inputs and must not borrow costs from a different vehicle candidate.

Open the exported records

The release journey offers a short summary, detailed result, source record, performance record and delivery draft. The summary is for orientation; the detailed result allows questions about specific scope and assumptions. The performance record shows whether the answer was calculated or reused. The delivery draft turns model quantities into proposed work and review tasks, without authorizing a purchase or a production order. The disabled business draft illustrates a possible later handoff while preserving the human approval boundary.

One can use this journey to teach a project team how to review a model: state the passenger question, identify the source, inspect all the costs, repeat an unchanged request, change one assumption, and explain the resulting

differences. It is a demonstration of traceable software behaviour. The actual Samawah corridor still needs demand evidence, surveyed access and alignment, site structures, physical train data, supplier prices and independent engineering review.

Carry one question into the field

A sensible next action is to select one station or street crossing whose real geometry could alter the model's result. A field team can measure walking routes, available frontage, platform approaches and utilities, then return with a documented alternative. The Workbench can compare the new proposal with the earlier one under the same service and candidate. The important measure of progress is whether the decision becomes clearer: perhaps one position has better accessible access, or perhaps a cheaper structure creates a longer walk. The software helps keep those effects together; the field visit supplies knowledge the software cannot invent.

2.5 Baghdad and retained comparisons

Baghdad gives the published platform a second, very different city to inspect. Its retained atlas packet includes fourteen lines and a calculated route and service comparison. A reader can move from the map to line assumptions, fleet and cost, then ask which findings might survive a local survey. In v0.6.1 this is a read-only example. It should not be described as a new calculation performed during the Samawah demonstration.

A larger city changes the questions

A fourteen-line network raises different issues from one corridor. Transfers may matter as much as the running time on a single line. A station placement can change access for one neighbourhood while affecting several journeys through the network. The service plan has to coordinate line frequencies, train availability, interchanges and terminal capacity. A useful comparison therefore asks not only how much guideway is drawn, but what passenger journeys the network offers.

The retained atlas can display proposed routes and their service and cost sensitivities. It can show where a line's length, station pattern or fleet assumption influences the result. It cannot establish actual demand, land take, right of way, utilities, ground or construction disruption. A generic allowance for such work is a way to keep a missing account visible, not evidence that the work has been surveyed and priced.

A retained result has a particular value

The Baghdad packet is useful because it preserves a complete, inspectable design-stage calculation. A reader can see the assumptions that made it and compare them with Samawah or with another proposal. But its source identity and publication conditions belong to that retained study. Reading it today does not update its geography, software method or rates. If the city provides a more recent survey or changes its planning policy, a new evaluation should be made with those inputs.

This distinction is especially important when a platform carries many historical studies. A number on an old chart may be numerically exact for its old model and still be irrelevant to a current investment question. Keeping the source and evaluation attached lets the reader use historical work as context, while preventing it from becoming unearned current proof.

The native-map boundary in the published release

The v0.6.1 atlas does not offer a supported editable RustGIS project export for Baghdad. The GIS and outline publications at that release do not have matching source epochs, so the platform holds the native export. The read-only map and source-bound result remain available. Later v0.7 work implements and tests an edit, return and reprice path under new source conditions; those later results are discussed separately.

A city team can still use the published Baghdad study to plan evidence collection. Which station catchments deserve field walking surveys? Which right-of-way or utility corridors are uncertain? What service frequency is worth testing against observed demand? Which quantity accounts need local rates or supplier input? Each answer should lead to a clearly named new proposal and evaluation, rather than an invisible alteration to the retained packet.

Baghdad thus shows both the strength and the limit of an open city atlas. The strength is that many connected questions can be inspected together. The limit is that a map and model cannot substitute for the local knowledge, survey, public engagement and physical engineering needed to choose and build a corridor.

Compare two cities without copying one answer

Samawah and Baghdad can help test whether a planning method generalizes. The same questions about passengers, stations, fleet and cost can be asked in both places. The answers should not be imported across cities. The larger Baghdad network, its land and utility situation, and its actual travel patterns need their own inputs. A useful comparison might ask whether a similar headway produces a very different fleet requirement, or whether station access dominates in one corridor while construction conditions dominate in another. A model that explains such differences is more useful than one that merely prints comparable totals.

For a public discussion, the retained Baghdad map should be presented alongside a short list of what people can challenge. Are the destinations right? Are interchanges accessible and safe? Would the route divide a neighbourhood? Which assumptions reflect local observation, and which were brought from a generic model? The answers can form the brief for a new, locally supported study. The retained atlas is the starting page of that conversation, not its conclusion.

2.6 Exploring candidate research

The platform is designed to compare railway candidates, not only to display one preferred train. The published development reference is a passive wheel-supported six-car formation. It leaves installed propulsion open and retains a passive interface for linear-motor research. Earlier and separate candidates explore powered wheels, station drives, recovery equipment and different structures. A useful comparison asks how each candidate delivers the same passenger service with its own complete installed system.

Keep the three identities apart

A physical candidate describes what the railway is: train, running gear, propulsion interfaces, civil form and relevant installation details. An evaluation describes a calculation or test of that candidate under declared loads, assumptions and tool conditions. A selection is a decision to carry one candidate forward for a stated purpose. A change in motor model might create a new evaluation of the same geometry. A new bogie, body or station layout changes the physical candidate itself.

This matters when a research concept looks promising. A candidate with a low preliminary beam mass cannot inherit another vehicle's braking, cost or fleet result. A different propulsion distribution may move weight from the train to stations, change electrical duty and change maintenance and recovery work. Comparing only the attractive component would miss the whole railway.

What the published release makes available

The v0.6.1 Workbench shows the selected passive-wheel development reference. Its current interactive portfolio selector is held because the integrated engineering dossier no longer matches its recorded source. A reader should not expect a live drop-down of accepted current candidates there. The source-pinned coupled-railway archive remains available for read-only research comparison at its historical source conditions. It includes a station linear-motor case and a separately identified powered-wheel case.

The powered-wheel record is a definition study. It lacks an available-force map and has unresolved installed mass and price for the additional equipment. It does not prove that powered wheels can meet the timetable or that they cost less. The station-drive case also has model and hardware limits. The archive is valuable for seeing how the project frames alternatives and preserves their ancestry; it does not elevate either concept to a qualified installation.

Read a comparison without choosing too early

Begin by stating the passenger service: route, demand, places, journey time, availability and accessibility. Then compare each candidate's vehicle mass, station equipment, energy, braking, civil loads, manufacturing route, repair access and whole-life cost. Where an account is unknown, mark it unknown or give a justified bound. Do not treat the missing motor, cooling or protection package as a zero-cost benefit.

The project also contains architecture searches, adaptive design and generative structural experiments. Its outline portfolio names 112 railway bundles, 54 body studies and 18 civil spans. These figures describe the breadth of digital exploration, not the number of physically accepted trains. A screen can identify a promising family or a physical contradiction. It cannot substitute for detailed installation, manufacturing and testing.

A good research result may be a failure with a clear reason. An axial mechanism can fail a pin-jointed load path without disproving a bonded shell that carries force differently. A motor calculation can close numerical energy without representing the complete inverter and thermal duty. The record should say what was tested, what failed and what next experiment would change the decision. This makes the archive useful to future teams without turning old calculations into current approval.

Decide what experiment comes next

When two concepts have overlapping cost ranges, another complete campaign may add little. The better next experiment may be a supplier mass estimate, a thermal test of a short drive section, a full-size door removal trial or a surveyed foundation. Its value depends on whether it could change the choice. Conversely, if a candidate violates a genuine load or clearance requirement, the team should revise its geometry before paying for a fine mesh. This staged approach preserves resources for the promising families without discarding an unfamiliar idea because a simplified model cannot represent its real load path.

The research archive also has a civic value. A local manufacturer can see which components might be made or repaired nearby and which need qualification or foreign sourcing. A city can see whether a proposed saving in train hardware shifts cost into stations or construction. An independent reviewer can reproduce a bounded calculation and identify where physical evidence begins. Open design knowledge is most useful when it keeps alternatives and failed attempts as carefully as the current favourite.

2.7 CAD and engineering views

The drawings and three-dimensional views make the proposed railway easier to understand. They show where a wheel bogie sits, how a body shell could be assembled, where doors and seats might fit, and how a beam could be handled at a construction site. The v0.6.1 package includes rotatable geometry and editable examples, alongside pictures meant only to explain a concept. A reader should ask which kind of image is being shown before drawing a conclusion from it.

Look beyond the attractive rendering

A concept rendering can make the passenger journey tangible. It can show a bright cabin, a compact elevated station or a precast beam being lifted into place. Its purpose is communication. It does not establish exact clearances, material thickness, structural strength, access for maintenance or a safe erection method. A CAD model can carry more geometry, but it too may omit fasteners, wiring, cooling, seals and working envelopes. A finite-element model can test a declared load and support condition, while leaving real joints or material behaviour uncertain.

The engineering gallery provides models of vehicles, bogies, doors, couplings, launch equipment and guideway elements. Some can be rotated in a simple viewer; others can be opened in editable CAD software. These formats serve different readers. A lightweight 3D view is useful for orientation. An editable solid is useful for testing dimensions and revisions. An analysis mesh or solver result is useful only when its load, boundary and material definitions are known.

Follow one component through the views

Take a passenger door. The cabin image shows how people might board. The vehicle model locates the opening in the body. The parts proposal breaks one operator into header, guide, carriage, drive, belt, leaves, seals, sensors, wiring and fixings. A detailed assembly would then show how these parts mate, how the leaves move and how a maintainer removes the guide. The v0.6.1 views and part definitions support the early stages; they do not close every mating datum, tolerance or removal path.

The same progression applies to a precast span. A picture shows its place in the city. A CAD solid establishes a proposed shape. An analysis model applies axle and environmental loads. The build design must still specify reinforcement or prestress, joints, bearings, piers, foundations, lifting points and temporary stability. Agreement between two solvers on an idealized beam cannot fill those gaps.

What a file format tells you

Rotatable STL and coloured GLB files can communicate geometry efficiently. Editable STEP and FreeCAD assets make it possible to inspect and change selected solids. IFC records can connect a geometry model to a construction or building-information context. The presence of a file in any of these formats says little by itself about whether every installed component is counted or whether the model is fit for manufacture. Some exports can lose source or material metadata; a companion engineering record may be needed to retain the meaning of a solid.

The release includes optional native-tool examples and comparisons using SolidCraft and FieldCraft, with independent FreeCAD and CalculiX routes where recorded. Their support is limited to the stated model domains and available tool versions. A reader can use them to see how a question was examined, not to infer that the whole train or guideway has passed a universal finite-element check.

Good visual review is therefore active. Ask what is missing, whether maintenance clearance is shown, how forces reach the ground, and whether the displayed vehicle and station are the same candidate as the quoted cost. A beautiful picture is most valuable when it helps someone find the next practical engineering question.

Use a view to plan maintenance

A model is especially valuable when it shows the space needed to remove and replace a part. A battery that fits under the floor may still be impossible to lift out with the bogie installed. A door guide may have room to operate but too little straight clearance to be extracted. A cabinet on a station platform may obstruct safe access to a drive. These questions can be asked while the design is still cheap to change.

The published models therefore belong in conversations with workshop, construction and maintenance teams as well as analysts. Their comments can reveal missing fixings, tools, lifting points or inspection surfaces. The next revision should carry those physical changes into mass, cost and service planning, not only into a more polished rendering.

2.8 Reading costs and delivery drafts

The platform tries to price a railway from the work it would require, not from a single allowance per kilometre. A city estimate connects route length and station pattern to trains, civil quantities, power, staff, production, maintenance and renewal. This is why the published Samawah journey has many outputs. Each answers a different part of the question, “What would it take to provide this service?”

Fourteen scopes, one candidate

The cost model separates rolling stock, guideway, stations and power, factory and tooling, construction, land, utilities and depot work, fleet, energy, operations, maintenance, renewals, recovery, reliability and decommissioning. These scopes help prevent a low vehicle purchase price from hiding a large station or maintenance cost. They also make a route edit easier to understand: a changed station may affect access, energy or operating work even if guideway length does not change.

Low, base and high figures are scenarios under declared quantities, rates and uncertainty. They are not a probability distribution by themselves, and they are not three competing contractor bids. A rate may come from a dated observation, an engineering calculation or a transferred assumption from another place. A reader should look for that distinction before comparing totals. Supplier quotations, local wages, actual ground and utility conditions and finance terms can all move the ranking.

There are also different cost boundaries. A part's factory cost covers stock and proposed process. Delivery adds transport and handling. Installation adds site labour, fixtures, wiring, testing and temporary works. Whole-life cost follows energy, staffing, maintenance, renewal, reliability and eventual removal. An early material subtotal is never a complete train price. A city capital estimate is never automatically a whole-life judgement.

A delivery draft is a proposal for work

The Samawah result can be accompanied by a delivery plan. It takes modelled quantities and produces possible work and resource tasks. This helps a team ask whether factory capacity, shifts, precast handling, construction access and inspection have been considered. It does not establish an actual crew calendar, signed purchase order or approved construction method. An estimate of process hours, for example, needs measured shop performance and the selected equipment before it can become a committed production schedule.

The optional business draft is deliberately reviewable and disabled. Its purpose is to show how candidate-linked engineering records might enter a project and purchasing system after people agree on the parts, suppliers, inspection and authority to transact. Turning a calculated design into a live order requires a separate human and organizational decision. The same boundary applies to any draft maintenance or workforce record.

Use exports to ask a decision question

The short summary makes the case legible to a stakeholder. The detailed result allows an engineer or planner to inspect service and scope. The source record identifies the chosen input context. The performance record explains whether the result was freshly calculated or safely reused. The delivery plan gives production and construction questions a first shape. Together they make it possible for two people to discuss the same case without passing around an unexplained screenshot.

Suppose a city considers a shorter station spacing. It should inspect passenger access and journey time, then fleet and energy, then station and civil quantities, then construction disruption and whole-life cost. If the result changes only a small capital account but changes operations over decades, that difference deserves attention. If a site cost is unknown, the model should keep the uncertainty visible and identify the survey or quotation that would most improve the choice.

The release's cost and delivery outputs are useful design-stage tools. They become stronger as local evidence replaces generic assumptions. They do not approve a budget, contract, manufacturing route or railway operation on their own.

Compare equal offers of service

A low-cost option may appear inexpensive because it has fewer trains, shorter operating hours or a longer passenger wait. A fair comparison states how many people must be carried, what access and journey quality they receive, and how reliably the railway must run. Each candidate can meet those needs differently, but its station, fleet, power, workshop and renewal consequences must remain in the same estimate. Mandatory safety and accessibility costs cannot be excluded to make a design look affordable.

If low, base and high ranges overlap, the result may be “more evidence needed” rather than a winner. The platform can then direct attention to the rate or physical quantity with the strongest influence. This is a productive outcome: a targeted ground survey or supplier quotation may resolve more uncertainty than a broad round of numerical refinement.

2.9 Business, supervision and controls

A railway is built and kept in service by organizations, not only by calculations. Parts must be ordered, inspected, assembled, fitted, maintained and replaced. Staff need training and work records. Equipment needs observation and fault response. The published platform connects these concerns to engineering candidates, while keeping drafts and simulation clearly separate from live transactions or train commands.

From design intent to a business record

The ERPNext and Frappe HR integrations are optional building blocks for project, purchasing, stock, workforce, manufacturing, quality and maintenance workflows. A candidate-linked bill of materials can inform a draft, but a real purchasing item needs a reviewed specification, supplier and source. An incoming part needs a quantity, origin and inspection result. An installed serialized component needs a position, revision and maintenance history. Those records are valuable because they let an owner know what is actually on the railway and why it was accepted.

The v0.6.1 package provides deployment recipes, adapters and reviewable examples. It does not create approved purchase orders or qualify a manufacturing process. A generated draft is a prompt for the responsible team to check the engineering definition and business meaning before conversion. This protects local capability: a workshop can develop and record its own processes without concealing the origin of specialist items it buys.

Observe equipment without confusing observation and authority

FUXA and gateway integrations explore how station and workshop equipment could report status, trends, alarms and maintenance events. A temperature trend might prompt inspection of a drive cabinet. A recurring door fault might lead to a design change or different spare-part policy. The observation record should identify the actual asset and its installed candidate so that a maintainer does not apply the wrong instruction.

These integrations are examples until devices, networks, permissions and procedures are commissioned for a real site. A displayed alarm is not a train-protection decision. The Workbench is a design and review interface, not a live operations desk. A railway's movement authority must come from a separately engineered and approved safety system.

Open control software as research

The control work examines station-led dispatch, automatic train protection, interlocking, doors, train supervision and fault response. Rust is used for deterministic contracts and simulations of state changes. The value of this software is that unsafe or inconsistent conditions can be described, tested and reviewed in the open. A refusal case may be more important than a smooth normal journey: the system should stop a contradictory movement, an occupied route or an invalid door state.

The software still needs real sensors, actuators, communications, braking characteristics, operating rules and independent safety assessment before any live use. A perfect test with an ideal brake curve cannot prove an actual wet-rail stop. Station-powered propulsion and onboard brakes must be studied together, but the onboard protective stop must remain independent of a receiving station drive.

Keep the lifecycle connected

Imagine a door operator whose sensor fails repeatedly. A useful platform should connect the service event to the exact installed module, its supplier or production lot, its drawing and inspection history, and the vehicle candidate. Engineers can then decide whether the problem is installation, maintenance, environment or design. An approved revision could flow into procurement and future train builds without erasing the history of the original part.

This complete loop is a project objective. The release establishes parts of its data and integration route, not a finished live lifecycle. The practical benefit of describing it now is that vehicle designers can include serial identification, accessible inspection and replacement from the first controlled drawing rather than trying to add

them after production.

Give the workforce a place in the design

Local capability requires people who can assemble, test, diagnose and repair the system over decades. Training records, safe work instructions, shift arrangements and tooling matter as much as the drawing of a component. Frappe HR can be part of a configured workforce process, but the platform cannot infer that a city has qualified staff or enough technicians from a modelled labour-hour total. A deployment team has to identify skills, local institutions, supervision and the actual work calendar.

This also changes the design choice. A specialized imported module may have a low first price but require an expensive distant repair channel. A locally serviceable module may cost more to make initially while reducing downtime and dependence on one supplier. The lifecycle record should make those consequences visible to the city and the workshop.

2.10 What v0.6.1 does not claim

The most useful software release is one whose limits are easy to understand. OpenSourceRail v0.6.1 publishes a planning, research and review platform. The release can run a Samawah study, display a city atlas, expose source-linked engineering examples and prepare cost and delivery drafts. It does not say that a vehicle, structure, station or railway has passed physical acceptance.

The map is not a survey

The 266-city catalogue is broad, but many routes, demands, rates and site facts are provisional. A corridor may cross an unseen utility or need different access from a map marker. A cost account may use a transferred rate instead of a local quotation. The atlas helps choose what to survey and compare; it cannot establish land rights, ground conditions, accessibility or local demand without local evidence.

The Samawah fresh study is a software demonstration of a bounded city calculation. The Baghdad packet is a retained, read-only study. They are not two newly engineered city projects. The published atlas also holds native RustGIS export because the GIS and outline inputs do not match. A user should not infer from an editable map concept that a complete native return and repricing journey shipped in v0.6.1.

A model is not an installed vehicle

The wheel-supported six-car reference defines passenger places and running gear for development. Its propulsion architecture, powered length, braking envelope and protected service still need candidate- specific engineering. Mass subtotals do not cover every installed joint, cable, cooling unit, protection item or supplier detail. Door, body, bogie and roof concepts need mating interfaces, fatigue, fire, repair and physical testing. A rotatable 3D view or matching CAD volume is valuable, but it does not certify the full assembly.

Likewise, a finite-element solver can finish a structural model and still leave the real rail seat, bearings, foundations, construction sequence or material behaviour outside that model. A propulsion calculation can close energy numerically without demonstrating a manufactured motor's force, current or repeated heat duty. Physical limits require appropriate measurements and independent review.

A draft is not an order or operating permission

The cost model's low, base and high cases are estimates under stated assumptions. Its delivery tasks are planning drafts, not a release to manufacture or construct. Business adapters can help prepare reviewed records but do not authorize live purchasing. FUXA and gateway examples can organize equipment observations but do not command a passenger train. Rust control simulations have no certified movement authority.

An older integrated engineering dossier remains visibly held at this release because its source binding is stale. The Workbench leaves global engineering cost and candidate portfolio views unavailable rather than showing a plausible but unsupported result. Historical archives remain useful for research at their own source conditions; they cannot fill the current dossier gap by being nearby in the repository.

How a reader can use the release well

Use the city atlas to frame transport questions. Use the Samawah journey to see how a changed demand produces a distinct result. Use the Baghdad study to inspect a retained network comparison. Use the engineering views to find fit, load and manufacturing questions. Use the cost scopes to identify missing local prices, and the delivery drafts to discuss factory and site capacity. At each step, ask what would have to be measured or reviewed before the conclusion could support a decision.

That approach preserves the release's real value. It gives cities and contributors an open, inspectable starting point and a common language for improving the design. The v0.7 programme described later in the book extends the connected workflow, but it does not retroactively change what v0.6.1 shipped or grant physical approval to its studies.

What would strengthen the claims

A city planning claim becomes stronger with observed demand, accessible walking surveys, verified alignment and local ground and utility information. A cost claim becomes stronger with measured quantities, supplier quotations, documented productivity and risk. A vehicle claim needs installed dimensions and mass, verified joints and equipment, physical motion and thermal tests, and independent review. A service claim needs a validated available-force map, braking and rescue envelope, protected control and operating procedures.

Those steps do not erase the value of the software release. They show what work the release is meant to organize. A reader can identify the next evidence item, connect it to the same candidate, and see which conclusions change. That is the path from a public research platform toward a railway a city could ultimately approve and support.

III. From city to corridor

Planning, service and a worked station revision.

3.1 From a city brief to an alignment

A city does not need an identical railway merely because another city has a promising train. Population, topography, heat, flood risk, streets, utility networks, land rights, transfers and local production capacity change the answer. The planning tools begin with geography and a transport need, then make each route and station proposal explicit. The objective is a comparison that a city can question and revise, rather than a map whose apparent precision hides missing surveys.

Start with people and places

An initial brief describes where people travel, when they travel, what connections they need and which journeys are currently difficult. It should include walking and wheelchair access, climate, travel time, comfort, safety and an affordability range. Many of these inputs are incomplete early on. A useful outline can still be made if the missing demand, prices or site facts are labelled as assumptions with a clear plan to improve them.

The city catalogue brings together route lines, stop and transfer markers, street and place context, and other available geographic layers. A planner can propose an alignment, move a stop, alter a dwell assumption or change the service. The revised proposal is then evaluated with a declared vehicle and propulsion policy. Its fleet, energy, civil quantities, cost and dossier can be compared with the parent proposal. An edit to one station need not recalculate an unchanged carbody; a change in a common material price may affect many cost accounts while leaving the geography alone.

Early question	Evidence needed before a build decision
Where should the line go?	Surveyed terrain, land, utilities, structures and local access
Where should a station sit?	Walking routes, platform fit, transfers, foundations and evacuation
How often should trains run?	Demand, dwell, protected separation, terminals and spare fleet
What does it cost?	Candidate quantities, local rates, site works, risk and renewals

Maps are a starting point

A visible route is not yet a buildable alignment. Railway horizontal and vertical geometry, gradients, drainage, bridge interfaces, foundations, access roads and temporary construction space have to be checked. A station marker may be close to a desired destination but difficult to reach across a major road. A neat viaduct line may intersect a utility or require costly ground treatment. The Workbench can identify such questions and show which results depend on them; it cannot create a survey that has not happened.

Native GIS editing is being connected to this process. An editor can export a city project, change a station and return the edit for recalculation. The Samawah and Baghdad development journeys have exercised two consecutive native edits each, including repricing and a refreshed project. Those tests show that the software can carry an edit through a defined workflow. The survey accuracy and engineering acceptability of the route remain separate.

Keep the proposal attached to its assumptions

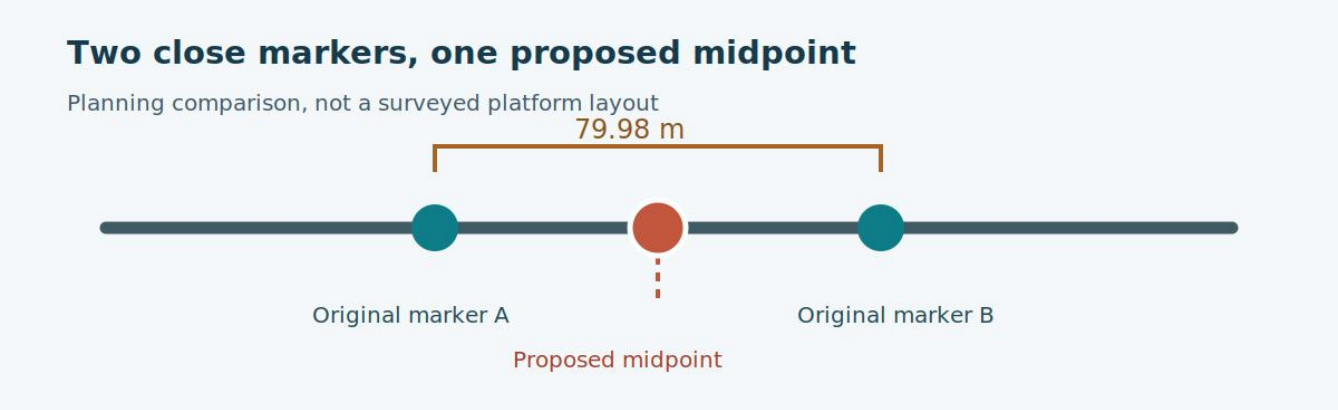
Each corridor revision should retain the route geometry, stops, chosen train, service assumptions and price basis that produced its result. This prevents a later design from borrowing an attractive cost or journey time from a different candidate. It also lets a reader see which inputs changed when two outcomes differ. If the edit reduces one platform or changes one dwell, the report should show the effect on station count, travel time, trainsets, energy, civil scope, capital cost and whole-life cost. A zero change is informative when it is verified; it should not be silently omitted.

The best first result is therefore a questionable, traceable outline. It should be detailed enough to expose the major trade-offs, honest about missing site evidence, and easy to revise when a city supplies better information.

3.2 Samawah: one edit through the system

Samawah provides a concrete example of why railway decisions must remain connected. Two raw station markers on the selected line are about 79.98 metres apart, while the planning case assumes a 120-metre platform. A station merge was proposed to resolve the apparent conflict. The important question is not whether one marker can be deleted from a map. It is what the edit changes for passengers, vehicles, structures and cost, and what it still cannot establish.

The v0.7 comparison replaces the two markers with one midpoint marker without changing the line's route coordinates. The published example had already grouped the pair into one operational station complex. The raw marker count therefore falls from 26 to 25, while the operational count remains 25. The route stays about 57.87 km and the modelled fleet stays at 81 trainsets. Neither outcome means the platforms or walking routes have been surveyed.



Schematic of the two nearby Samawah markers and proposed midpoint

Schematic only. The separation is from the planning record; platform and pedestrian geometry have not been surveyed or approved.

Measure in the planning comparison	Before	Midpoint proposal
Raw station markers	26	25
Operational complexes	25	25
Modelled trainsets	81	81
Base installed capital	about $1.141491 billion	about $1.141481 billion
Base whole-life present cost	about $2.258426 billion	about $2.258656 billion

The model's installed capital falls by roughly $10,000, yet its whole-life present cost rises by roughly $230,000. The selected line's one-way time grows by about 1.5 seconds. Most of the whole-life increase comes from the energy account. The access allowance also changes: one notional repair length is replaced by a shorter midpoint access hypothesis with a separate risk allowance. Those figures are useful because they show the direction and scope of the model response. They are not a local construction quotation, a walking survey or proof that the station is feasible.

What stayed the same matters too

The other four lines were not redrawn. The vehicle and manufacturing reference did not change. The civil quantity identity was preserved because this particular edit did not revise the modelled guideway. All fourteen cost scopes were compared, including scopes with zero change. Reusing unaffected results keeps the study focused and makes it easier to understand why a cost moved.

The comparison also keeps its adverse findings. A line-1 coast-reach question, actual propulsion and braking evidence, transferred prices, platform fit, catchment and walking access still need work. A separate compact-drive research case once examined a similar midpoint idea with a different body candidate. Its fleet or cost result is not transferred into this passive-wheel comparison.

The next on-site decision

The midpoint should be tested against the real pedestrian network, frontage, crossings, utilities, station entrance and platform geometry. A suitable station might be placed at the midpoint, at one original marker or elsewhere; the model alone does not decide. The value of the software journey is that each new proposal can be carried through the same service and cost questions, while the unresolved site evidence stays visible to the people making the decision.

3.3 Networks, service and city variety

OpenSourceRail contains planning outlines for 266 cities. This breadth is useful for exploring how different urban patterns might influence a railway. It is not a claim that 266 independent surveyed, engineered and priced systems are ready to build. Many city outlines use shared assumptions or transferred rates. A city dossier can help an owner ask better questions while remaining clear about the evidence that is still local and absent.

One city, several possible services

An alignment is only part of a network. The number and position of stations, interchanges, service hours, frequency, train formation and spare fleet affect the usefulness and cost of the railway. A dense centre may reward shorter walks and frequent stops; a longer peripheral corridor may need a different spacing or an express pattern. The same train may not be the lowest-cost choice for both. Passenger demand by time of day matters as much as a total city population.

The software can model station counts, route lengths, timetables, fleet and energy under stated assumptions. It can compare low, base and high cost cases across capital, operating, renewal and financing accounts. These are design stage scenarios. They cannot replace observed demand, protected headway, supplier prices, actual ground conditions or approved operations. Where a route is drawn through uncertain terrain, the corresponding civil and risk cost should remain provisional.

Geography and economics must move together

A change in route may alter length, curves, gradients, bridges, station access and land take. That can change travel time and energy as well as structures and cost. A change in demand can alter the fleet and power requirement even when the line stays in place. Conversely, a change in local wages should reprice the applicable labour work without pretending the railway geometry changed. This is why the planning platform keeps a relationship between inputs and results, instead of publishing a new opaque total after every edit.

Baghdad is a useful second test after Samawah because its corridor and city context differ. A native GIS workflow has returned a sequence of station edits from both cities to the calculation and back to a refreshed project. That checks the continuity of the software route. The engineering case still needs its own surveyed alignment, site-specific structures, station access, demand and candidate-specific operating assumptions.

From outline to city decision

The first decision might be which corridor deserves a survey, where an interchange should be studied, or which train and civil family merit a detailed comparison. It need not be a commitment to construct. A strong city proposal states its transport objective and alternatives, records the inputs, shows the cost and service sensitivities, and lists the missing evidence likely to change the ranking. It also includes the people who will use the system: wheelchair users, older passengers, children, shift workers and those whose journey crosses the proposed line rather than following it.

The 266-city catalogue is a broad planning library. The detailed v0.7 work concentrates on one Samawah corridor, then one Baghdad corridor, so that a complete change can be followed through design, cost and delivery before the same method is offered more widely.

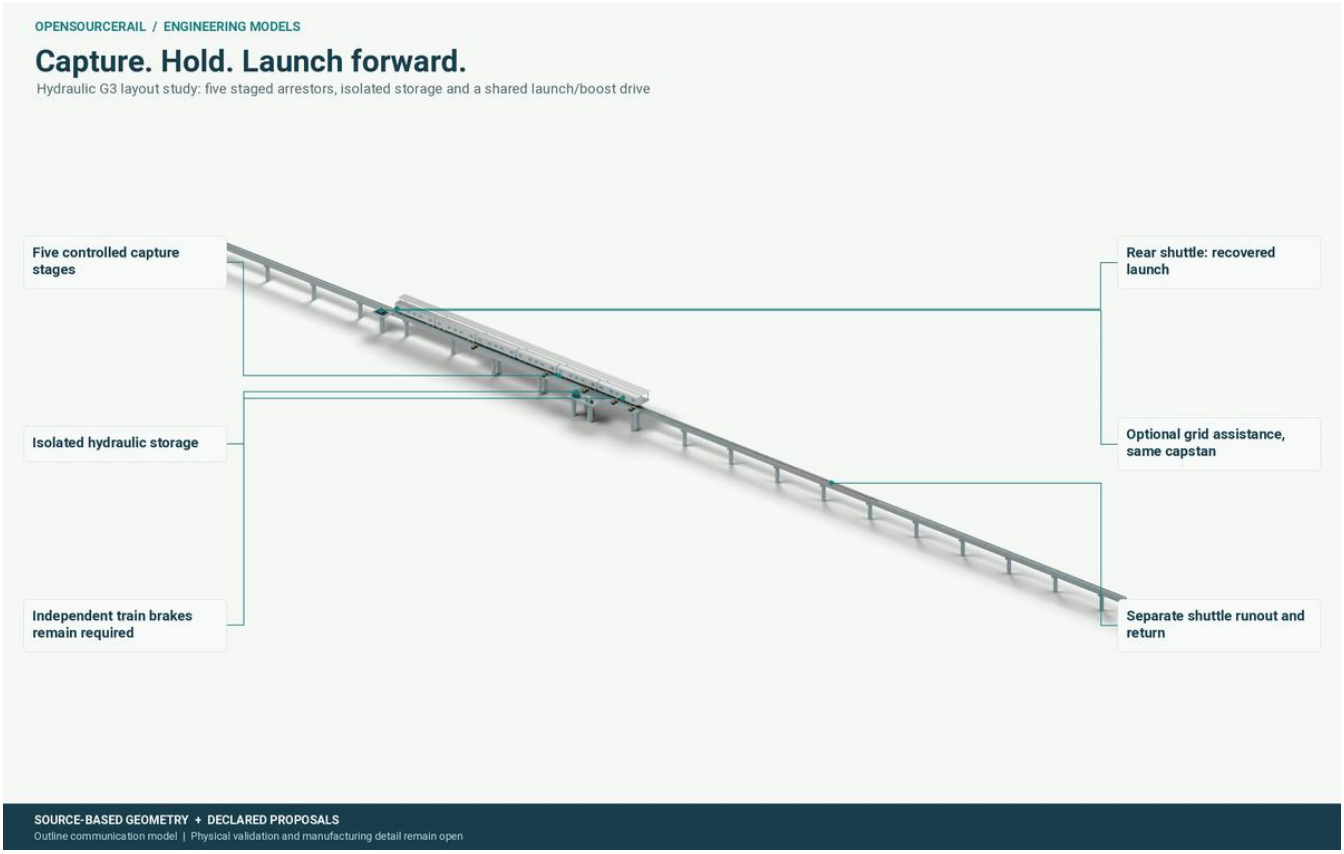
3.4 Stations and the streets around them

A station is a connection between the railway and the city, not simply a point on a line. Its success depends on whether a person can find it, reach it safely, enter without unnecessary effort, wait comfortably and transfer to the next part of the journey. The arrangement must also give staff safe access to the railway, its power equipment and its structure.

The journey begins outside the platform

An elevated line can save street space, but height creates access work. Entrances, lifts, stairs, ramps where feasible, wayfinding and emergency routes have to connect both sides of any road barrier. A station placed a short distance from a market may still be a poor choice if the legal and accessible walking route takes a long detour. Heat, shade, lighting, drainage, crossing safety and the availability of local transport belong in the station brief from the start.

Platform length must follow the selected train and its stopping tolerance. The current reference formation is roughly 104.5 metres long under the proposed inter-car gap assumption; a 120-metre platform is the planning control used in the Samawah comparison. That allowance cannot be treated as a complete platform design. End clearances, doors, accessible boarding positions, sightlines, egress, structures and future operating choices must still be resolved. Changing the formation after setting the station layout would change much more than a vehicle drawing.



Station and street concept

The concept shows a connection between rail and street. It does not settle the site's land, access, fire or structural design.

A stop and a station complex are not the same thing

City source data may contain several markers for what people understand as one destination. In Samawah, two markers were found about 80 metres apart, closer than the platform planning assumption. Their merger reduces the raw marker count from 26 to 25 while the operational station-complex count remains 25. That distinction matters: the data cleanup can be useful even when a model says that the number of built stations is unchanged.

The midpoint of two markers is a comparison device, not automatically the best entrance location. One side may have better walking access; the other may have poorer ground or a utility corridor. A real station choice requires an accessible-catchment study, land and ground checks, utility information, public-realm design and engagement with people who will use it. The calculated capital and travel-time changes in the Samawah chapter help frame that inquiry; they cannot decide it alone.

The station as a maintained asset

Platforms, lifts, lighting, fire systems, drains, signs, communications and power cabinets need inspection and replacement space. A design that can be built only by closing a major street for months, or maintained only by blocking both tracks, may cost more over its life than a slightly larger initial structure. Access for cleaning, repair vehicles, emergency services and safe isolation should be tested before the layout is fixed.

A station proposal can move through increasing levels of evidence. First comes the city outline and the passenger purpose. Then come surveyed geometry, access routes, ground and utilities, a matched train and service, and a civil concept. Detailed fire, egress, structural, electrical and operational reviews follow. The Workbench's value is to retain those links and show what a station edit changes, while displaying missing evidence plainly. It should never make an unsurveyed map look like an approved site.

IV. Ten city route examples

Calculated maps from different city contexts, with their planning limits.

4.1 How to read a city route map

The city maps in this part are invitations to ask transport questions. A coloured line shows a proposed route through available geographic context. Station symbols show operational platform groups, and interchange references show intended connections between lines. Water, streets, land use and place names help a reader understand why a route might serve one corridor and miss another. None of these marks means the alignment has been surveyed or that a station entrance can be built where its symbol sits.

What a map can already tell us

A network outline makes several questions visible at once. Is the proposal a few long radial corridors, a dense cross-city web, or a web with a ring? Where would passengers change lines? Do the longest journeys pass through several interchanges, and could a shorter local journey be made without crossing the city centre? A map can reveal a possible service gap or an unexpectedly close pair of station markers. Samawah's proposed merge began with precisely that kind of observation.

The legend deserves attention. Different colours identify proposed lines, while small symbols distinguish an operational stop from an original access marker or an interchange reference. Several symbols can refer to one possible station complex. A line shown near a named place indicates map proximity, not a verified entrance there. The scale bar is useful for orientation, but an accessible walking route should be measured along actual streets, crossings and level changes.

The catalogue also attaches modelled service and cost records to each city. Route length, number of lines, operational platform groups and fleet size can be read together with the map. In the examples that follow, fleet figures include a modelled reserve and assume one declared six-car reference and service policy. They are neither an order for trains nor a prediction of actual demand. The broad catalogue gives a common starting point for comparing cities while leaving site-specific decisions to each city.

What the symbols do not establish

A short straight line between a station and a street can be a schematic access connector. It is not a measured walking route, step-free entrance or legal right of way. An operational platform group may combine several raw source markers. It is not necessarily one built platform or one land parcel. An interchange reference says where two services might connect; transfer distance, accessible circulation, fire escape and timetable coordination remain to be designed.

The background geography has its own limitations. An open map may be very helpful for locating rivers or broad street patterns, yet omit a utility, recent building, retaining wall or land restriction. Digital screen distances are not construction chainages. Curves, grades, drainage, bridge clearances, foundations, temporary works and the space needed for a station can change the route materially. This is why a planning outline should lead to survey and local engagement before an attractive line hardens into an engineering promise.

Compare cities without equating them

The ten examples range from a five-line Samawah outline to much larger Baghdad, Karachi and Nairobi networks. Their published maps use a common visual language, but they should not be ranked by total route length or fleet count alone. A larger mapped network may reflect a larger study area, more radial branches or an assumed service level rather than stronger evidence. Population context in a dossier is not a travel survey. A common peak-flow brief is useful for testing a calculation but cannot substitute for trips by origin, destination and time of day.

The most productive comparison asks what evidence would change a city decision. For a compact network, it might be access to two closely spaced stops. For a regional ring, it might be whether the proposed orbit serves real cross-town demand or merely adds a long line and fleet requirement. For a river crossing, it may be bridge location, flood level and land take. For a hillside city, gradients and station access may govern. The map is the beginning of that discussion, not its conclusion.

From picture to project

A city can start by choosing one useful corridor, gathering observed demand and walking routes, and marking the places where ground, land or utilities could prevent construction. It can then compare several alignments under the same passenger brief. Each revision should retain its own route, stations, service, vehicle, civil assumptions and price basis. A changed station can alter travel time and energy while leaving the train unchanged; the resulting report should say both. The later engineering and delivery work has to follow the selected revision, not an appealing figure borrowed from another city.

The atlas can also help communicate uncertainty to people who do not read engineering calculations. A public meeting can discuss why a proposed line serves one district, what would be needed to cross a road safely, or which destinations have been omitted. The map should be presented with those questions and the missing evidence, so that feedback can alter the proposal before a more expensive design is commissioned.

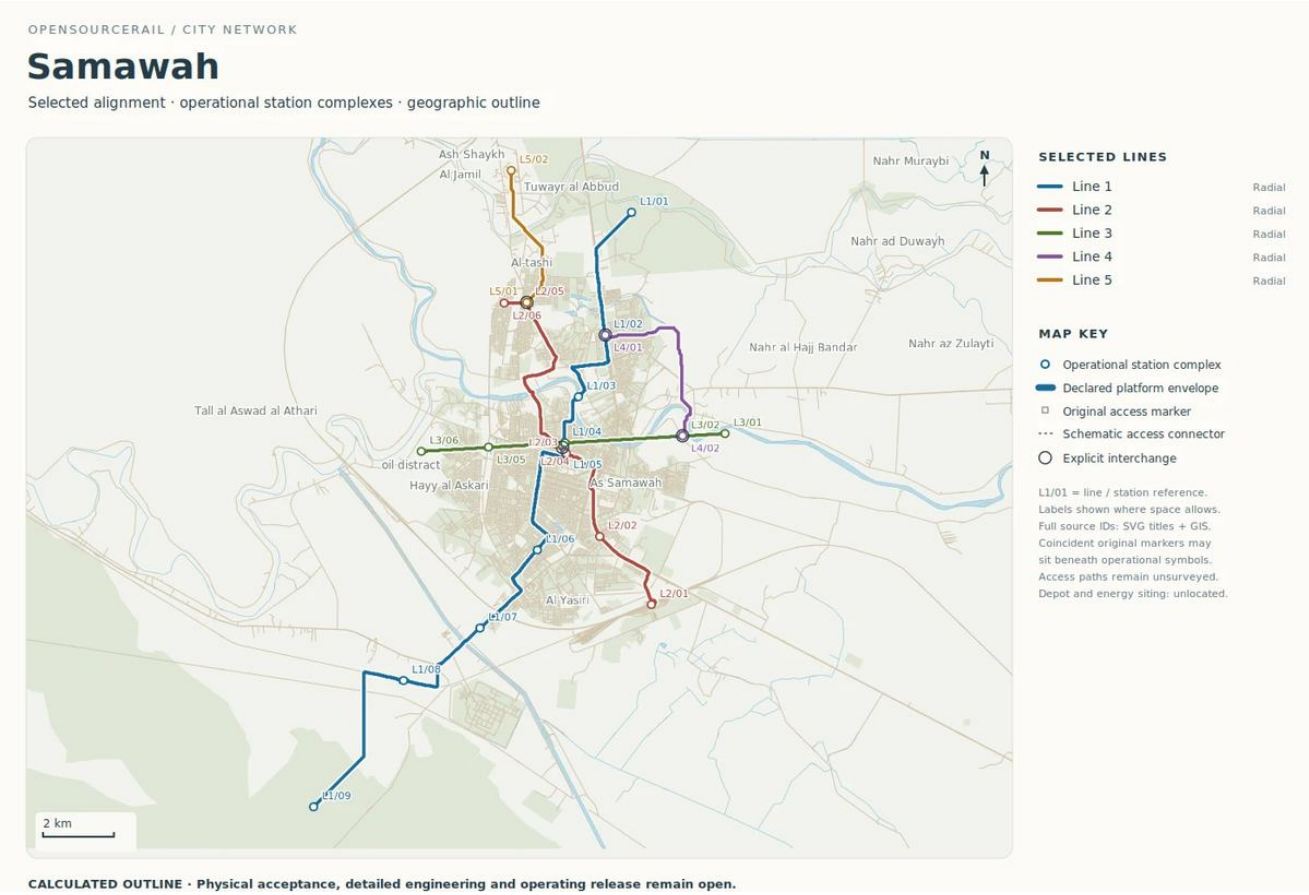
The following maps are cropped reader editions of the catalogue's original route drawings. Their source drawings and detailed assumptions remain available in the project. Captions mark them as calculated planning outlines. They help us see the scale and shape of a possibility while keeping the missing local evidence in view.

4.2 Samawah and Baghdad

Samawah and Baghdad are the clearest pair for understanding how the same platform can serve different questions. The Samawah outline is small enough to follow a single station revision through the entire city study. Baghdad tests a much larger network, with more transfers, competing route choices and greater pressure to keep each line's assumptions distinct.

Samawah: resolve one local decision

The retained Samawah map has five lines over about 57.87 route kilometres, with 25 operational platform groups and four explicit interchange references. Its 81 six-car trainsets include modelled reserves. The longest line is about 26.6 kilometres, crossing the study area from northeast to southwest; shorter routes connect other directions and bring the lines into one planned transfer group. These are calculated outline quantities. The map does not tell us that every station entrance or transfer path is accessible on the ground.



Samawah route planning outline

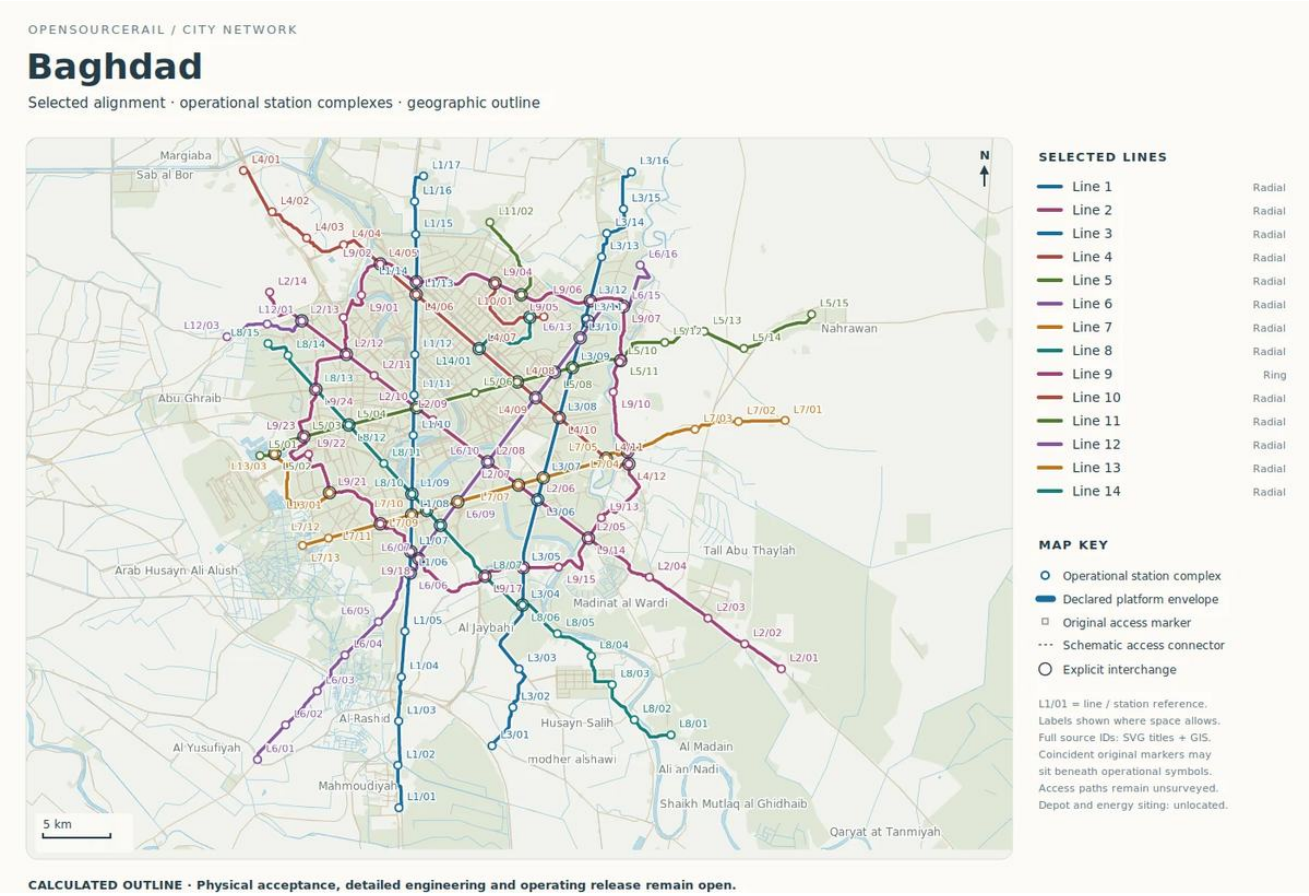
Calculated Samawah route outline. Alignment and access are unsurveyed; this is a comparison map, not a construction design.

Two raw markers on the selected line sit about 80 metres apart, closer than the 120-metre platform planning assumption. The v0.7 comparison merges them at a midpoint. It reduces raw markers from 26 to 25, but the earlier study already treated the pair as one operational station complex, so the operational count remains 25. The route and modelled fleet also remain the same. A reader can therefore see an important distinction: cleaning a source map may be necessary even when a headline count does not change.

The comparison's small capital decrease and larger whole-life increase show why first cost is only one part of the decision. Energy and access allowances respond to the changed stop hypothesis. A city should now test the actual pedestrian catchment, road crossings, land, utilities, platform fit and station entrance before preferring the midpoint to either original location. The software can make those alternatives traceable; people in Samawah must establish which place works.

Baghdad: keep a large network intelligible

The Baghdad catalogue outline is about 494.43 route kilometres over 14 lines, with 157 operational platform groups and 45 explicit interchange references. Its modelled fleet including reserve is 533 six-car trainsets. One ring is about 96.79 kilometres long, while several cross-city lines reach beyond the central cluster. The map makes the transfer problem visible: the apparent density of coloured routes is useful only if passengers can make practical, accessible connections and if the service pattern supports them.



Baghdad route planning outline

Calculated Baghdad route outline. Alignment and access are unsurveyed; the interchange symbols do not approve station construction.

The size of the Baghdad proposal multiplies uncertainty. A model can calculate line times, fleet and cost for its declared network, but a city decision would need observed travel patterns, river and bridge options, ground and utility information, station access, road impacts and land rights. A long ring deserves its own demand case: it may relieve journeys that would otherwise cross the centre, or it may be a costly route with few useful transfers. The outline alone cannot settle that choice.

In the published v0.6.1 software, Baghdad is a retained, read-only atlas comparison. It should not be described as a fresh Baghdad calculation from that release. Later development has exercised a native map-edit and reprice cycle on a Baghdad line, but that is still digital workflow evidence, not a surveyed alignment. The city's source-bound maps and calculations remain valuable because they expose exactly which next local investigation would make the planning result more credible.

For a city team, a workable Baghdad pilot would begin by naming one passenger corridor and its alternatives. The selected study would need measured trips, terminal and interchange opportunities, a chosen train formation and a station pattern. Only then would it make sense to compare construction quantities and fleet requirements under equal service. A cost from the larger outline cannot be divided by kilometres and called the price of that pilot: a short stage may carry depot, power or factory costs that the network total spreads across many lines.

What the pair teaches

Samawah rewards a detailed before-and-after examination of one stop. Baghdad tests whether the same discipline survives a larger network. Both need the same kinds of evidence before building: local demand, access, survey, ground, utilities, a selected train and protected service. The number of lines does not change the standard of proof. It changes how carefully the study must track which part of the network a revision actually affects.

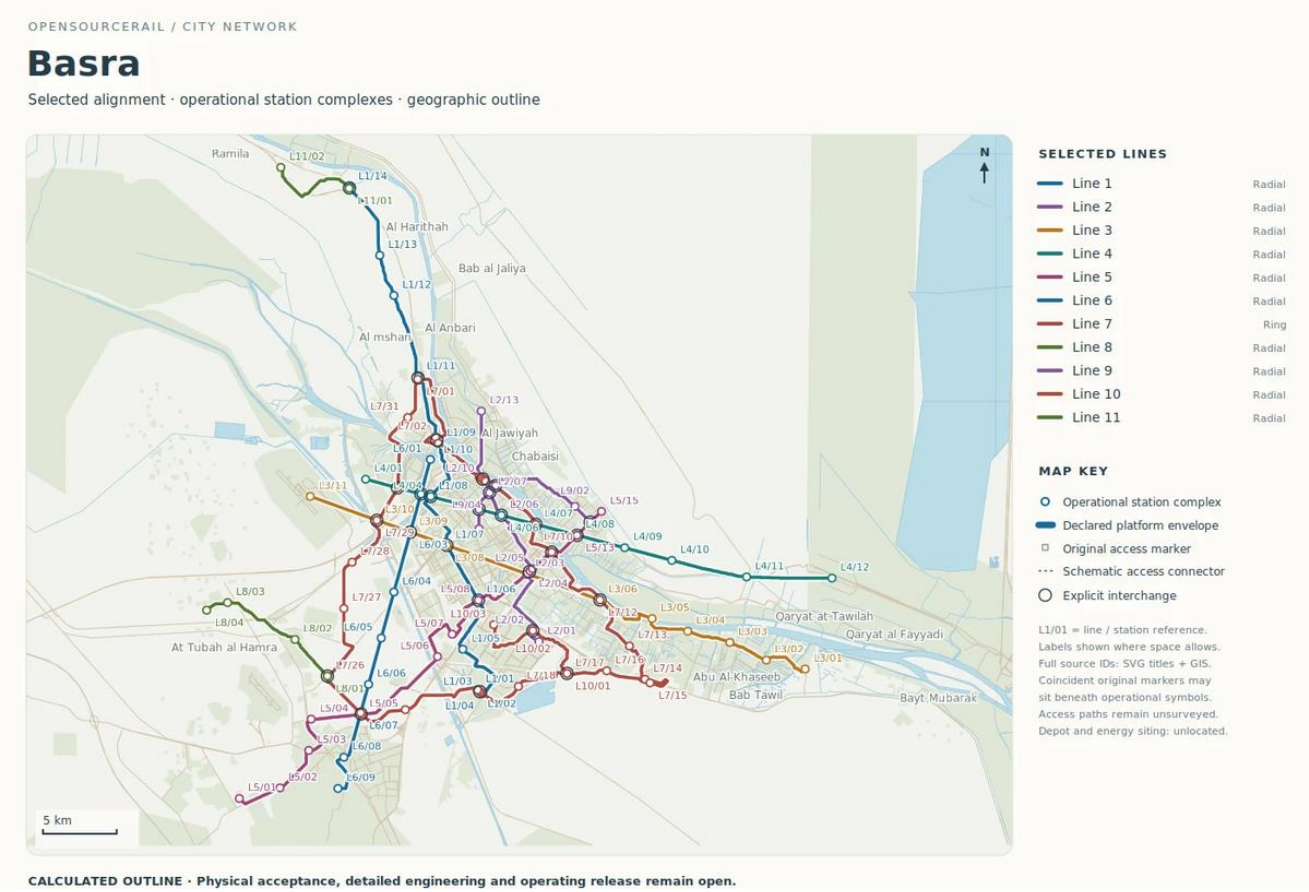
It also changes the conversation with the public. In Samawah, residents could compare walking routes to three nearby station positions. In Baghdad, several districts may be affected by a line choice, a transfer or a proposed ring. A clear map can bring those choices into view, but the people who would use the railway must help decide what counts as a useful and accessible journey.

4.3 Basra and Erbil

Basra and Erbil show why a standard train does not imply a standard city railway. Their retained outlines use the same broad planning vocabulary, yet the proposals differ in length, number of lines, transfer structure and the physical questions likely to dominate the next study. A useful comparison reads the map and the service assumptions together without claiming either city has chosen this system.

Basra: a long network with an orbital question

The Basra catalogue maps eleven lines over about 341.45 route kilometres. It groups stops into 120 operational platform locations and records 26 explicit interchange references. The modelled fleet is 369 six-car trainsets including reserve. The longest line is a ring of about 94.55 kilometres with 31 platform groups. The ring is visually prominent, but its value depends on real journeys: whether it connects districts people travel between, shortens transfers, or simply offers a long circumferential route under a generic demand brief.



Basra route planning outline

Calculated Basra route outline. Route, access and ground are unsurveyed; the map is not an approved construction alignment.

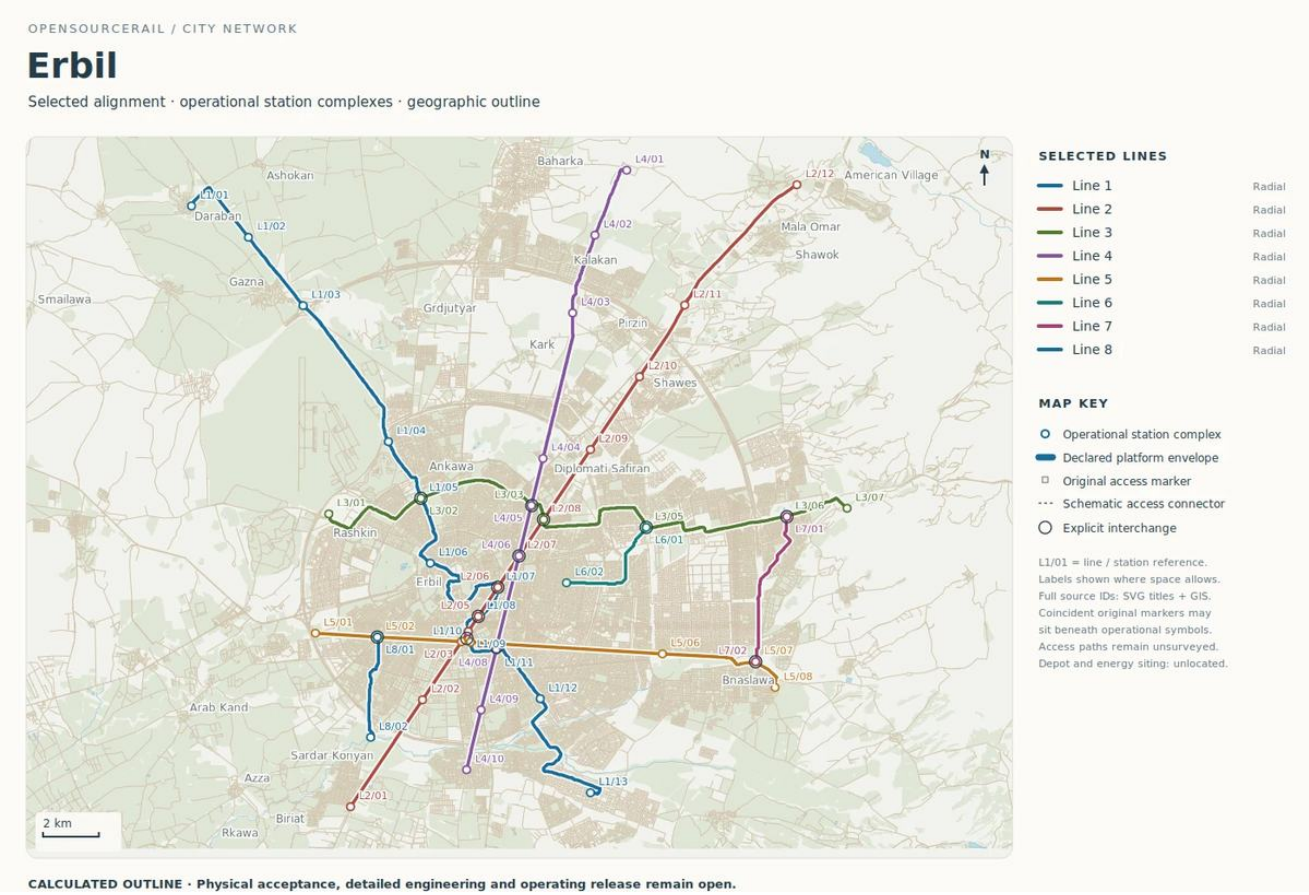
Basra's map invites a closer look at the relationship between routes and water, built districts and major transport corridors. A coloured crossing is a planning line, not a bridge design or a confirmation that a suitable right of way exists. A local study would need ground and flood evidence, utility records, crossing and drainage concepts, station frontage and the actual walking network. These facts can change the alignment, civil form, erection method and cost much more than an early vehicle price change.

The ring also asks an operating question. A long line takes time to complete a circuit; fleet size depends on that time and the proposed headway. If local demand is concentrated on a few radial journeys, a shorter staged build could serve people sooner. If many trips are cross-town, the ring may have a different case. The model helps calculate each scenario once the city supplies a credible demand and interchange brief; it cannot infer those journeys from a population total.

Construction staging is similarly specific. A useful first route might avoid the most difficult crossing, or it might need that crossing to reach the passengers it is meant to serve. The choice should include temporary traffic arrangements, transport of precast components, access for cranes and future maintenance. A map can show the crossing location; the cost and practicality come from a surveyed construction method.

Erbil: shorter network, close transfer decisions

Erbil's outline is about 145.24 route kilometres across eight lines, with 56 operational platform groups and twelve explicit interchange references. Its modelled reserve-inclusive fleet is 168 trainsets. Unlike Basra's long ring, the longest listed Erbil line is a roughly 38.02-kilometre cross-city route. This gives a different planning emphasis: where the crossing lines meet, which station entrances can work together, and how much local coverage the smaller set of platform groups offers.



Erbil route planning outline

Calculated Erbil route outline. Stations and access are unsurveyed; the map does not establish buildable platforms or transfers.

The dossier records the mapped network and its modelled service, but the actual transfer quality still depends on platform arrangement, level changes, safe pedestrian crossings, accessible circulation and the surrounding street. A line drawn across difficult ground may require a different civil section or station approach. Such choices belong to a surveyed corridor study, not a generic per-kilometre allowance.

Erbil also illustrates why a city should distinguish total network size from the first useful project. A smaller integrated corridor can test real demand, walking access and local construction capability. Its train, station and guideway candidate could then be extended if the evidence supports it. The platform's catalogue is a menu of questions for that staging conversation, not a mandate to build every coloured line at once.

The station study should also ask what a transfer feels like. A short map distance between lines can hide a long climb, a road crossing or a platform change without step-free access. Those details alter the passenger benefit and may require a larger station than the planning allowance. A corridor comparison should include the entrance and interchange arrangement, not just the line centreline.

A shared next step

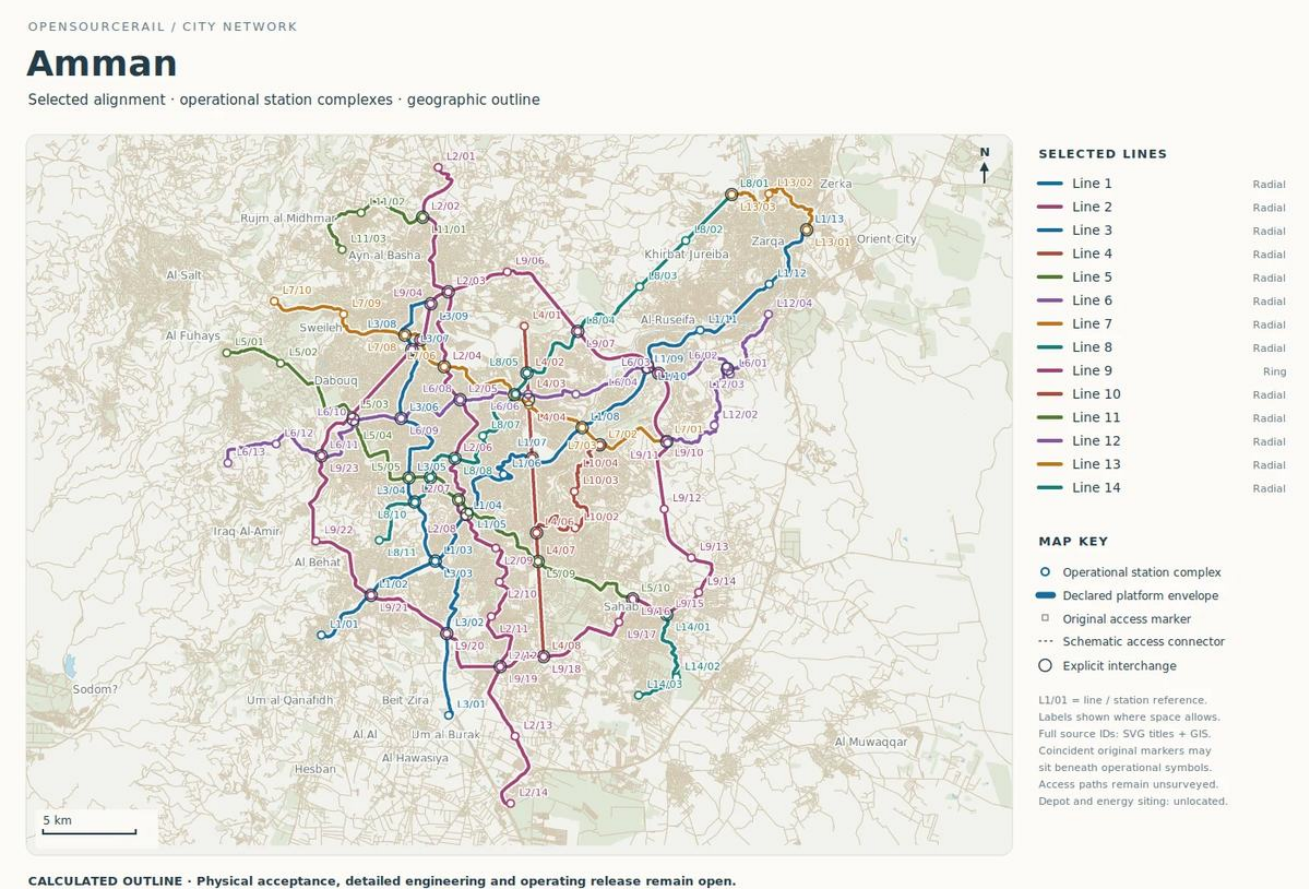
Both cities need local origin–destination evidence, surveyed alignments, ground and utility information, accessible station concepts and a selected service. Their published diagrams are calculated planning outlines under declared assumptions. Comparing them is useful precisely because it shows that the same engineering method must remain sensitive to different city forms rather than force every place into one layout.

4.4 Amman and Karachi

Amman and Karachi occupy very different physical and urban settings, yet their catalogue outlines can be read with the same questions: which trips are served, how the lines connect, what a station needs from its street, and which geographic obstacles could change the engineering. The mapped figures are intentionally provisional. They can expose a design decision before a city commits money to detailed investigation.

Amman: a network through a demanding landscape

The Amman outline records fourteen lines and about 412.73 route kilometres, with 128 operational platform groups and 39 explicit interchange references. Its modelled fleet including reserve is 469 six-car trainsets. One ring is about 82.33 kilometres long and crosses the network at several points. The map helps a reader see possible connections between radial and orbital journeys. It does not show that every drawn route can meet railway grade, curve, station and construction requirements in Amman's actual landscape.



Amman route planning outline

Calculated Amman route outline. Grades, access and ground are unsurveyed; this is not a construction route or station plan.

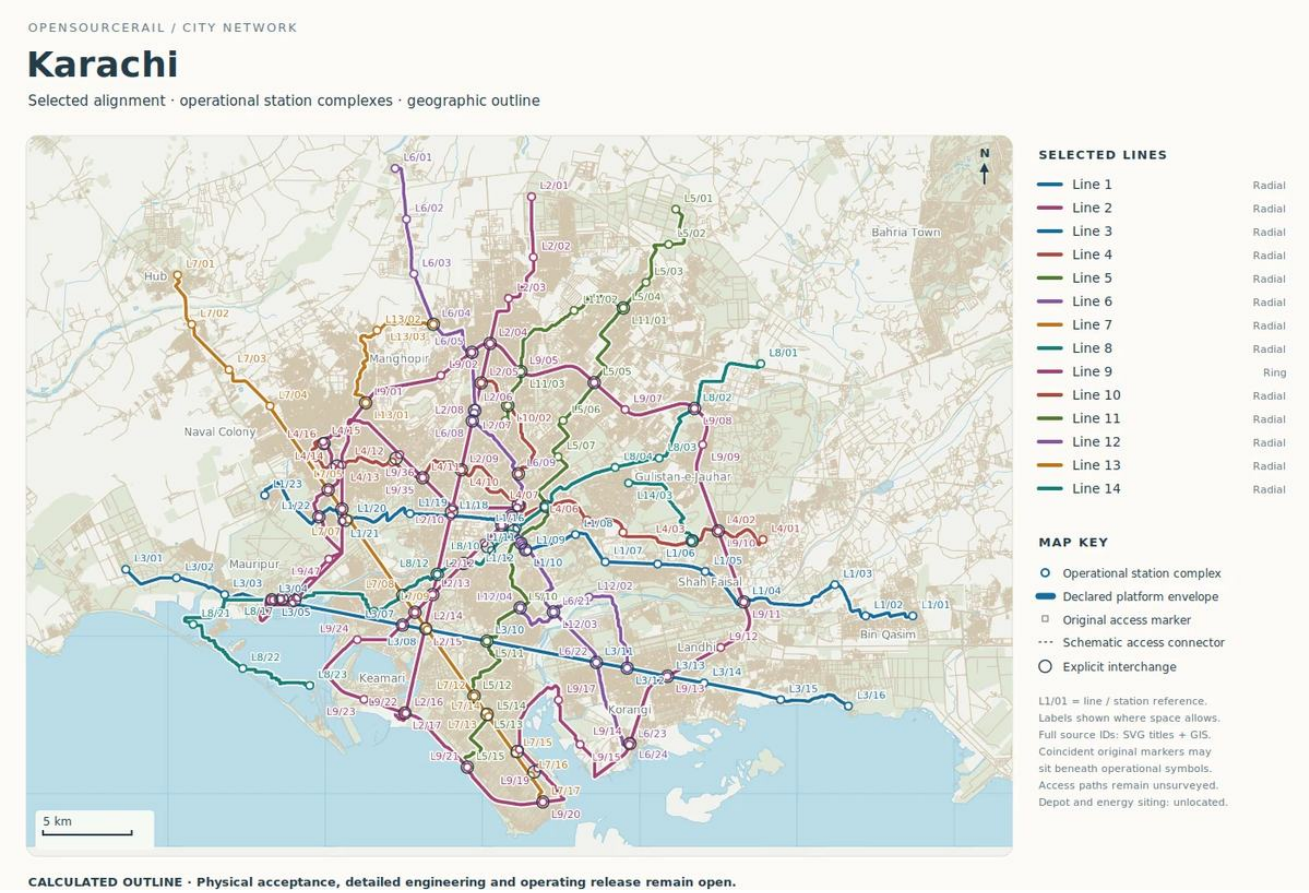
For a hilly corridor, the vertical profile may matter as much as the plan view. A proposed station could be close in map distance to homes but far away by an accessible walking route. A guideway might need long approaches, retaining structures, difficult foundations or a different arrangement at an interchange. The candidate vehicle's acceleration, braking and recovery also depend on grade. These questions should be tested on the same surveyed alignment rather than answered separately by a flat-map cost and a generic journey model.

The ring invites an equal-service comparison. A different pattern of transfers or a shorter set of lines might offer similar useful journeys with less guideway, or it might leave important cross-town trips unserved. The city needs observed travel demand, walking and feeder connections to decide. The outline gives a structured starting point for comparing those options.

The selected vehicle matters to that comparison. On a steep or closely stopped route, a propulsion concept that performs well on level ground may need a different powered length, supply or recovery plan. Independent braking must be shown for the actual gradients. A price estimate based on one vehicle and a journey time based on another would make the ring look more credible than either complete candidate permits.

Karachi: a much larger transfer problem

Karachi's retained map has fourteen lines over about 551.34 route kilometres. It lists 221 operational platform groups and 58 explicit interchange references, with 618 modelled trainsets including reserve. The longest line is a ring of about 143.62 kilometres with 52 platform groups. Those are large planning quantities. They are not a procurement programme, and the population context in the dossier is not a measured rail demand forecast.



Karachi route planning outline

Calculated Karachi route outline. Route, station access and capacity remain unsurveyed and unapproved; the map supports comparison only.

The density of route and station marks raises questions that a simple network total hides. Which interchanges would actually allow short, accessible transfers? Where would a station fit without cutting off a busy street or market? What are the effects of a long ring on journey time, fleet and maintenance? Could a staged corridor meet a defined transport need before the whole mapped network is studied in detail?

Climate, drainage, land and utility records also affect the design. Assuming one civil cost per kilometre would conceal places where a foundation, crossing or terminal is much harder than a typical span. Similarly, applying one peak demand brief to every line cannot describe different times and directions of travel. The platform can compare candidate service packages, but Karachi's local travel and site evidence must set the brief.

At this scale, the order of building becomes a service decision. A partially built network may have few useful transfers even when its final map has many. The city would need to test each stage as a stand-alone travel offer: where people board, how often trains arrive, which depot and power facilities are required, and what remains to be built later. Such staging can avoid presenting the cost of an unfinished but unusable network as an affordable first step.

What the comparison is for

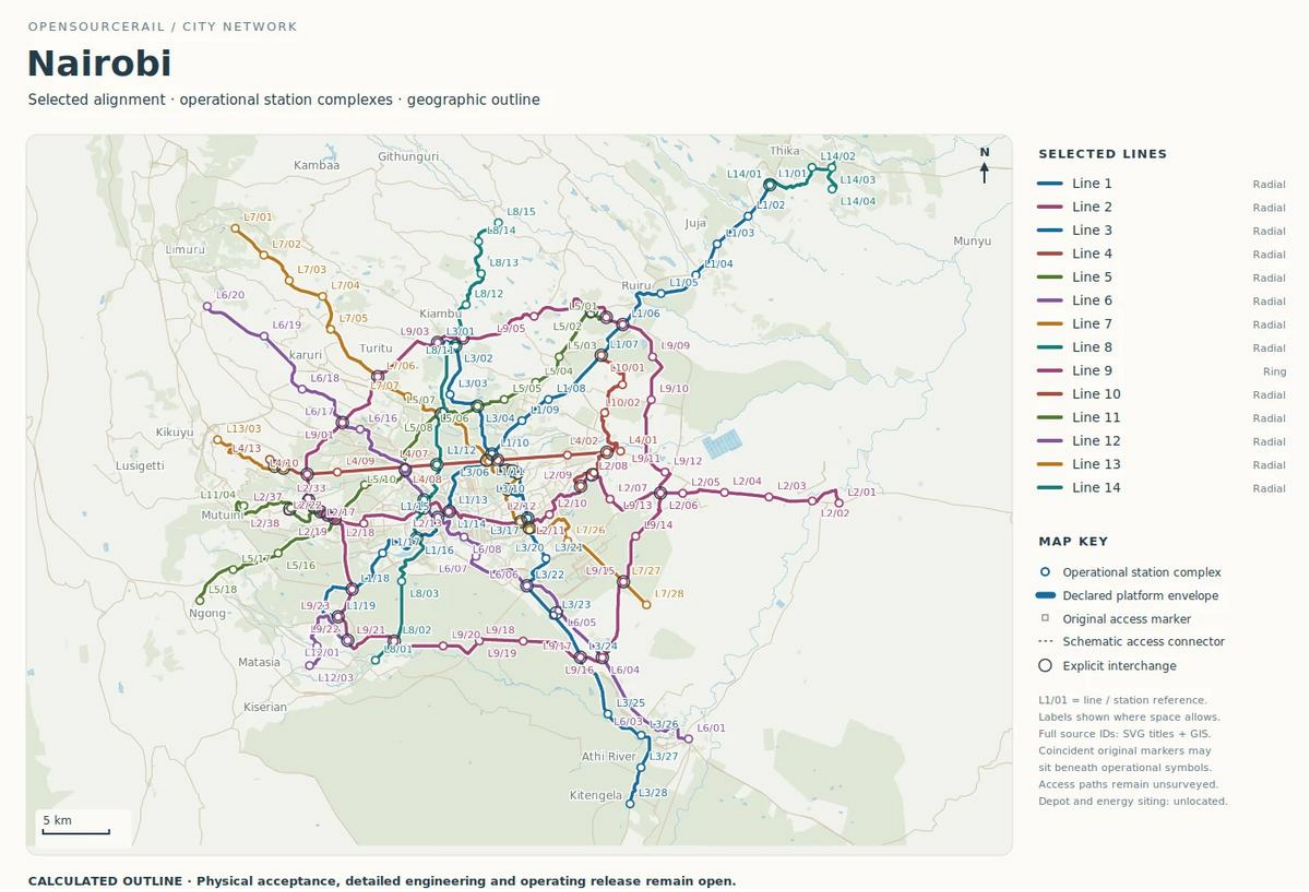
Amman stresses three-dimensional alignment and access; Karachi stresses network scale and interchange practicality. Both challenge the idea that a catalogue figure is a finished city design. A meaningful next project would select one corridor in each city, validate its passengers and site, then carry the same candidate through train, structure, power, station, construction and whole-life cost. The map tells a reader where to begin that work, while its missing facts remain visible.

4.5 Nairobi and Lusaka

Nairobi and Lusaka show how a route drawing can prompt different questions about reach, frequency and transfer. Both maps extend beyond one central district. Their line lengths and fleet totals are calculated from a common planning method, so the numbers are useful for examining consequences. They do not prove the corridors will attract the assumed passengers or fit every road, river and neighbourhood they cross.

Nairobi: many destinations, many transfers

The Nairobi catalogue outline covers about 578.33 route kilometres across fourteen lines. It has 243 operational platform groups and 51 explicit interchange references. The modelled fleet including reserve is 655 six-car trainsets. One ring is roughly 107.86 kilometres with 44 platform groups, while long radial and cross-city routes reach many parts of the study area. The resulting map is a rich hypothesis about connectivity, not a verified construction programme.



Nairobi route planning outline

Calculated Nairobi route outline. Alignment, access and interchange quality are unsurveyed; no route shown is a construction design.

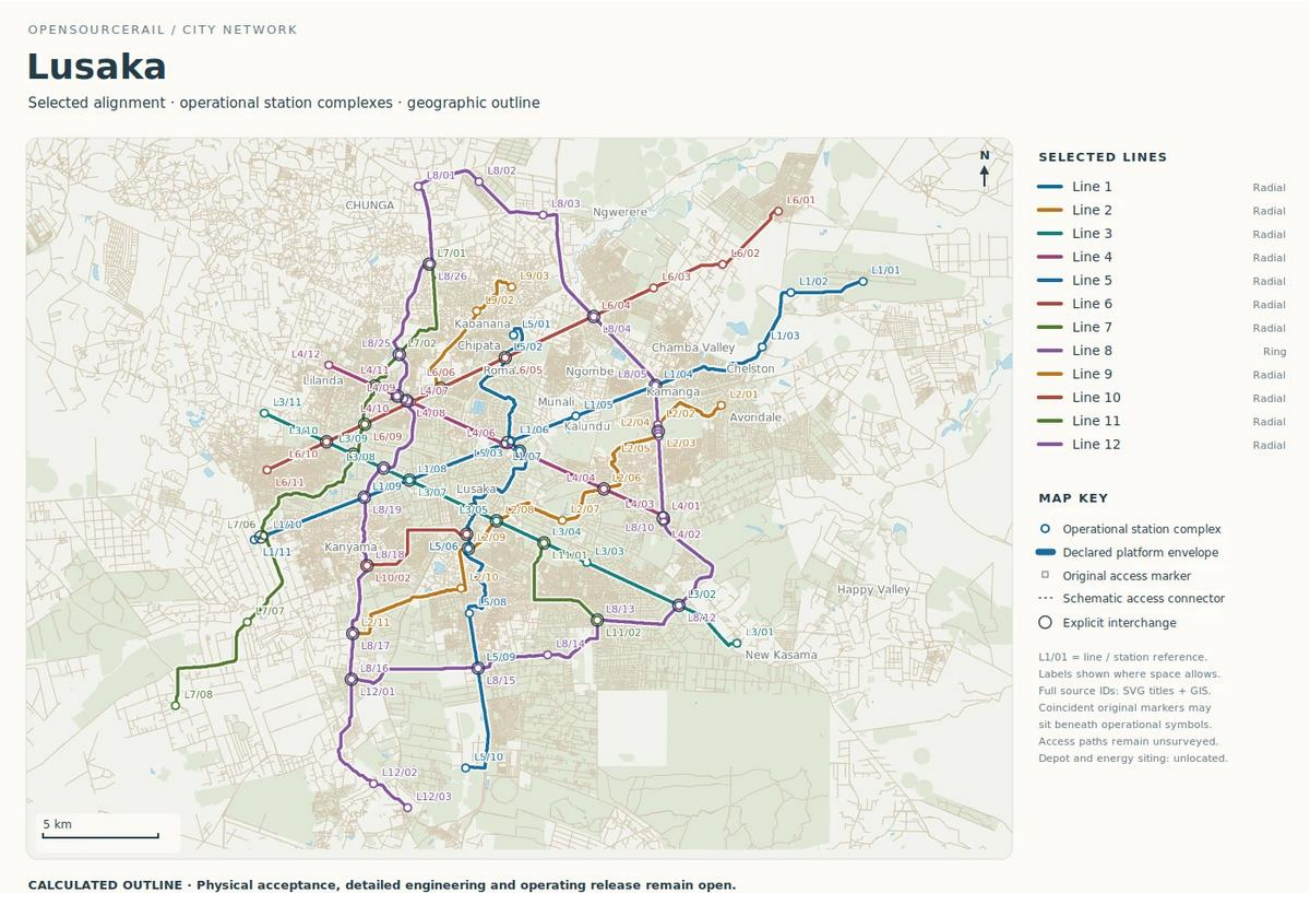
The high count of platform groups makes station access especially important. A passenger wants to know whether a stop is reachable on a safe, direct walking route, whether a transfer can be made without a long detour, and whether the offered frequency matches when they travel. A symbol at a road junction does not answer any of those questions. Local origin–destination data, pedestrian networks, feeder services and accessibility review are needed before a station pattern can be selected.

At the engineering level, a long network crosses different ground and street conditions. Repeating one span or station price everywhere is a starting approximation; bridge sites, utilities, construction traffic and depot access must be surveyed. The modelled fleet is sensitive to round-trip time and headway. If a line is shortened, joined or given a different stopping pattern, its train requirement and passenger usefulness should be recalculated together.

Nairobi also highlights how a ring and radial lines should meet in time, not only on a map. A passenger who waits a long time for the second leg may receive little benefit from a geometrically short interchange. A study of a first corridor should therefore include transfer time and reliability, crowding at the connection, and a practical response when one line is disrupted. These are service choices that can change the value of an expensive station.

Lusaka: a smaller outline with an orbital link

Lusaka's retained proposal has twelve lines and about 293.18 route kilometres. It identifies 110 operational platform groups and 32 explicit interchange references, with a modelled fleet of 340 six-car trainsets including reserve. A ring of about 74.93 kilometres gives the proposal an orbital component, while shorter lines connect to it and across the mapped area. It is a different scale of study from Nairobi, even though the same kind of passenger and civil questions apply.



Lusaka route planning outline

Calculated Lusaka route outline. Alignment and station access are unsurveyed; this drawing does not approve a civil or operating plan.

The map can help the city ask whether an orbital journey should be served by one continuous ring, by transfers between shorter lines, or by another transport mode at an early stage. Each option changes waiting time, reliability, fleet and construction scope. The answer depends on where people actually travel, not on how symmetrical a network appears on the page.

A first corridor could also be a test of local production and maintenance. Which structures can be made with available workshops? How would spans be transported and lifted? Where could a depot and power equipment be maintained without disrupting surrounding streets? The catalogue's costs and schedules are planning scenarios; local suppliers, ground and workforce evidence are needed to turn them into a delivery decision.

The orbital choice can be tested with several passenger stories. A journey between two outer districts may be shorter on the ring; a trip from an outer district to the centre may not use it at all. A rider with limited mobility may prefer a slightly longer in-train journey to a difficult transfer. Those examples should be checked against observed

travel rather than inferred from the circular shape alone.

Read the shared assumptions openly

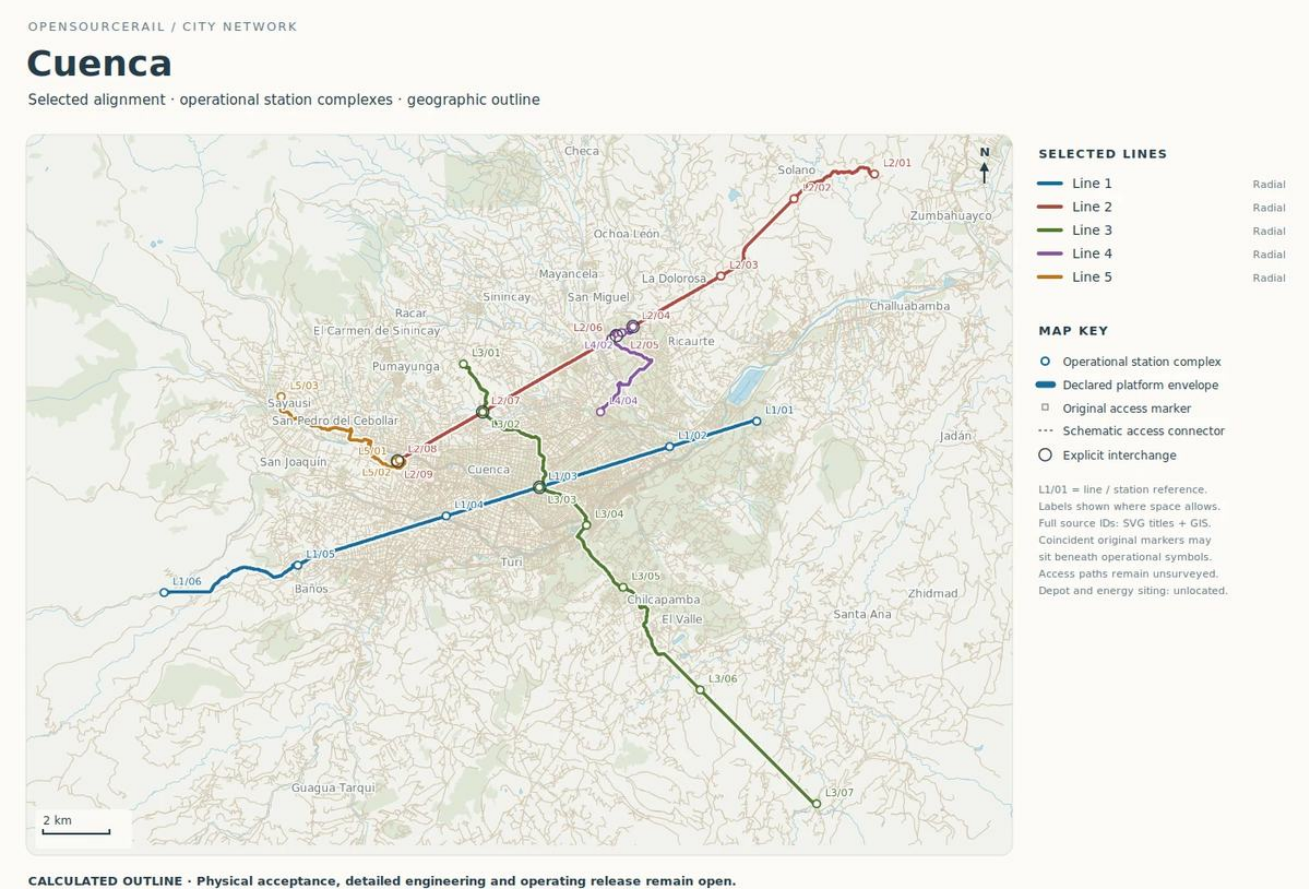
The Nairobi and Lusaka dossiers use declared demand, occupancy, availability and service-span assumptions. They are not local travel surveys or approved rosters. Their value is that a reader can see what follows when those assumptions are held consistent across a map. A city can then replace them with better observations and see whether the preferred routes, stations and train fleet change. That is how a broad catalogue becomes a useful local conversation rather than a claim that hundreds of cities already have finished railways.

4.6 Cuenca and San Salvador

Cuenca and San Salvador are useful final examples because their mapped networks differ sharply in scale. Cuenca's outline is a compact set of five lines; San Salvador has a larger nine-line proposal and a long orbital route. Neither city should begin by treating every coloured line as a committed project. A first corridor ought to solve a clear passenger problem and be testable against local land, access and construction facts.

Cuenca: a compact, legible proposal

The Cuenca catalogue records five lines over about 74.15 route kilometres, with 29 operational platform groups and five explicit interchange references. Its modelled reserve-inclusive fleet is 105 six-car trainsets. The longest line is about 21.92 kilometres, and several shorter routes meet the main corridors. This size makes the network relatively easy to read on one map, but the small number of interchanges also makes each one important to the usefulness of the whole proposal.



Cuenca route planning outline

Calculated Cuenca route outline. Alignment, grade and accessible access are unsurveyed; this is not a construction or operating design.

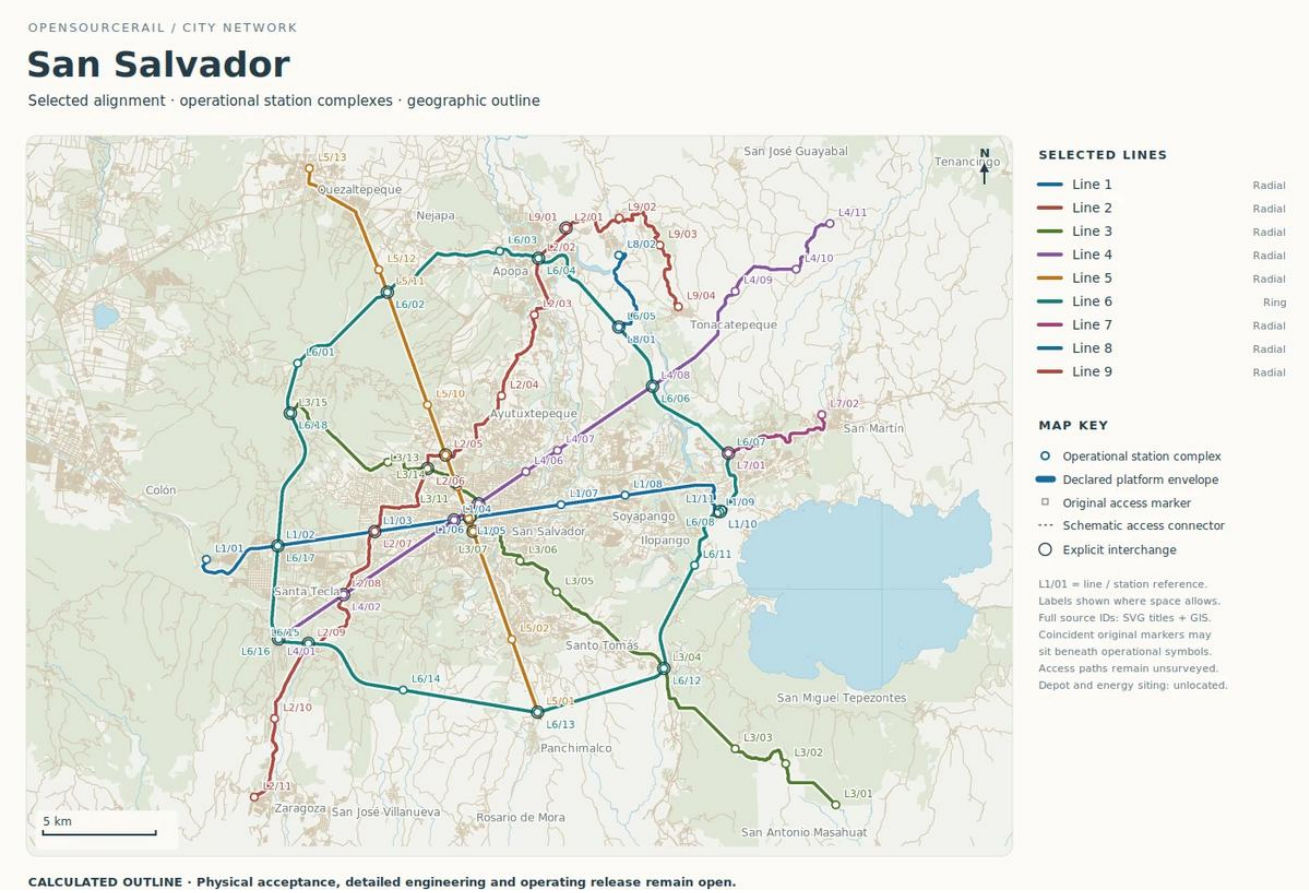
Cuenca's map invites a closer look at terrain and the actual walking routes to stations. A station can be near a destination in plan yet require a difficult climb, crossing or level change. A surface feature that seems minor at network scale can set the height, length and cost of an elevated approach. The city would need surveyed vertical geometry, land, utilities, drainage, accessible routes and passenger origins before preferring one alignment or platform location.

Because the network is compact, a sensible first study might compare two corridors with different station patterns while holding the passenger objective constant. A more direct route could shorten trips but miss an important destination; extra stops could improve access while lengthening ride time and adding station work. The software can expose the fleet and cost consequences of each option. It cannot decide which journey local people value most without their evidence.

The compact scale also makes a physical prototype question easier to frame. A representative station, span and vehicle module could be tested against one selected alignment and a real site. That would reveal whether the proposed interfaces can be built and maintained locally. It would still be one corridor's evidence, not a shortcut to approving every other line in the mapped city network.

San Salvador: a larger pattern to stage

San Salvador's retained outline has nine lines and about 271.61 route kilometres. It groups stops into 87 operational platform locations and records 22 explicit interchange references. The modelled fleet including reserve is 302 six-car trainsets. The longest line is an approximately 74.76-kilometre ring with 18 platform groups. The map shows several cross-city connections as well as this orbital proposal.



San Salvador route planning outline

Calculated San Salvador route outline. Route and station access are unsurveyed; the diagram is not an approved construction programme.

For a first project, the most useful question may be whether a selected line or interchange can demonstrate a strong travel benefit and a buildable site. A complete ring can appear attractive as a finished network, but its capital, fleet and operating commitments are large. An early stage might serve different trips at lower first cost, or it might fail to deliver the connections that make the later ring worthwhile. Demand by corridor and time of day, accessible transfers and a realistic construction sequence are needed to compare those choices fairly.

The map also cannot resolve local ground, utility or road interfaces. Those site questions belong in survey and civil design before one candidate's whole-life price is presented as a city budget. A modelled service can still be valuable now: it shows what each extra station and kilometre asks of the fleet, energy and maintenance plan.

The same discipline applies to construction sequence. A line that looks simple as a coloured path may need a depot, a power connection and workshop arrangements before its first passenger can ride. If those common assets are assigned only to a later network stage, the early stage's cost will be understated. Conversely, a shared facility should not be charged repeatedly to every future corridor. The scope of each stage must be explicit.

From an atlas to one decision

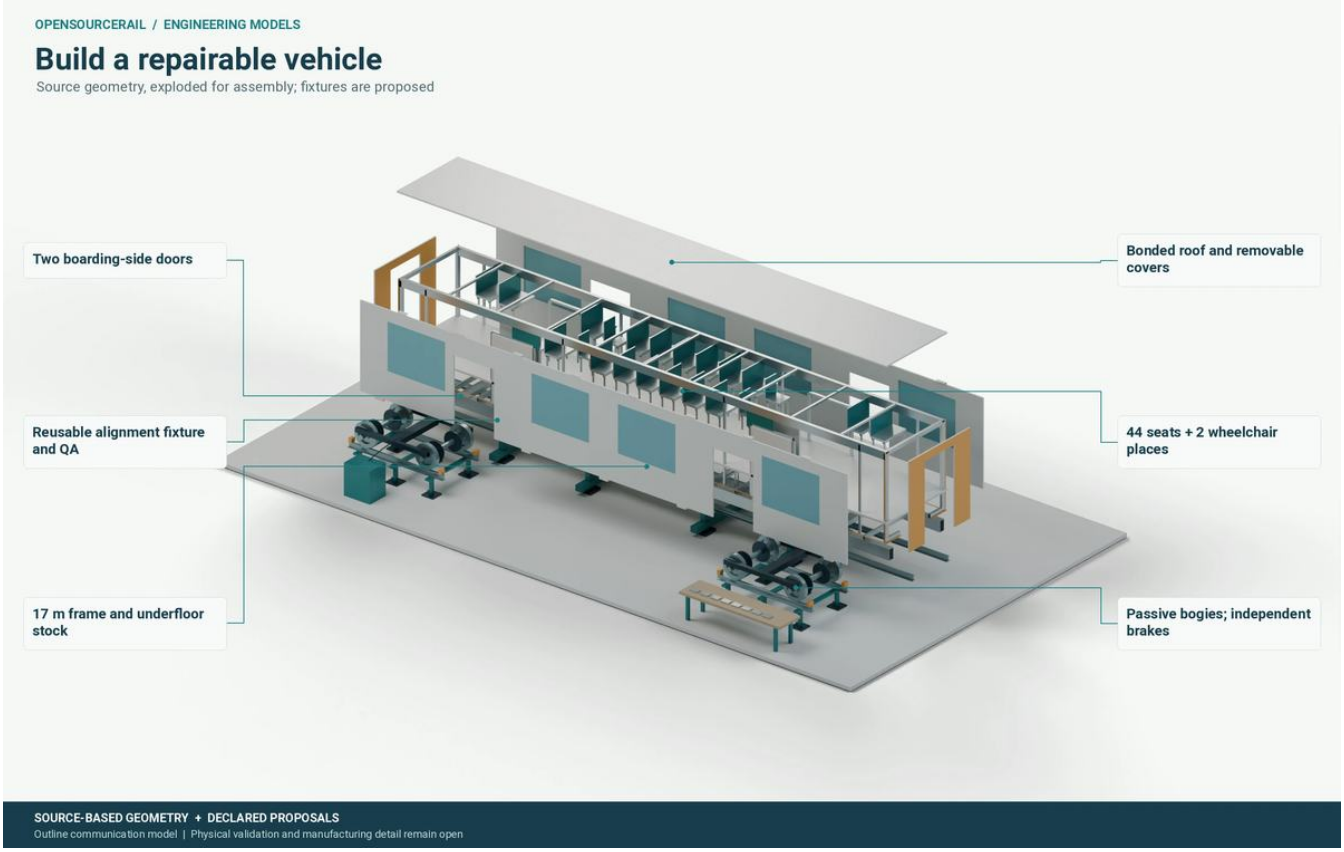
Cuenca and San Salvador illustrate a common discipline. Start from the passenger journey, then choose the smallest corridor study that can test the most important uncertainty. Keep the planning map, survey, train, service, structure and cost linked to that one revision. The catalogue can broaden imagination and make quantities visible; it should not substitute for the city's own evidence or approval.

V. The physical railway

Vehicle, passenger interfaces, power, stopping and civil works.

5.1 Vehicle and running gear

The reference train has six 17-metre cars. A passive two-axle bogie supports each end of each car, giving twelve bogies, 24 wheelsets and 48 axleboxes for the formation. Steel wheels run on conventional rails carried by the concrete guideway. Motors and gearboxes are absent from this reference running gear; springs, damping, bearings, brakes, monitoring and emergency support still have to be installed, inspected and maintained.



Source-based vehicle assembly view

Engineering communication model. It separates major assemblies and shows their proposed relationship; it is not a complete installed-mass or approved manufacturing drawing.

Design the car as an installed system

A carbody is not just a shell. Its floor, sides, roof and joints must carry passengers, equipment, bogie reactions and accident loads while leaving room for doors, windows, ventilation, cables, batteries, seating and maintenance access. A material can appear light in a small structural test and still require heavy joints or reinforcement around openings. The project therefore compares body families with their complete load paths rather than treating a single beam stiffness result as a qualified car.

The proposed manufacturing breakdown names 605 leaf-part types in 40 assemblies across the six-car train. It counts major structural stock, wheelsets, bearings, doors, seats, batteries, HVAC, lighting, wiring, service equipment and many fasteners. This is a substantial outline inventory, but the list is not a measured installed train. Gross stock and assumed bought equipment masses must be reconciled with actual fitted parts, centre of gravity, inertia and available space.

One detailed proposal estimates 78.49 to 106.60 tonnes across the train from stock and component allowances, with 89.81 tonnes as its base subtotal. These are bounds within that particular proposal, not an approved tare or proof of a weight saving over another candidate. Unlocated equipment, overlapping reserves and supplier internals must be resolved before an installed mass can be stated. Those quantities matter to axle load, braking, energy, beam design and fleet cost, so an optimistic assumption cannot remain isolated in the vehicle chapter.

Formation and interfaces

The six cars need five inter-car mechanical couplers and separate passenger gangways. A gangway gives people a route between cars; it does not replace a load-bearing coupling or an emergency rescue connection. Air, signals, lighting and other essential services also cross the joint. Their bending, movement, fatigue and removal paths have to fit the operating curves and the workshop process.

A revised manufacturing proposal increases the inter-car gap from 500 to 650 millimetres, taking its body-plus-gap envelope from 104.50 to 105.25 metres. The extra space may ease certain interfaces, but platform margins, coupler projection and stopping tolerance must be checked on the same candidate. A formation change cannot quietly inherit the parent platform or fleet result.

The bogies themselves present a similar trade-off. A familiar wheel-and-rail interface offers an established body of engineering knowledge, but wheel wear, rail contact, noise, bearing service, wheel-slide protection and wet-weather braking remain real tasks. The project needs dynamic and physical checks, not only a visual fit of wheels beneath a CAD body.

5.2 Inside the vehicle

Passengers see a floor, seats, windows and doors. Beneath those surfaces is an arrangement that must carry people safely and let maintainers reach the equipment. The body, bogies, batteries, ventilation, wiring, couplers and interior are one installed vehicle. Their combined geometry and mass decide what the track and structures experience, how the train stops and how easily it can be repaired.

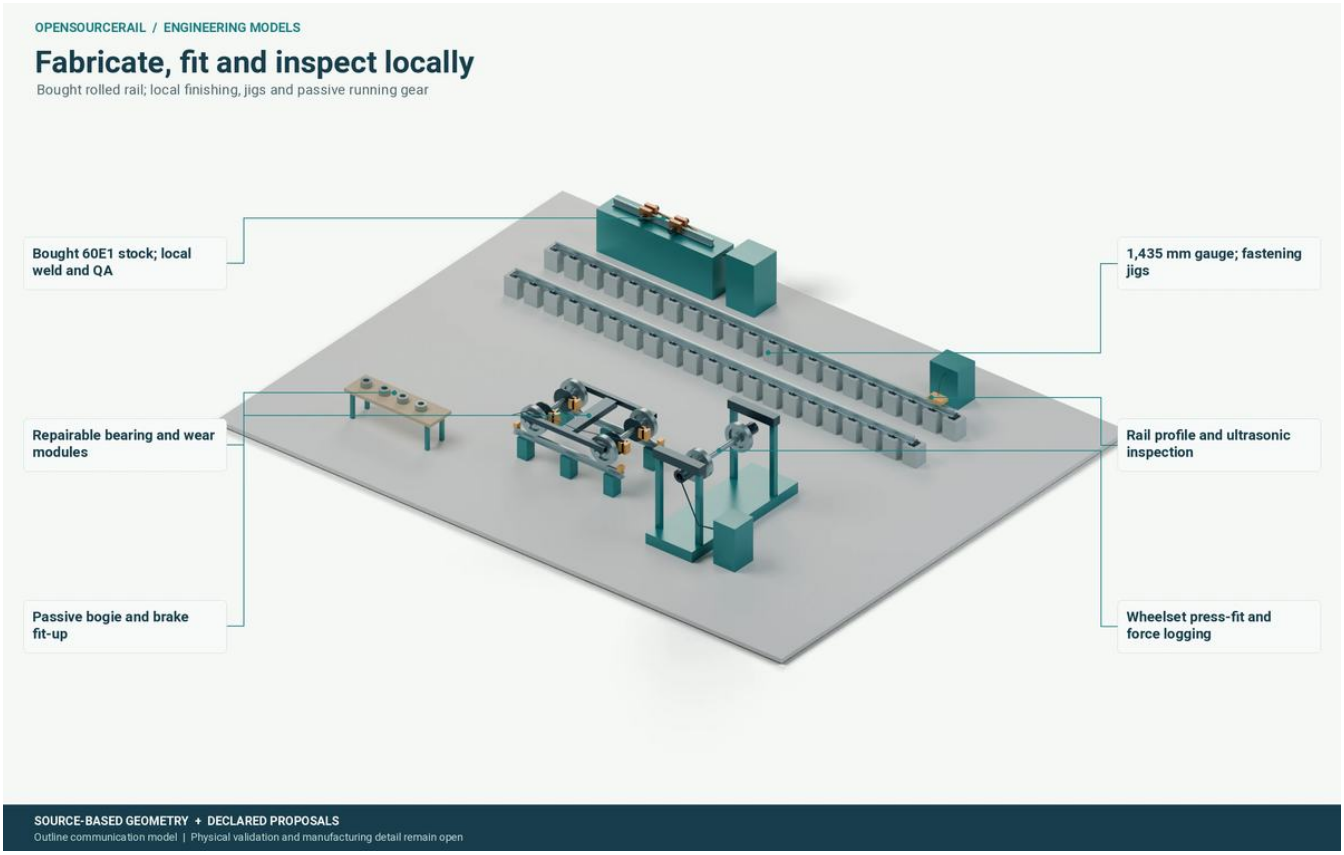
The body is a load path with openings

A railway car must transfer vertical, longitudinal and lateral forces among floor, sides, roof, bogie connections and couplers. Doors and windows remove material from the sides just where a simple continuous beam model might expect it. The remaining structure may need frames, skins, joints and local reinforcement. Seats and equipment also impose attachment loads, while thermal movement and fatigue test the joints over time.

OpenSourceRail explores different structural grammars and materials rather than fixing every member before analysis. A braced frame, a bonded shell and a rigid jointed body cannot be judged by exactly the same simplified mechanism test. Their manufacturing routes also differ. A lighter panel may need more costly curing or a more demanding repair process; a familiar metal route may be easier to inspect and source locally. The appropriate comparison includes installed stock, joining, fire response, fatigue, corrosion and replacement, not a single material's strength-to-weight ratio.

The bogie carries more than weight

For the reference six-car train, twelve passive bogies and 24 wheelsets support the formation. They guide the wheels along the rail, transmit braking and lateral forces, and isolate the body from track irregularities. Bearings, springs, dampers, brake components, sensors and safety restraints need their own mounting and access. Wheel profile and rail contact affect wear, noise and adhesion. A body that fits over a bogie in a static drawing may still foul it on a curve, during suspension movement or when a wheel is worn.



Rail and running gear study

An engineering model of the running interface. Dynamic clearance, wear, braking and maintenance need their own checks.

An alternative powered-wheel bogie may simplify station-side propulsion while adding motors, gearing, converters, cooling and protection beneath each car. That changes unsprung or sprung mass, clearances, energy, braking and workshop tasks. The passive reference and the powered alternative should therefore have separate, complete vehicle and service results.

Space is an engineering resource

Seats, wheelchair places and circulation compete with equipment cupboards, ducts, structural members and emergency routes. The same underfloor region may be proposed for batteries, cables and removal access. A large unit that fits in its installed position may still be impossible to take out without removing a bogie or blocking the platform. A useful three-dimensional model shows both the operating envelope and the maintenance envelope.

The current parts proposal names 605 leaf-part types in 40 assemblies and over 110,000 expanded pieces across a formation. These counts expose the many connections behind a simple visual concept. They are not a released manufacturing bill: some masses are allowances, some fit positions remain generic and some supplier internals are not fully counted. A complete installed mass must reconcile the physical item, its quantity, location, fixings, protection and replacement route. The low, base and high stock allowances are useful for sensitivity, but the selected vehicle must be weighed and checked at the relevant build stages.

The interior deserves its own tests

An accessible passenger place is more than a marked rectangle. Boarding gap and step, turning area, securement, handholds, reach, emergency egress, lighting and communication all matter. The one-sided two-door arrangement must suit the chosen platform side and station layout. A crowded structural test does not set the normal seating policy, but it does remind the team to consider misuse, unusual loads and evacuation. Passenger feedback and physical mock-ups can reveal problems that a detailed CAD model misses.

5.3 Doors, cabin and access

The proposed train has twelve one-sided passenger door openings, two per car. Doors are a small part of the vehicle's visible area but a large part of its daily reliability and passenger protection. They must open and close without striking the body or platform, detect an obstruction, remain secure in motion, allow emergency release and be repairable without dismantling the car.

One opening as a representative module

The current door study counts 74 mechanical and electrical part types and 694 individual pieces for one opening. The make-or-buy proposal assigns 97 pieces to local fabrication, 568 to standard purchase and 29 to purchase subject to qualification. It includes the header, rails, carriage, belt and pulley, drive, leaves, glazing, sensors, seals, wiring, fixings and inspection items. These counts help expose the work hidden by a simple animation of two moving leaves. They do not mean the door is ready for manufacture.

The next design step is to establish mating datums: where the header attaches to the body, how the rail and carriage align, how the belt and drive meet, where sensors sit, and how the assembly can be lifted out. Several proposed operator parts still occupy a generic central header zone instead of fully resolved fitted positions. A two-metre removal sketch does not represent the full 2.9-metre guide stock. These are specific reasons to keep installation and maintenance open, even though the inventory is detailed.

A cabin that works in service

Each car's 44 fixed seats and two wheelchair places provide a clear normal accommodation policy. Their location must still allow boarding, circulation, evacuation, cleaning and access to service equipment. Windows, seat fixings, handholds, lighting, ventilation and interior panels must meet their own fire, impact and maintainability requirements. A bright concept interior is an invitation to test the layout with passengers, not a substitute for those tests.



Passenger interior concept

The accessible interior is a design intention. Actual circulation, doorway, platform gap and emergency arrangements remain engineering tasks.

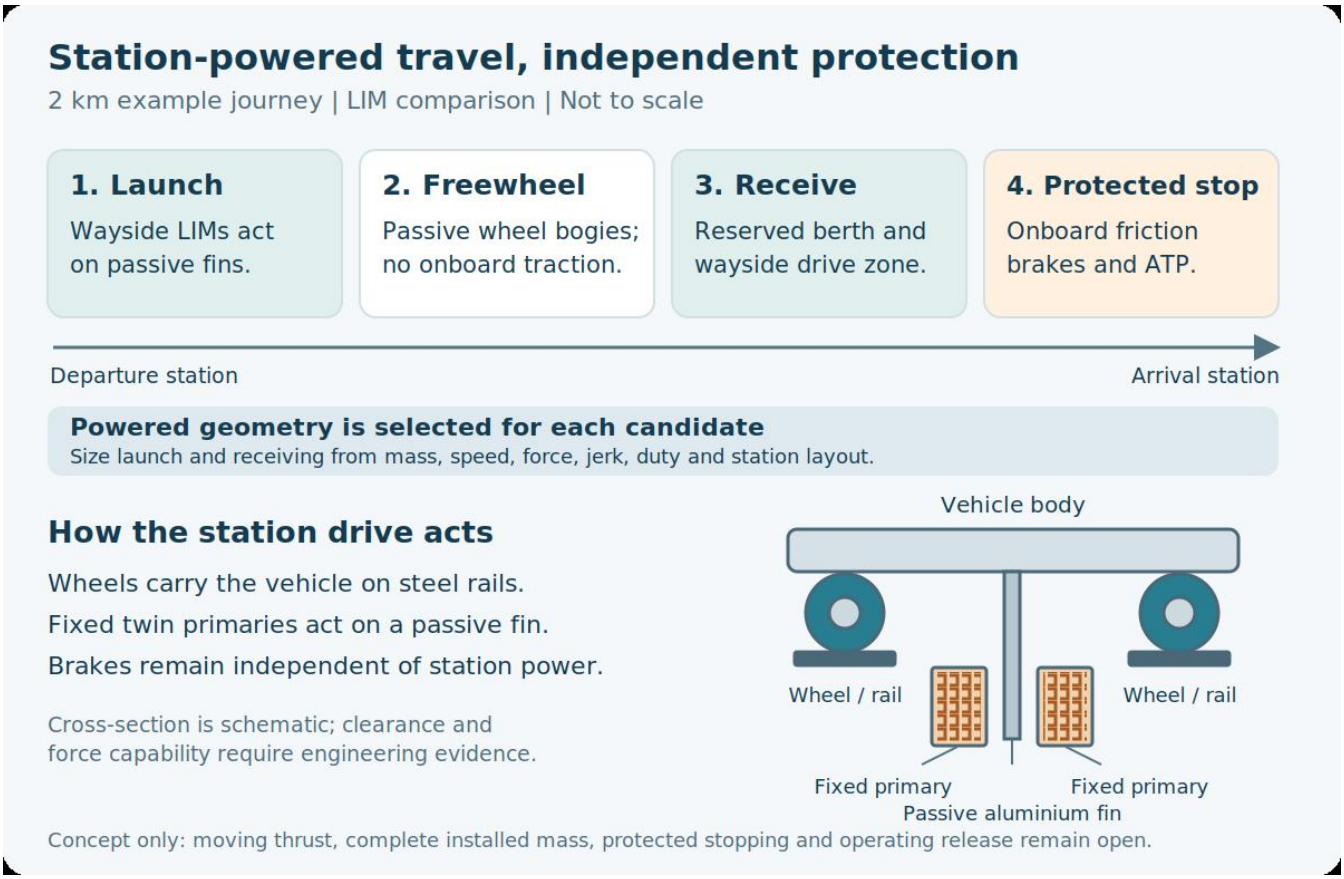
The lighting proposal uses replaceable LED modules in two rows per car, with separate protected circuits and independent emergency fixtures. The counted normal lighting demand is modest relative to propulsion, but the

design must still establish illumination, glare, electrical protection, electromagnetic compatibility and hot-weather performance. The same principle applies to HVAC: a catalogue load or a CAD envelope is a starting point, while real airflow, pressure, temperature and service access must be verified.

The door operator is chosen as the v0.7 representative manufacturing and maintenance module because it forces mechanical, electrical, body and passenger questions to meet. A controlled package should show dimensions, tolerances, materials, assembly sequence, tooling, QA limits, removal path and replacement record for one exact vehicle candidate. A business-system record can then follow the physical module through receipt, inspection, fitting, service and eventual replacement.

5.4 Propulsion, power and braking

The wheel-supported reference separates carrying the train from driving it. Wheels and rails bear the load. Station-side equipment is being compared for launch and arrival, with a passive aluminium interface on the vehicle for linear-motor research. Other candidates may use mechanical launch equipment, powered wheels or different recovery arrangements. None is selected as an installed system merely because a schematic looks simple.



Launch, freewheel, receive and protected stop concept

The diagram explains one operating idea. The actual force, powered length, energy transfer and protected braking distance depend on the chosen candidate and route.

What a station drive must achieve

A station-powered train could receive its main acceleration near departure, coast through much of the interstation distance and be received at the next station. This may reduce train-borne traction hardware. It also concentrates power and thermal duty at stations and introduces equipment gaps between stations. On an uphill route, a coasting train may not reach its destination within the proposed speed limits. On a downhill route, it may need additional speed control. A failed station or an unexpectedly stopped train makes rescue and independent stopping essential.

A linear motor needs an attainable force map across speed, current, voltage, air gap and temperature. A model that matches an ideal travelling field is a useful method check, but it does not include every slot, end effect, inverter or moving support of a manufactured motor. The passive fin, its mounting, thermal expansion and wheel/rail movement must fit the same clearance envelope. Powered length is consequently a result of design, not a fixed 100-metre or 900-metre prescription.

Mechanical capture and release is another research family. It asks whether a station can recover some arriving kinetic energy, store it briefly and help launch the next train. The physical mechanism would need contact geometry, energy storage, losses, guards, fault handling, maintenance and a reset sequence. Energy recovered in a calculation does not automatically become available to the next departure. Powered-wheel alternatives must

likewise carry their added motors, gearboxes, converters, cooling and bogie changes.

Independent stopping and service power

The reference train requires onboard service, emergency and parking brakes that do not depend on station power. Stopping must be checked at low adhesion, on gradients, under crowd loading and after relevant faults. One retained adverse calculation from 20 m/s uses a low adhesion assumption and a downhill grade; it gives a stopping distance of roughly 700 metres including response. That is a sensitivity, not a validated wet-rail brake curve or an approved separation distance. Wheel-slide protection, thermal duty and degraded operation need their own evidence.

Service batteries support HVAC, lighting, doors, communications and protection. A retained 82.8-kWh-per-car pack is a sizing hypothesis, not a qualified installation. Heat can dominate the design: a cell with a 45°C ambient limit cannot be kept below that limit in 50°C surroundings by a radiator alone. Cooling, enclosure, charging, emergency endurance, fire isolation and replacement access must be designed and tested together.

The main question is not which drive sounds most elegant. It is which candidate supplies the required service with credible force, safe stopping, bounded heat, repairable equipment and the lowest justified whole-life cost.

5.5 Stopping, faults and rescue

The safest journey is one whose unusual moments have been designed as carefully as its normal motion. A train can lose a launch, stop between stations, encounter poor adhesion, meet an occupied route or need to be evacuated in heat. A proposed drive and timetable are credible only when the train can be brought to safety and the passengers cared for in these states.

Braking belongs to the train

Station-side equipment may help accelerate or receive a train, but the reference requires service, emergency and parking braking independent of that equipment. This gives the train a means to stop after a station power fault or missed receiving event. It does not remove the need to prove the brake's actual force and repeated thermal capacity. Wheel and rail adhesion varies with water, contamination and load; wheel-slide protection has to preserve steering and stopping rather than assume a fixed friction number.

Stopping distance includes perception or detection, control response, equipment build-up and actual deceleration. It depends on initial speed, gradient, mass, available adhesion and the faults considered. One retained downhill, low-adhesion calculation from 20 metres per second gives a distance around 700 metres when response is included. That is a warning about the size of the problem, not a validated minimum separation. The protected route and service headway must ultimately use the accepted worst relevant cases and approved margin.

A failed drive changes more than arrival time

If a station cannot launch a train, passengers may be held at a platform. If reception fails, the train needs a safe independent stopping location. If it stops between stations, the project must decide how it will be secured, communicated with, reached and moved. Gradients, curves, weather and the absence of a continuous powered rail can make recovery difficult. A candidate may need a rescue vehicle, a tow interface, local storage or another provision. Whichever path is chosen has mass, civil access, maintenance and operating costs.

Emergency planning also changes stations and guideways. An elevated track must have a way for responders to reach people and for people to leave a disabled train, including those with reduced mobility. The guideway needs working space and protection from adjacent traffic or power equipment. Communications, lighting, ventilation and battery endurance must last long enough for the chosen procedure. These are candidate requirements, not generic promises from a diagram of a train at a station.

Faults can combine

A hot day can increase cooling demand at the same time that a power system is stressed. A vehicle fault can lengthen dwell and disrupt the following train. A crowded carriage can change brake load during poor adhesion. Engineering should therefore test combinations that matter, while keeping the scenarios bounded and explainable. It is equally important to show what has not been tested. An apparently safe result under normal dry conditions cannot be borrowed for a wet, loaded, downhill fault.

The current platform has software controls and conditional journey models that can exercise parts of these scenarios. Actual propulsion availability, brake behaviour, thermal endurance and human evacuation need measurements, procedures and independent review. Recording that boundary is a practical part of safety: it tells the next engineer what evidence is still needed before a timetable or design can be accepted.

5.6 Heat, ventilation and electrical life

Power reaches a passenger railway through more than a propulsion device. Stations and vehicles need lighting, ventilation, doors, communications, controls, fire protection and safe supplies when the normal source is lost. Each system occupies space, adds mass and creates heat. Their combined installation can change the vehicle, structures and operating cost as much as a prominent motor choice.

Separate traction from essential services

The wheel-supported reference avoids onboard traction motors and traction batteries. This does not mean the cars have no electrical system. Door operation, emergency lighting, communication, control, ventilation and other essential functions need a protected supply and a defined response to faults. A retained service-battery hypothesis uses about 82.8 kWh per car. Its actual duty, mass, chemistry, cooling, isolation and replacement interval have to be sized against the selected equipment and emergency scenario. The number is a research input, not an installed procurement specification.

An onboard recovery or wheel-traction alternative remains possible. It would add different running gear, protection, charge or fuel logistics, mass, thermal loads and maintenance work. It must therefore be carried as a new candidate with its whole-system effects, rather than inserted into the reference train as a small component substitution.

Hot weather changes the design

Samawah and Baghdad make climate a central engineering input. Cabin comfort depends on passenger load, sunlight, glazing, insulation, ventilation, infiltration at doors and equipment heat. Electronics and battery modules can have narrower temperature limits than the cabin. A battery study at a 45°C module ambient does not demonstrate operation at a 50°C site condition without a cooling model and suitable measurements. Shading, airflow, equipment placement and service access should be assessed together.

Cooling has a cost on both sides of the ledger. A larger unit consumes energy, mass and roof or underfloor space. An undersized unit can shorten component life, cause service failures and increase replacement cost. A plausible heat balance must follow energy from its source through the equipment to the air or structure that removes it. The same discipline applies to station drive equipment, braking resistors, transformers and power electronics during repeated departures, not only during one short demonstration.

Current, force and heat must agree

A proposed station-powered linear drive can be attractive because it keeps major traction equipment off the train. Its available force depends on geometry, speed, current, magnetic field, temperature and the secondary conductor. Electrical input becomes motion, stored field energy and heat; the balance has to close for the actual operating duty. A calculated peak force at one position cannot by itself support a timetable.

The next practical evidence is a force-speed and thermal map with a bounded electrical supply, repeated launch cycles and measured or justified material behaviour. It should include losses and cooling at the drive, its structure and the vehicle interface. Braking remains independently sized, with stopping and rescue possible when the receiving drive is unavailable.

Design for isolation and replacement

The electric railway must tell maintainers which equipment can be safely isolated and replaced, and how a failed module is identified. Cable routing, connectors, water ingress, fire separation, access panels and working space are part of the design. A locally repairable system needs a parts list and inspection method that a local workshop can actually use. The Workbench can connect a failure record to its component and candidate, but the approved maintenance intervals and procedures must come from qualified equipment and operating experience.

5.7 Stations, structures and construction

The reference railway uses conventional rails carried by concrete structures. Mostly elevated, prefabricated construction is intended to reduce occupation of busy streets and to make repeatable local production possible. A structure still needs to be designed for its particular ground, water, utilities, access, loads and construction method. Repetition can lower cost only after the repeated unit itself is complete and safe.



Concept of a precast guideway installation

Concept illustration. The lift, crane, restraints, traffic management and temporary stability must be designed for each installation.

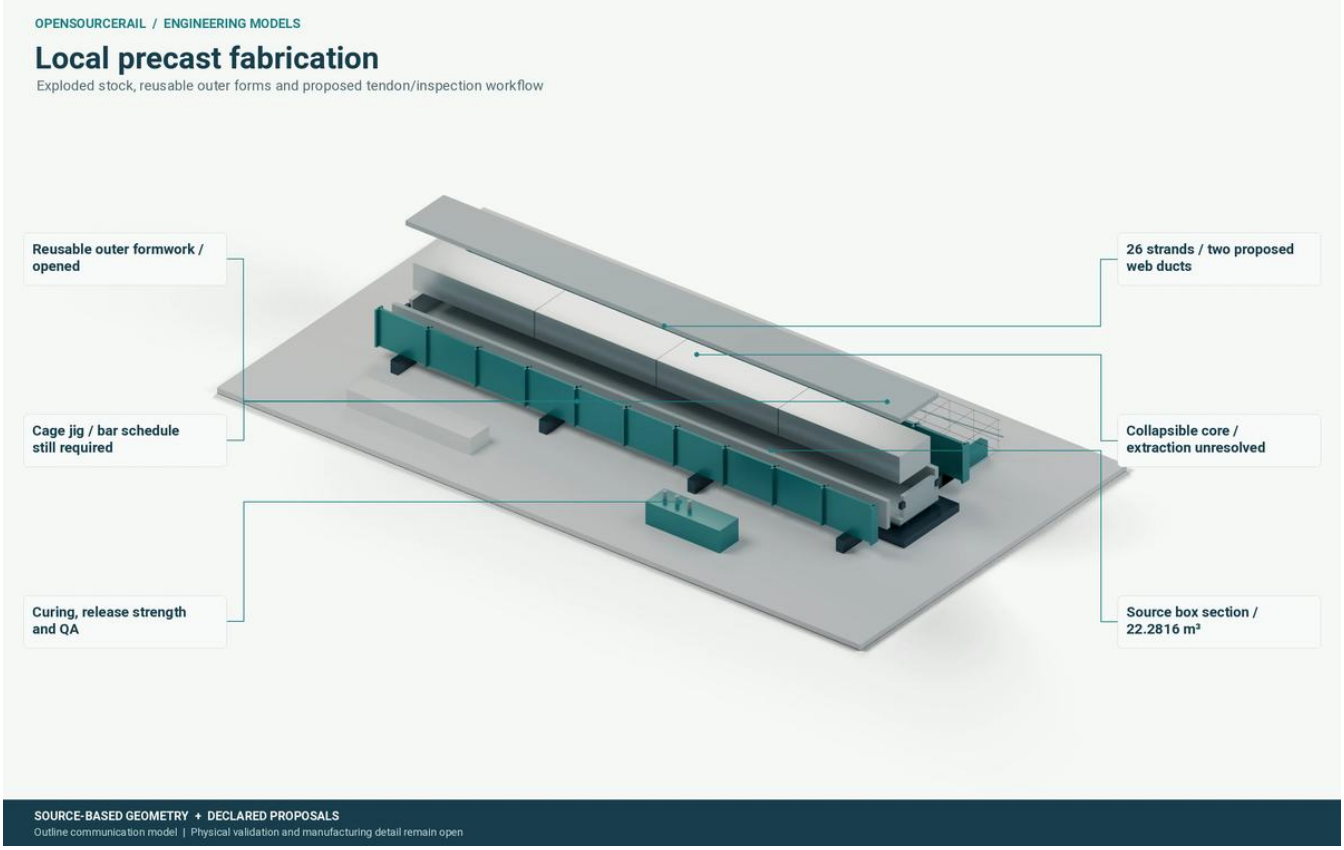
Follow loads to the ground

A train load reaches the rails, fasteners, beam or deck, bearings, piers and foundations. Each interface can govern the answer. A span that looks efficient in bending may have a difficult rail seat, bearing reaction or joint. A light beam may need expensive erection equipment or a larger maintenance closure. Moving axle positions, crowd cases, braking forces, temperature, fatigue, drainage and access must be considered together.

The project has explored a representative 20-metre precast span and several alternative civil forms. One recent numerical control used the reference train's 24 axle positions, a declared dynamic factor and finite bearing patches. Two independent mechanics routes gave very close displacements in that model; the fine result was about 10 millimetres. A refinement difference of roughly 2.7% passed a diagnostic threshold, while mesh independence, material and joint behaviour, actual rail support and site foundation evidence remained open. Agreement between solvers is valuable, but both can agree on a model that omits a real load path.

Construction is part of the design

The precast concept asks a local yard to make repeatable beams with controlled forms, reinforcement, prestressing where selected, curing, lifting points and inspection. Transport then constrains length and mass. At the site, the crane must have enough capacity at the actual lift radius, after allowing for rigging, ground bearing and the temporary state of the span. A nominal crane capacity at zero radius does not answer that question. Bearings, joints and rail installation follow, along with the inspection and replacement route.



Precast fabrication and reusable tooling concept

The illustrated yard shows reusable forms and proposed handling. It does not establish production rate, qualified prestressing or finished beam cost.

An overhead launching gantry is an alternative where a conventional crane cannot work easily. It changes temporary works, plant cost and construction sequence. Both methods need a complete erection study before one is selected.

Stations add their own structures and interfaces: platform loads, lifts, stairs, fire and evacuation routes, shelters, power equipment, maintenance rooms, drainage and public-realm connections. Their footprints cannot be inferred from a single platform length. A compact station is only affordable if access, safety and constructability survive the full design. The Samawah station pair illustrates why even a small map edit can open these larger questions.

VI. Product and assembly views

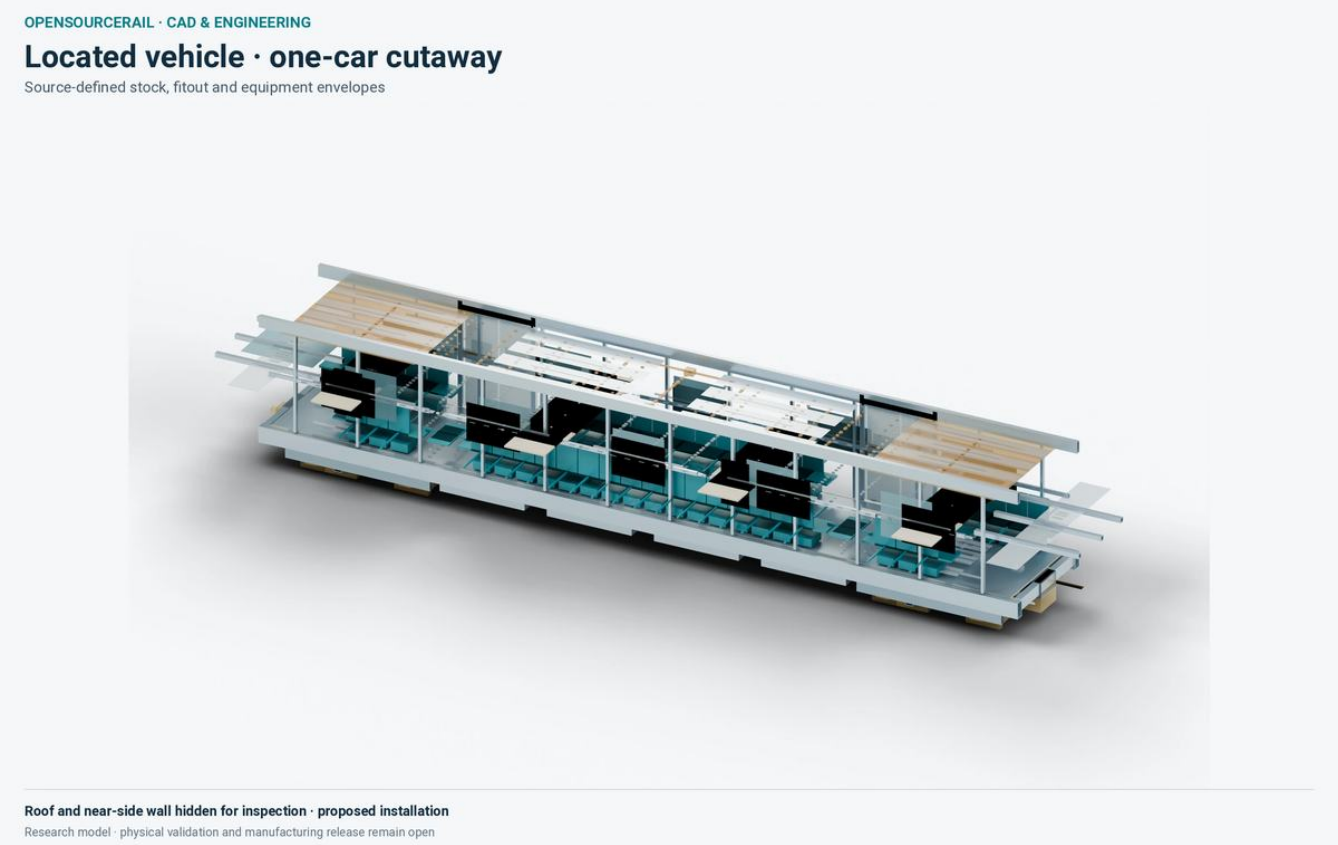
Drawings and models of distinct railway research candidates.

6.1 The train in engineering views

A concept rendering shows what passengers might see. An engineering view should let a designer ask a harder question: Which physical part occupies this space, what supports it, what does it weigh, and how can it be fitted or removed? The project has both kinds of images. Keeping them distinct lets a reader enjoy the visual proposal without mistaking it for a finished train.

One car and its counted stock

The cutaway below comes from a located research car. Its roof and near side are opened for presentation so the reader can inspect the main body, interior and equipment arrangement. The displayed cuts are not construction openings. Source stock, purchased items and provisional equipment envelopes have different meanings even when rendered in similar colours.



Cutaway of one located research car

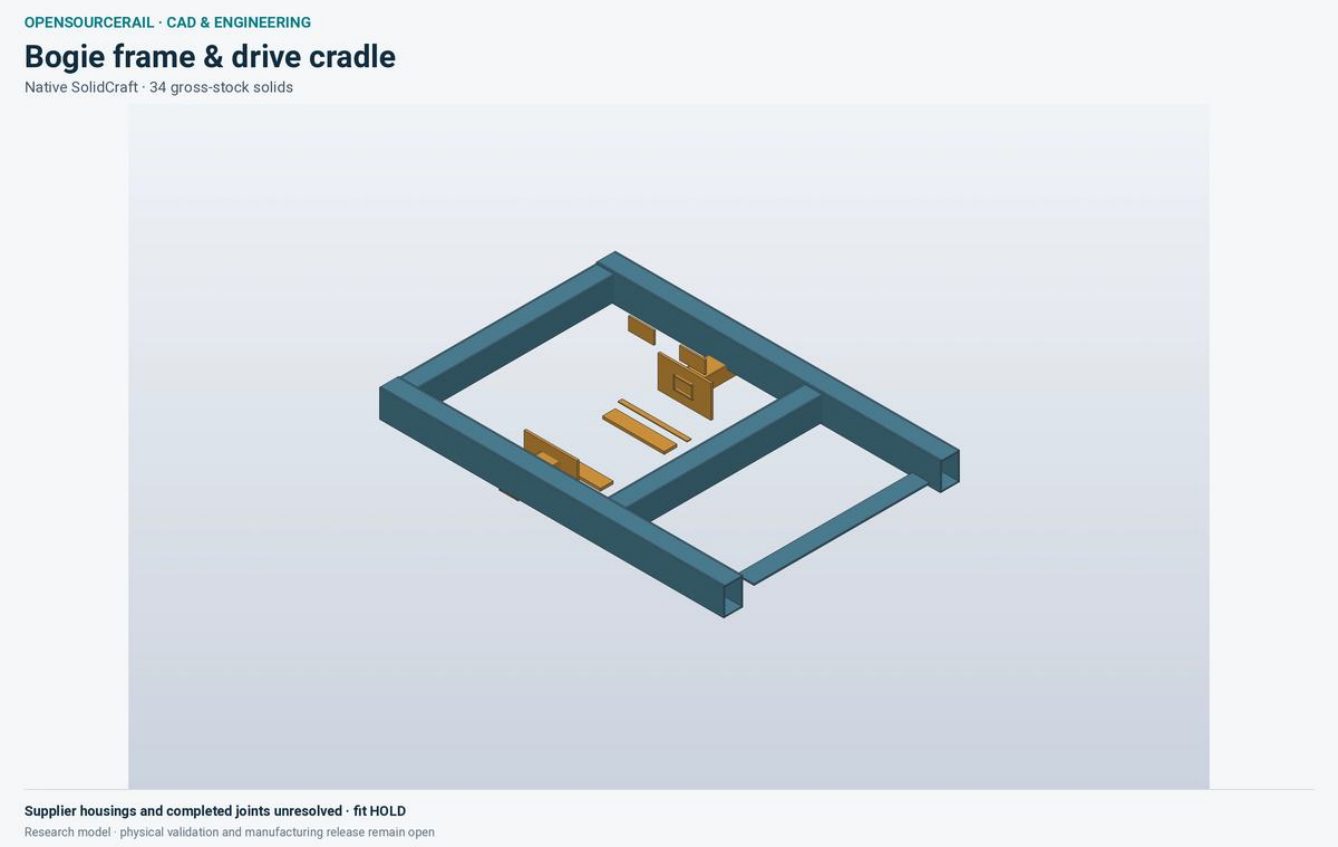
A source-derived research cutaway, with display-only openings. Installed equipment, joints and physical fit are not approved.

The reference railway uses six 17-metre cars and passive bogies on conventional rails. Its passenger brief prioritises fixed seats, wheelchair places and no planned standing. The body must hold those interiors while transferring loads to the bogies and couplers. Door and window openings interrupt the sides; underfloor equipment competes with access and the running gear envelope. A structural body family cannot be selected from an isolated stiffness number if its joints, floor fixings and maintenance routes are still unknown.

An exploded view can make a complicated car understandable, but the explosion is a presentation device. Downloaded geometry may preserve the source positions instead. The engineering gallery identifies thousands of located source parts in one vehicle view, while warning that their stock mass is not complete car tare. Unlocated HVAC, wiring, internal supplier hardware, mounts and protection must still be added or bounded. A car's centre of gravity and inertia cannot be inferred from a neat inventory subtotal alone.

The bogie is an interface, not a decorative support

The next view is a proposed frame and cradle subset for a powered-bogie research candidate. It is shown here to explain the kind of mechanical fit that a complete running-gear design must resolve. It is not the passive bogie installed beneath the selected reference train, and its mass or performance must not be transferred to that reference.



Gross-stock bogie cradle research view

Native geometry for a research bogie from the same located vehicle study. Supplier housings, joints and complete fit remain unresolved.

A wheelset needs a defined contact with the rail, bearings and axleboxes with inspectable loads, suspension motion, brakes and restraints. Wheel diameter changes through wear. The bogie moves relative to the body on curves, gradients and rough track. A box that fits on a static drawing can interfere with cables or a door mechanism when the suspension travels. The relevant question is the swept envelope in operation and during maintenance, not simply whether two parts overlap in the saved CAD pose.

The reference has twelve passive bogies, 24 wheelsets and 48 axleboxes across the six-car formation. That count helps define a maintenance programme: every wheelset and bearing has an identity, inspection route, removal method and replacement consequence. A different bogie family may offer benefits, but it changes mass, drive, braking, electrical, parts and service work for the whole candidate.

Move from view to design decision

A useful design review puts the image beside a list of questions. Can the carbody carry its real installed loads? Do equipment and passengers fit without blocking evacuation? Can the workshop lift the car, roll out a bogie and reach its fasteners? Are fire, corrosion, fatigue and crash scenarios represented? Which dimensions are controlled, and which are currently only presentation envelopes?

The project can now produce impressive source-derived scenes and editable models. Those views are valuable because they expose conflicts early. Their engineering value grows when a particular candidate, part location and test result are connected. A photograph-like view by itself neither qualifies a body nor fixes the train that a city

should buy.

Read the space around the train

The train has to fit the railway in more than one position. At a platform, the passenger step, door threshold and platform edge define a narrow accessibility interface. On a curve, car ends swing outward while the centre moves inward. Under braking and suspension travel, the body and bogie move relative to each other. A drawing of the straight, unloaded train cannot show all of those clearances. The selected corridor must therefore be tested with a swept vehicle envelope, including worn wheels and credible installation tolerances.

The roof and underfloor space have similar conflicts. Air equipment needs an intake and a place for condensate to go. Batteries need restraint, ventilation, isolation and a removal route. Braking equipment needs heat rejection and inspection. Cables need segregation, protection and slack where parts move. The volume occupied by a coloured box in a concept view is only the start of that layout. Its mounting, service clearance and connections may be larger than the device itself.

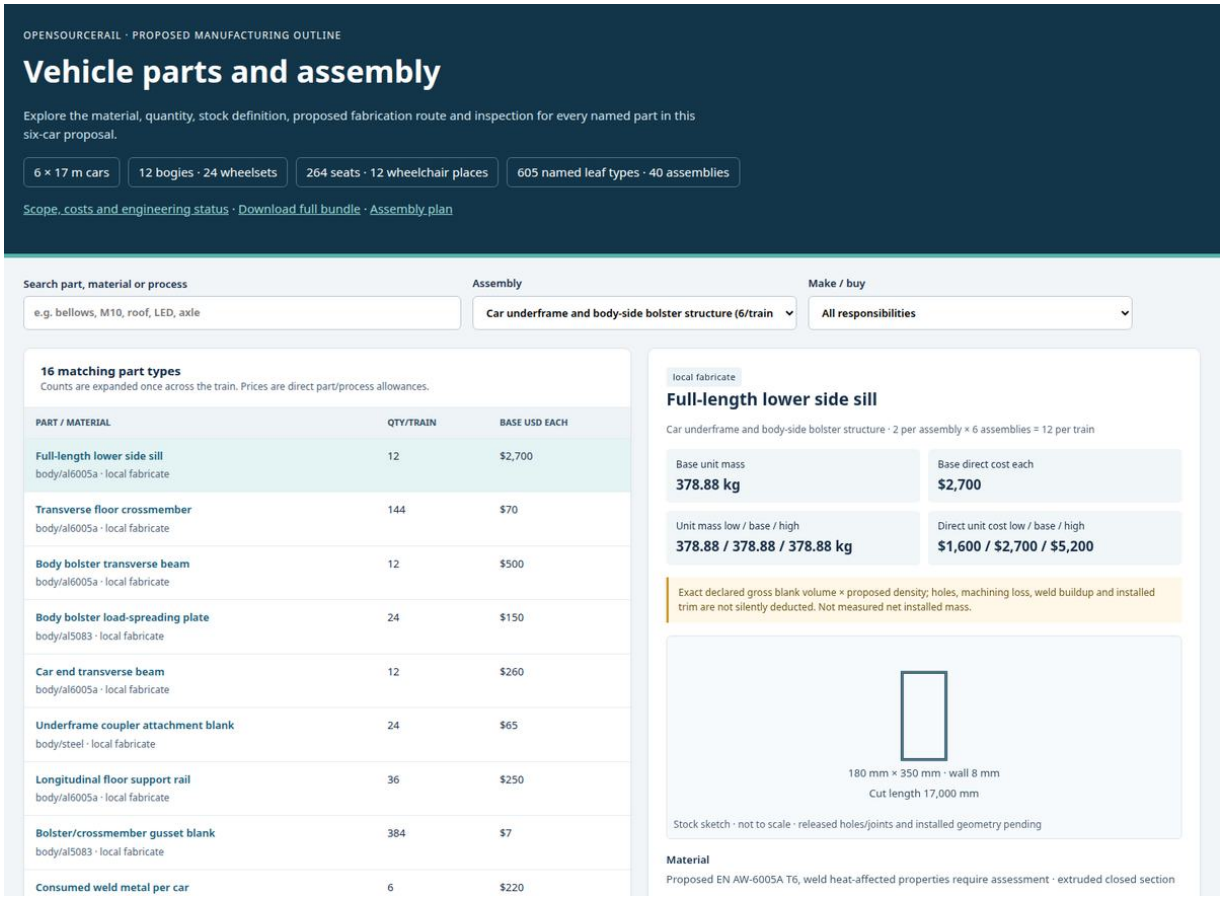
This is also why the six-car formation matters. A car design is not six identical isolated objects: end cars carry different equipment and cab interfaces; intermediate cars share couplers, gangways and supply paths. Changing an end-car mass or door arrangement can affect stopping position, platform fit and maintenance stocking. The reader should expect every revision of the displayed train to carry a formation-level mass and interface account, not merely an attractive replacement picture.

6.2 Doors, seating and accessible interiors

The interior is where the engineering meets the passenger. The proposed layout must support comfortable ordinary travel, wheelchair use, simple boarding, circulation, emergency movement and quick cleaning. Behind a calm cabin view are hundreds of door parts and many seat and service attachments. A successful design must fit both the passenger and the mechanic who will repair it.

Count what the image hides

The parts catalogue below shows how a visual vehicle can be opened into named assemblies and individual items. A counted proposal for one passenger door opening contains 74 part definitions and 694 pieces. They include leaves, header, rail, carriage, drive, belt, pulleys, sensors, glazing, seals, fasteners and electrical work. The make-or-buy proposal assigns some pieces to local fabrication, others to standard purchase, and a smaller set to purchase subject to qualification. These counts expose the production and spares problem; they do not make the door build-ready.



Searchable vehicle parts catalogue preview

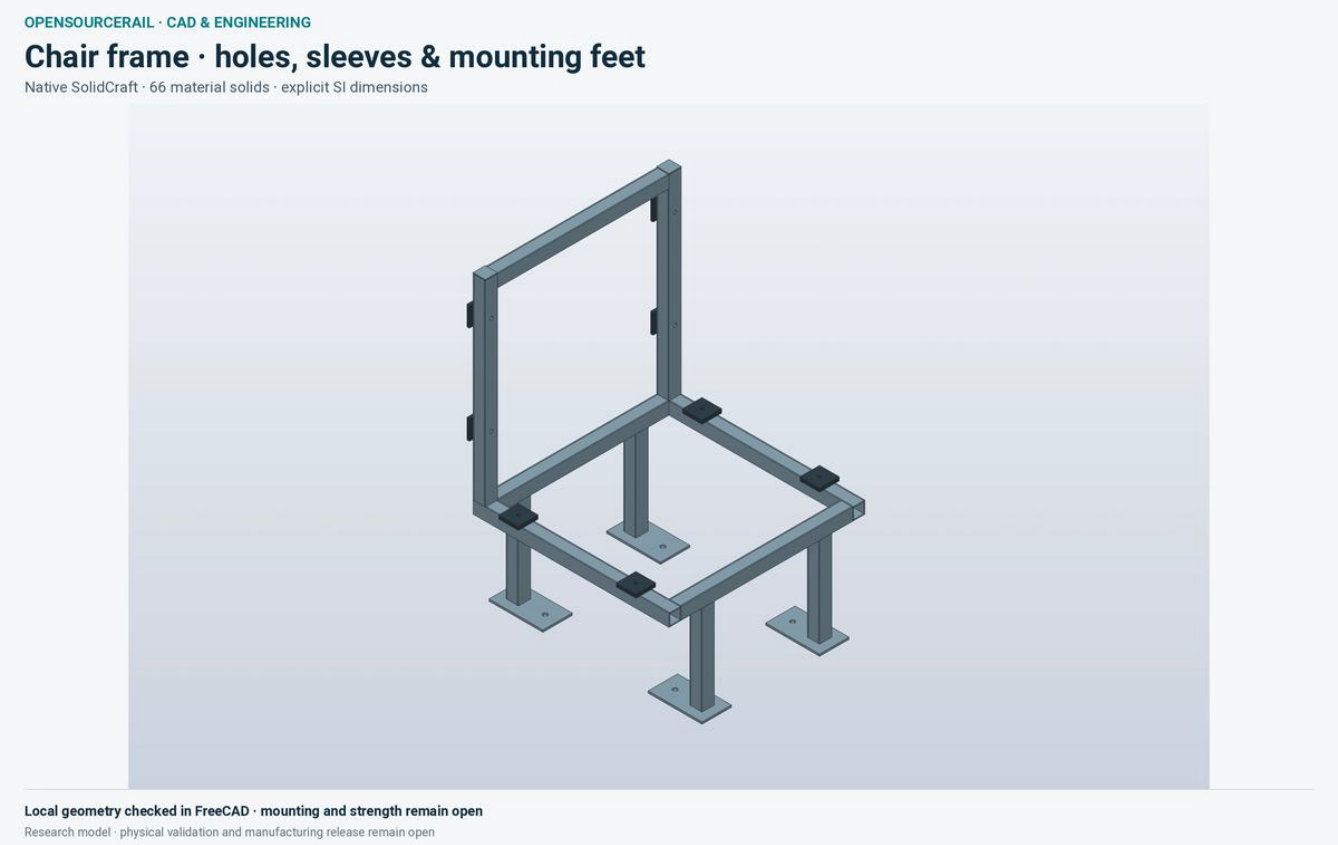
A real interface for inspecting proposed parts. Counted items and search views are planning aids, not released fabrication drawings.

The door operator currently illustrates an honest engineering hold. Several items have generic positions near the middle of the header rather than fully mated locations. The proposed two-metre extraction route is shorter than a 2.9-metre guide stock item. A car may show a beautifully animated pair of leaves and still lack the space to replace its guide. Before controlled manufacture, the design needs shared datums, tolerances, drive and sensor placement, motion envelopes, cable routing, inspection limits and a physically plausible removal sequence.

Accessibility is a spatial design

The six-car reference names 44 fixed seats and two wheelchair places per car. Their numbers form a clear normal accommodation policy, but the cabin must also support the route between the platform and those places. The boarding gap, doorway clear width, turning space, securement, nearby handholds and reach of communication equipment matter. A one-sided door layout must match every station platform in the chosen service. If the layout or train formation changes, the platform, stopping and civil consequences need to change with it.

The chair pictured below is a detailed local manufacturing proposal. It can help a team discuss structure, upholstery, joining and replacement, but it is a separate item study until tied to a controlled vehicle seat mount and passenger tests. Seat count alone does not resolve fire, vibration, impact, cleaning or accessibility.



Locally made passenger chair proposal

A proposed finished component. Its mounting, qualification and actual vehicle installation require separate evidence.

The cabin is a service environment

Lighting, ventilation and communication must work during an ordinary journey and in a degraded one. Doors opening repeatedly in a hot climate can dominate cooling loads. Windows and roof surfaces admit solar heat. The location of ducts, fans and batteries affects both passenger space and maintenance access. An illustration of a bright cabin cannot establish illumination, glare, air movement or hot-weather endurance; these require defined limits and tests with the selected equipment.

Emergency conditions also challenge the interior. A crowd structural case can exceed the intended seating load, yet it does not change the normal passenger policy. An evacuation must work for people with a range of mobility, perhaps when the train is stopped away from a station. The door release, step, gangway, communications and rescue route need to be studied together. Clear passenger instructions and accessible staff procedures are part of the system design.

The v0.7 door module work is deliberately specific: one opening, one vehicle candidate, controlled geometry and a route through manufacture, inspection, installation, service and replacement. Closing that route would teach the project how to apply the same discipline to seats, HVAC, lighting and the other items passengers rely on every day.

Design for daily use and replacement

The opening cycle is a useful way to examine the whole door. People approach from different directions, sometimes with luggage, mobility aids or children. The door must detect an obstruction, communicate its state and avoid trapping a passenger. It must also hold its required position when power is lost and allow a controlled release when evacuation is necessary. Each feature needs a defined test condition, a limit and a way to inspect the result after years of wear. A polished animation cannot answer how the seal behaves with dirt, heat or a slightly displaced leaf.

The same design review should follow a fitter replacing a worn roller or drive belt. Can the part be reached while the vehicle is in the depot? Which panels must be removed? Are lifting points, isolation and test access provided? Can the replacement be adjusted using a documented datum, or does it depend on the memory of the person who built the prototype? These questions turn a general aim of repairability into actual vehicle space and controlled assembly work.

Interior finishes deserve this discipline as well. A seat should be comfortable and cleanable, but its mount also carries normal, emergency and impact loads into the floor. Repeated removal must not damage that floor. Finishes and adhesives must be selected for fire behaviour and durability, while replacement parts remain obtainable. A locally made chair can be an excellent starting point once these interfaces and tests are included in the same vehicle candidate.

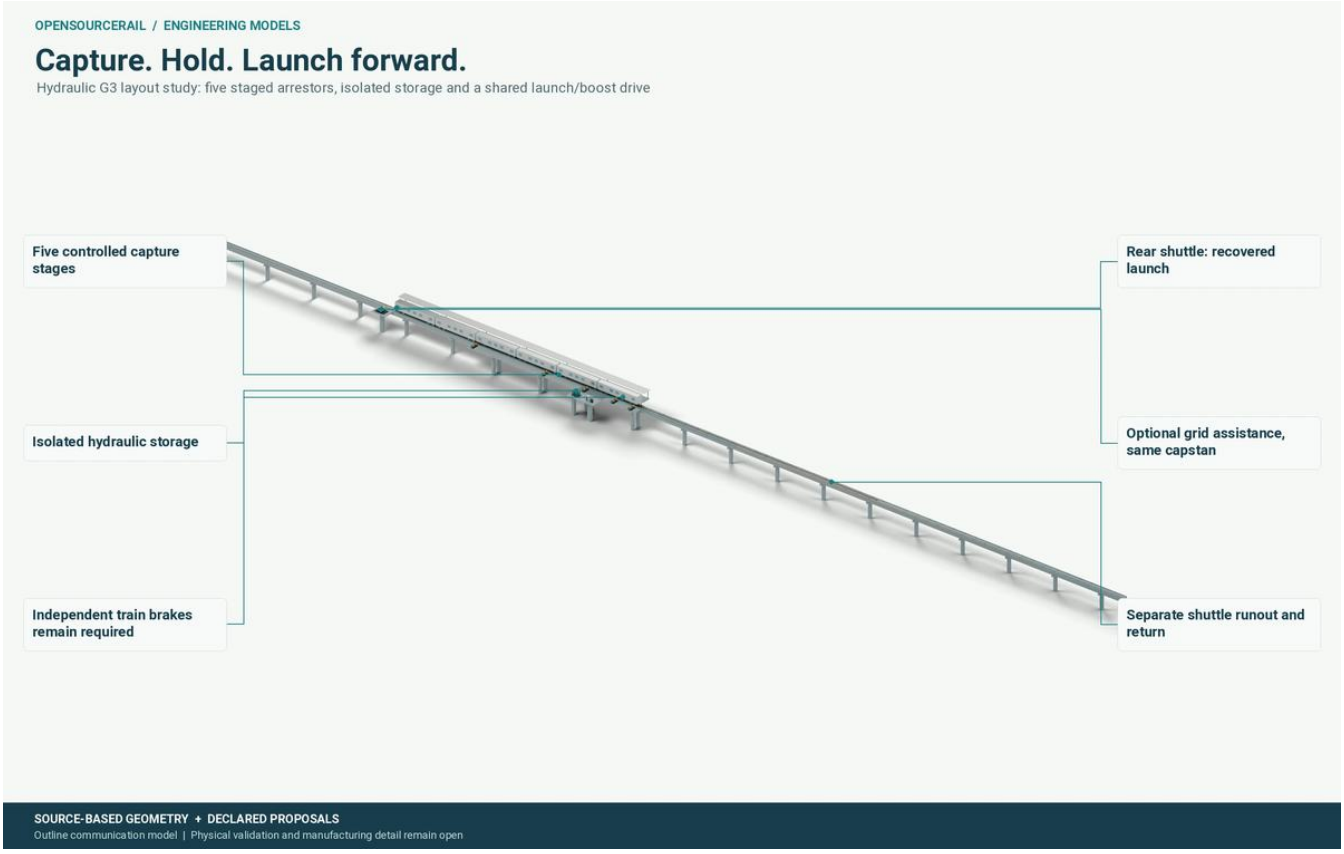
6.3 Station drive and launch alternatives

The wheel-supported reference places the train's weight on ordinary rails and explores ways to supply traction near stations. Keeping heavy traction equipment off the car may simplify the vehicle, but it moves power, cooling, maintenance and failure handling into the station. The project keeps several propulsion families open: wayside linear motors, mechanical launch and recovery, powered wheels, rollers and other explicit research configurations. Their pictures are alternatives, not parts of one combined installation.

A station is an energy exchange

In a station-powered concept, a departing train receives energy while passing a driven zone, then coasts or uses another declared means to reach the next station. Near arrival it must be received or stopped safely. The idea invites useful questions: How much force is needed at each speed? How long must the active equipment be? Where does loss heat go? How does the train stop if the receiving equipment has failed? What happens on a hill, with a crowded train or during a power interruption?

One visual research layout considers a controlled mechanical capture, temporary energy store and a shuttle that helps launch the next train. The view below shows a proposed set of assemblies and maintenance envelopes. Its calculated travel distances are not the lengths of validated hardware. Guards, sensors, brakes and energy isolation would need to be part of a complete machine.



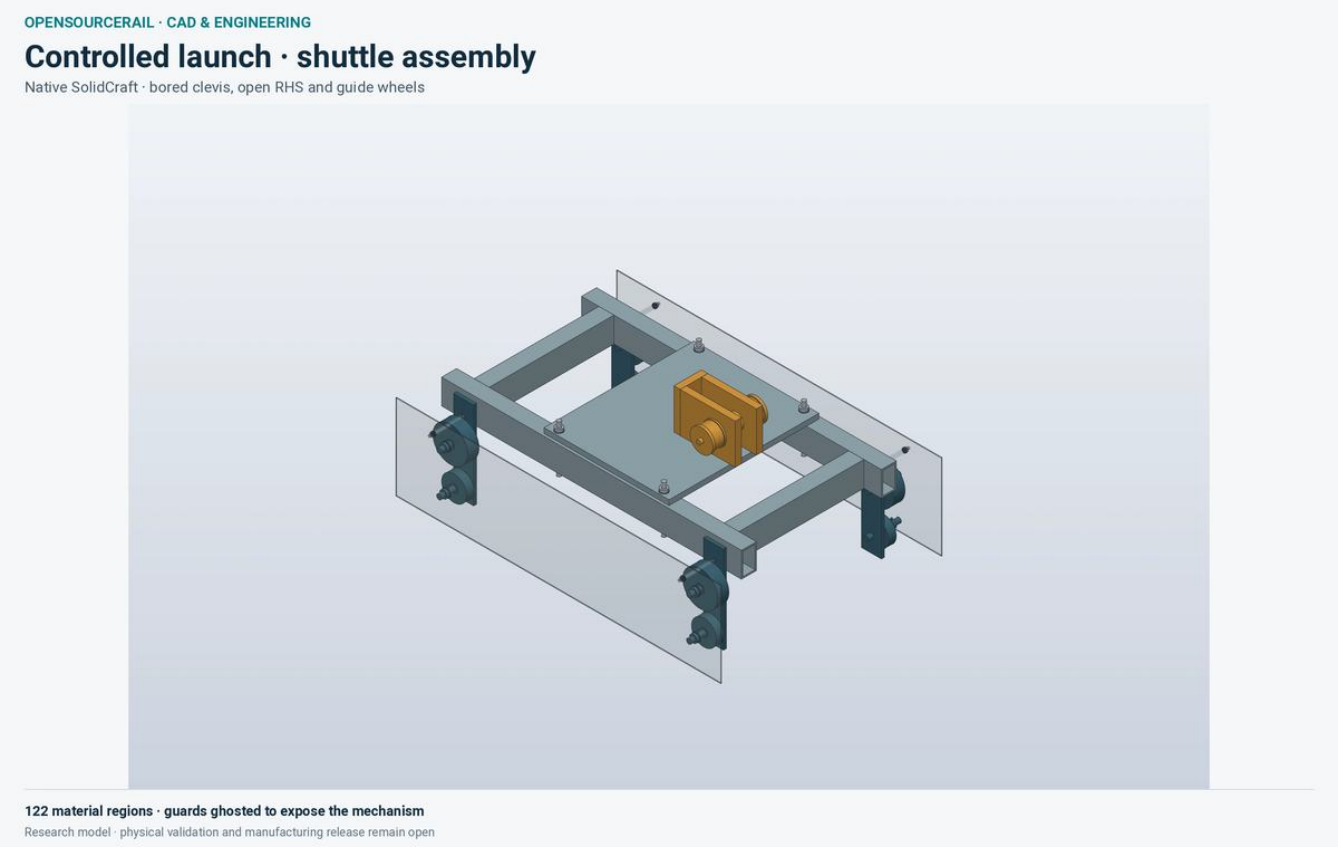
Station capture, storage and launch arrangement

A source-derived research layout for one mechanical station concept. It does not select the installed drive for the passive-wheel reference.

Energy recovery has practical limits. The arriving and departing trains must be timed so that a recovered store can be used. The capture process loses energy in friction, pumping and conversion. Storage must be safe while charged and while isolated for maintenance. If the next departure is delayed, the railway needs another energy path or a way to dissipate the stored energy. A complete economic comparison counts those losses, standby equipment, inspection and replacement.

What a detailed launch view can show

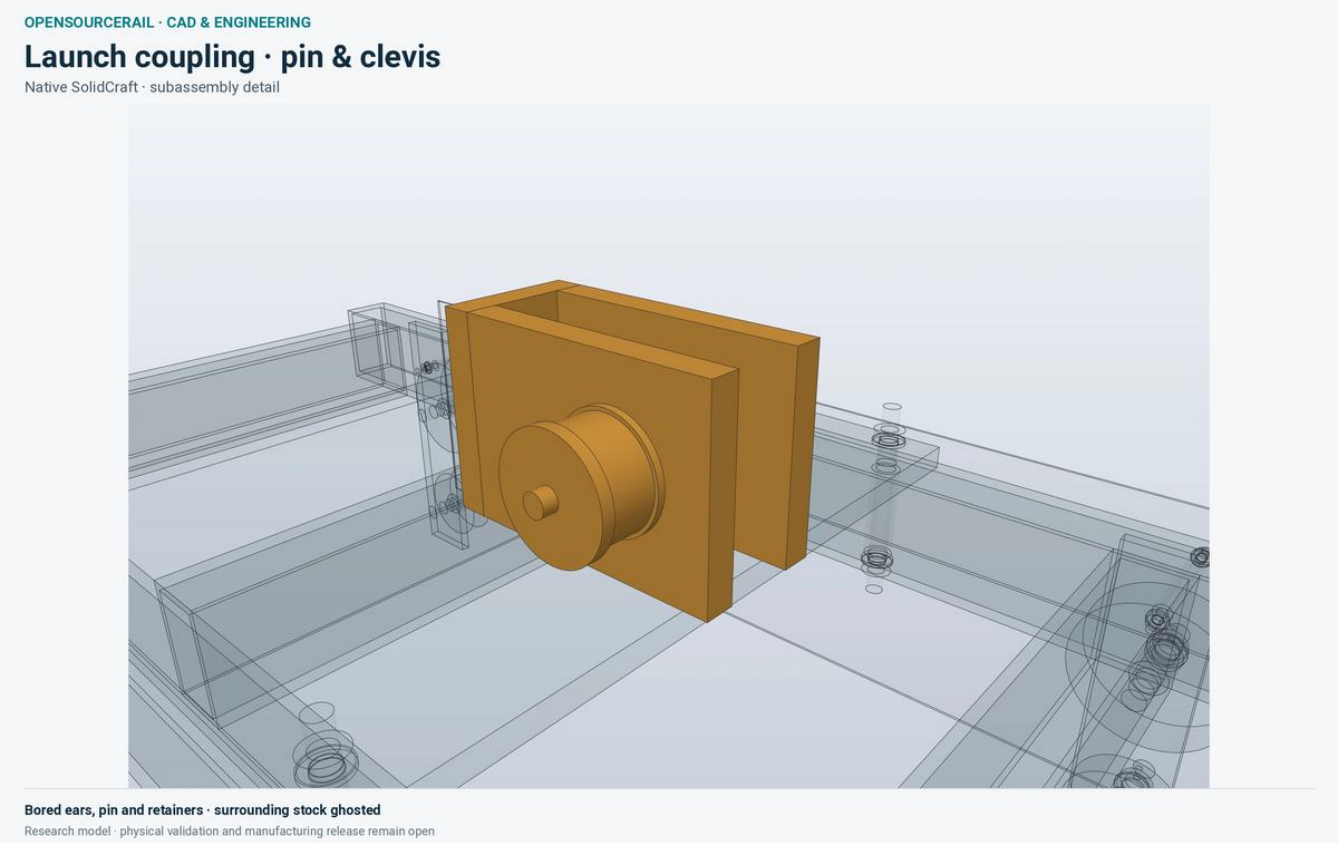
The launch-shuttle study below is a distinct physical research candidate. Its native geometry resolves many material regions, bores and wheel surfaces. That makes fit and manufacture questions easier to identify. The pictured guards are ghosted so the mechanism can be seen; they are not absent from a future safety requirement.



Native launch-shuttle assembly proposal

Detailed native CAD for a separate launch proposal. Mated contact, fatigue, protection and operating capability remain unqualified.

The clevis and pin detail focuses the question further. A pin can carry large repeated forces, but its actual load path depends on contact, clearance, retention, material, joining and lubrication. A valid solid is the beginning of a mechanical study, not its final answer. Surrounding ghosted parts in a presentation image do not prove their joints.



Clevis, pin and retainers in the launch proposal

A mating-area study for the launch candidate. Weld, contact and fatigue qualification remain open.

A wayside linear-motor alternative has different questions. Its passive vehicle element must fit between the rails throughout wheel and body movement. A useful motor map has force, normal force, current, voltage, heat and cooling across speed and position. A short ideal-field method cannot stand in for a manufactured unit with slot ends, inverter limits, feeders and a complete thermal system. Powered length is therefore a candidate result, not a fixed number drawn from a concept image.

Each alternative should be compared at the same passenger service and with independent braking and rescue. A low onboard mass is attractive only if the station, electrical, civil and whole-life consequences are also credible. This is why the image gallery presents mechanisms as research questions rather than a ready-to-install propulsion package.

Make the comparison at railway scale

The drive is only one part of a departure. A station also has to admit the arriving train, hold it, open the correct doors, exchange passengers and dispatch it with safe separation. If an accelerator requires a longer straight platform, additional protection or a larger electrical room, those requirements belong in the same corridor comparison as the motor. If a train relies on several powered zones, the spacing and condition of those zones govern what happens after a failure. The selection must be made for a whole service, including recovery, rather than by comparing isolated peak thrust figures.

Energy is equally dependent on the timetable. A brief peak may be easy to demonstrate once and difficult to repeat every two minutes in summer. Transformer, converter and cable losses become heat at a specific place. Power demand must be compared with feeder capacity, cooling and the availability expected of a public service. Where braking can return energy, the electrical network needs somewhere safe and useful for it to go; where it cannot, the heat and wear have to be carried elsewhere.

The pictured mechanisms make these trade-offs tangible. The shuttle asks whether moving contact hardware can survive repeated duty and be replaced without long closures. A fixed linear motor asks whether an air gap and force map can be maintained across real wheel wear, track tolerance and weather. Both need protection from

people and debris in a station. These are testable questions for separate candidates, and the answers may depend on the chosen city alignment and service pattern.

6.4 Civil assemblies and construction

An elevated railway becomes a line of structures through a city. The guideway supports rails, vehicles, stations and maintenance access; it must also be buildable above streets, utilities and people. The civil gallery makes the work tangible by showing precast spans, bearings, supports, transport and erection ideas. It should be read as a set of candidate studies, not a completed bridge package.

A span is more than its beam

The load path begins at the wheel and continues through rail, fastener, deck or beam, bearings, piers and foundations. Each handoff can govern the design. A beam with a good bending result may need a difficult rail seat. A compact deck may make bearing access or drainage harder. A lighter span may demand more temporary restraint during erection. The comparison therefore needs moving loads, crowd and accidental cases, thermal movement, fatigue, rail alignment and maintenance, not just the material volume of the beam.

The segmented civil image below shows a proposed two-track assembly with segments, rails, bearings, piers and footing envelopes. Several joint, reinforcement and foundation details are still hypotheses. This image belongs to a separate civil candidate and is not the assembled form of the ideal continuous beam in the finite-element image in the next chapter.

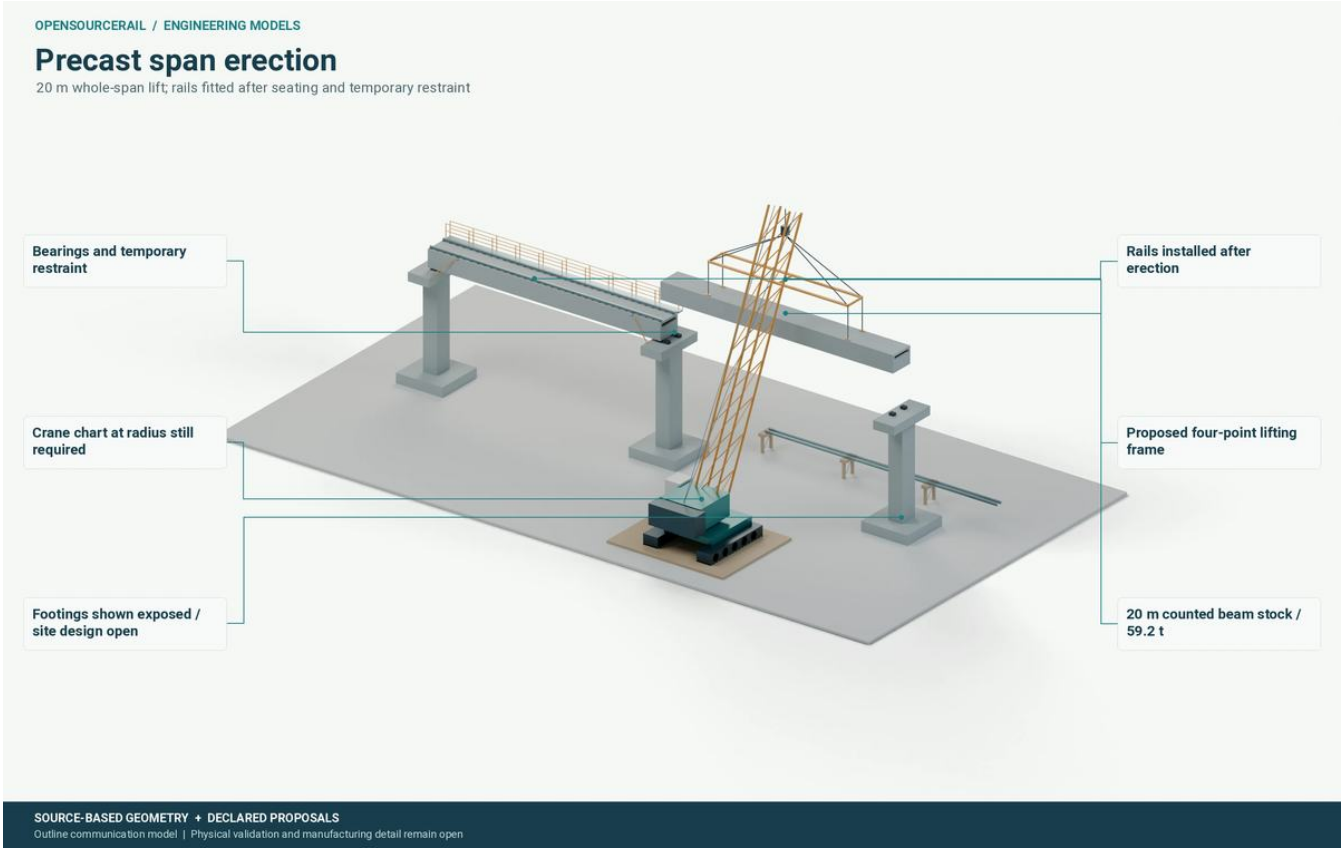


Segmented two-track civil assembly proposal

Located research geometry for one civil proposal. Segment joints, bearing contact and site foundations remain unverified.

Erection changes the design

Precasting can shift labour from a street site to a controlled yard. A yard can use repeatable forms, inspections and planned curing. The resulting unit still needs a road route, lifting points, a crane or gantry, safe temporary support and a final connection. Its size and weight should follow what the city and its workshops can actually handle. A crane's useful capacity is its capacity at the required working radius, with rigging and ground bearing accounted for, not the largest number printed on the machine.



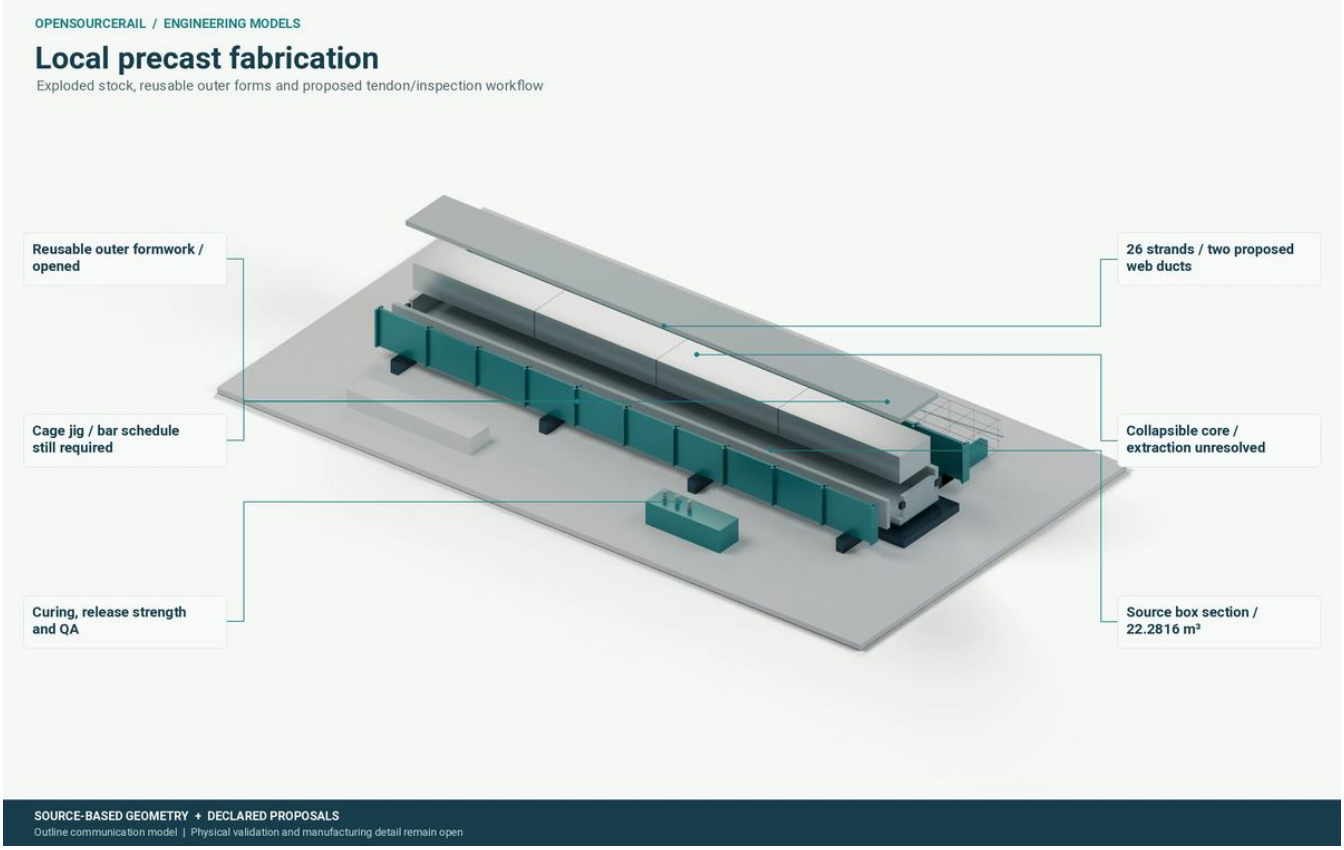
Crane erection concept for a precast span

A proposed erection sequence. Actual lift, temporary restraint, traffic management and ground capacity need site-specific design.

An overhead launching gantry is another possible method. It can reduce dependence on a large crane at constrained streets, but introduces its own plant, support reactions, launching sequence and inspection needs. The appropriate method may differ from corridor to corridor. Choosing it late can force changes to the span's lifting points and geometry, so construction should influence the first civil family comparison.

Forms and foundations have their own lives

Reusable forms can lower cost if the candidate is repeated enough times, the form can be stripped reliably and the factory can meet the curing cycle. Proposed tendon ducts and cage fixtures need real bar schedules, anchorage, cover and quality tests. The cost model should count forms, lifting gear, transport and rejected pieces, not just concrete and steel. Where soil and utility conditions change, a standard beam may still sit on very different piers or foundations.



Local precast yard and reusable tooling concept

A proposed production route, not an established cycle time or qualified prestressing process.

For a public owner, these images help ask what the elevated line will mean on the ground. Which roads will be occupied, for how long? How will stations connect to pavements and crossings? Can a damaged bearing be inspected and changed? Which details must be agreed before land or utility work starts? The engineering answer needs the exact corridor, ground and candidate, with construction and maintenance represented as part of the same design.

The corridor changes the structure

A repeated span can help a factory learn and improve, but repetition does not remove local design. Each support meets a particular street, utility network and ground condition. A road crossing may need a longer clear opening; a constrained junction may have no safe crane location. Near a station, the deck may widen, carry different rail equipment or meet a building entrance. These variations should be visible in the quantity and programme, so a standard-span price is not silently applied to every metre of route.

The order of construction changes what the beam must resist. It may be stressed in a yard, lifted from only two points, transported with temporary supports, set onto bearings, then loaded with rails and trains. The largest local stress can occur before the first passenger service. Temporary stability of a pier or half-finished station deserves the same care as the completed structure. A design package must tell the builder which supports and restraints are required at each stage.

Long life brings other interfaces. Bearings must accommodate movement and remain replaceable. Drainage should keep water away from reinforcement, fasteners and public space below. Rail supports need access for adjustment and renewal. Expansion joints can cause noise and recurring maintenance, so their number and position should be an explicit design choice. These details influence whole-life cost more than an image of a clean concrete deck can convey. The civil gallery is most valuable when it starts these conversations before a corridor is frozen.

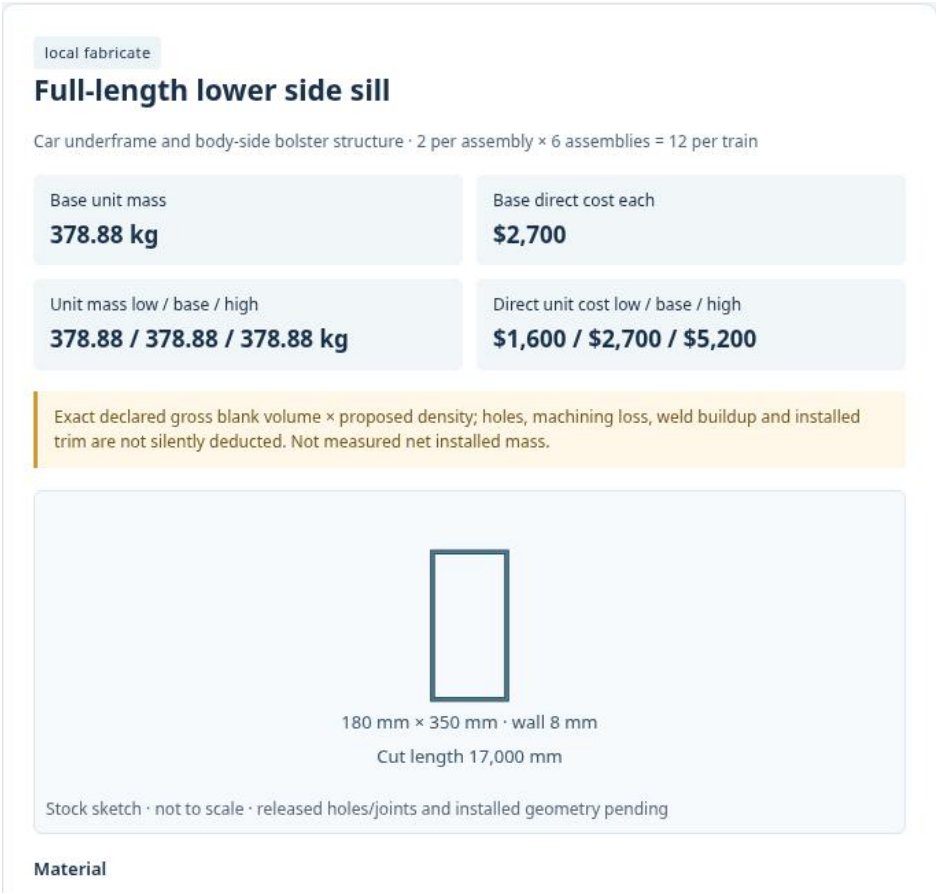
6.5 Manufacturing details and assembly

Making a railway locally requires much more than permission to download a drawing. The workshop needs stock forms it can source, tools it can use, operations it can repeat, dimensions it can inspect and a record of which part went into which train. A digital assembly becomes practically useful when those decisions can be made from it without guessing at hidden interfaces.

Start with a countable structure

The proposed vehicle catalogue names 605 leaf-part types in 40 assemblies across the six-car train. It reaches from bogie frames, wheelsets and body members to seats, doors, batteries, HVAC, lighting, couplers and service hardware. Individual fasteners, cut strips and bought subcomponents expand the bill into many more units. That is a serious attempt to account for the work behind a train, but it is still a proposal: bought equipment internals, finished joint details, installation positions and complete as-built mass need further evidence.

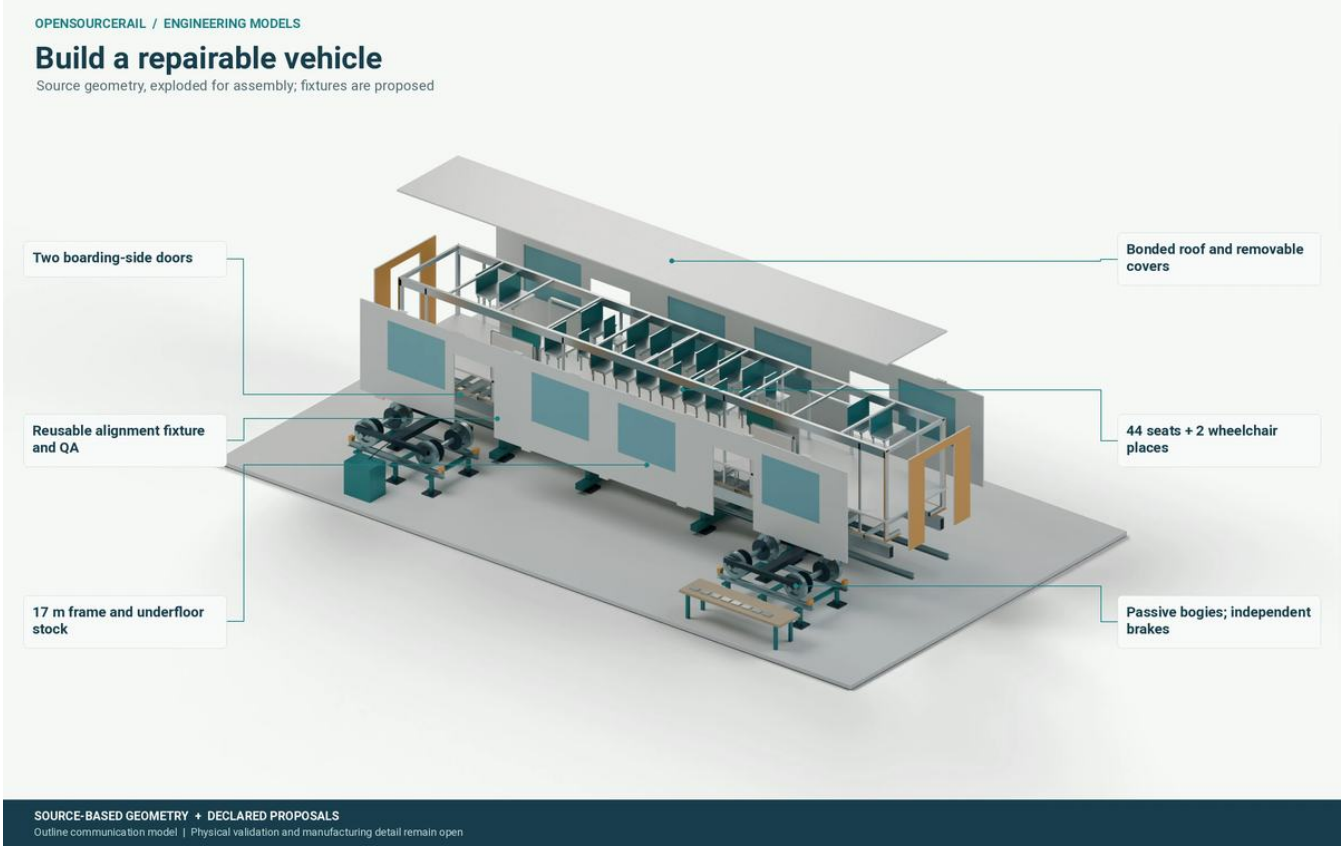
The stock-detail image below makes one important distinction visible. A blank or envelope can be counted before its holes, joining faces, clearances and inspection limits are released. The first count helps with sourcing and factory planning; the controlled drawing makes a repeatable part. A tolerance chosen without a matching assembly function can make a product expensive without making it safer or easier to maintain.



Vehicle part stock and proposed fabrication detail

A detailed stock and process view. Mating geometry and acceptance limits remain subject to controlled drawings and tests.

The assembly sequence begins with a datum-controlled underframe, followed by body members, roof and floor, bogies, equipment, panels, doors, glazing, passenger fitout and inter-car connections. Each subassembly needs the tools and inspection opportunities that suit its materials. A welded frame may distort; a bonded panel needs a qualified preparation and cure; a purchased bearing needs traceable receipt and fitting. An open design can enable local work only when these process controls are as clear as the nominal shape.



Proposed vehicle fabrication and inspection route

An assembly communication model. Fixtures and stations are proposed; the image is not a measured production rate or released shop instruction.

Make, buy and qualify are different routes

Local fabrication can include cut and formed members, body panels, interior fittings, some castings and assembly operations. Specialist items such as bearings, cells, electronics or railway glazing may be bought. Their local installation still creates value, but the source and certification of the bought item must remain visible. A supplier's sales address is not proof of domestic manufacture. The bill should distinguish standard purchase from purchase that needs additional qualification for the railway duty.

The door opening illustrates the mix. Some components can be made locally, many are bought to a standard description, and a smaller set needs specific qualification. A complete module then requires its own assembly drawing, tooling, electrical checks, motion test, incoming inspection and serial or lot custody. It must also be removable after years of service. A part that can be installed only before the body is closed is not necessarily a repairable railway part.

Cost the route, not just the material

The project calculates provisional low, base and high direct parts and process allowances. These help compare design families, but the amount for a stock item is not a finished train purchase price. Factory space, shared tooling, final assembly, rejected work, qualification, delivery, warranty and renewal may sit outside a narrow subtotal. Production hours are hypotheses until a real workshop measures them. A lower piece price may be offset by difficult fitting or a long imported repair channel.

A candidate-linked manufacturing package should make these boundaries readable to a buyer. Which parts have a drawing? Which materials have an approved grade and source? Which operations are measured and which are estimates? Where will quality hold the work? What equipment can be changed without disassembling half a car? Such questions turn an engineering image into a practical programme for local production and long-term repair.

Build a route from prototype to fleet

An early prototype can reveal access and tolerance problems while changes are still affordable. A pilot build then tests whether ordinary workers, tools and suppliers can repeat the process with controlled quality. Only after that should a fleet plan rely on stable production hours and yield. The stages need different records: prototype measurements and anomalies, pilot work instructions and inspection results, then serial histories and change control for production vehicles.

This route also protects local capability. If a factory can make a body member but must import a long-lead bearing, the schedule and spares plan should state that dependency. If a supplier changes a material or connector, the affected geometry, tests and maintenance documents must be reviewed before installation. Open drawings enable more people to contribute, but shared knowledge is most valuable when it includes the reasons for tolerances, test limits and assembly choices.

A workshop should be able to answer two practical questions from the finished package: how is this train built, and how is it restored to service after a failure? Designing those answers together can avoid special tools that are needed only to reach one buried part or an assembly sequence that prevents later replacement. The result is a railway that can be maintained locally, not only assembled locally.

VII. Making and keeping the railway

Local production, whole-life decisions, control and maintenance.

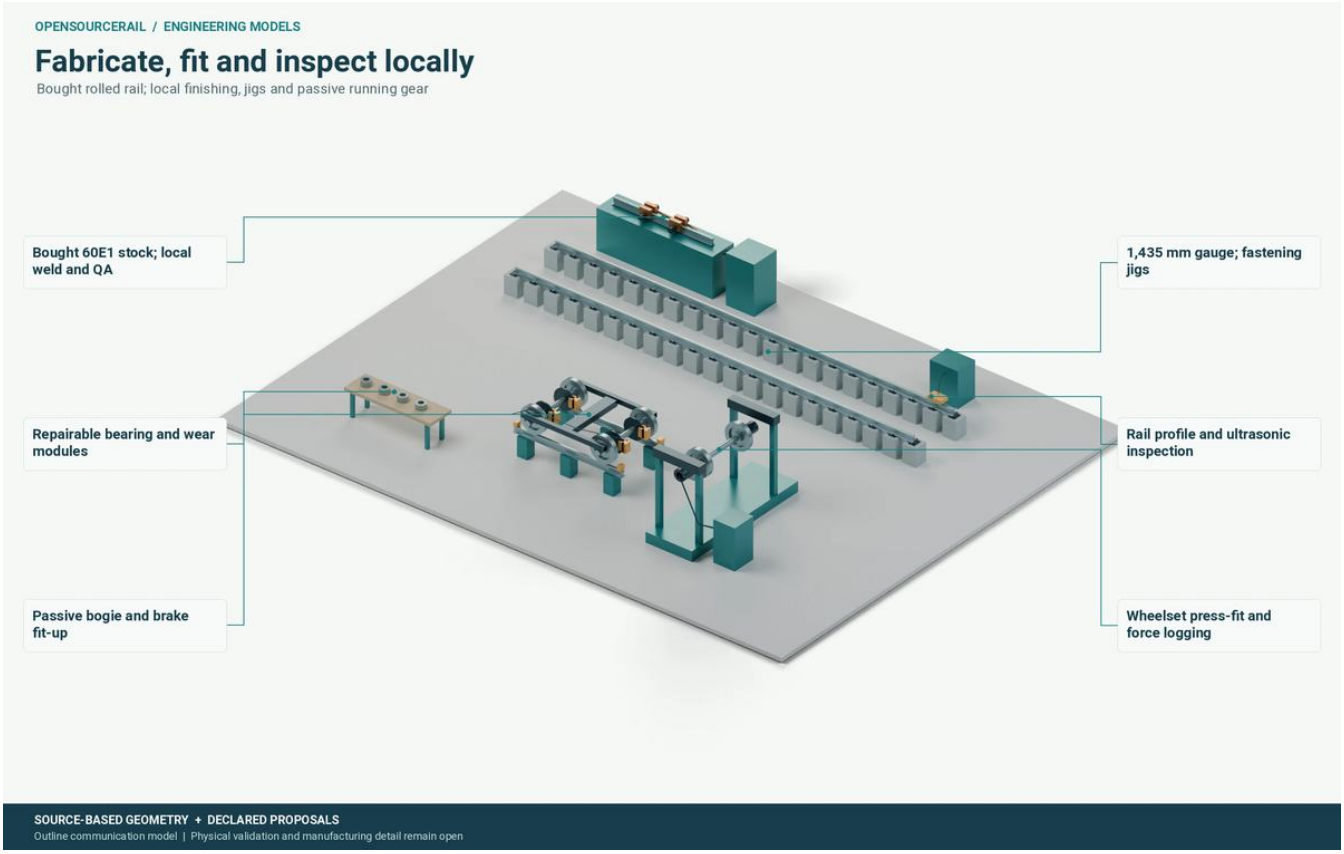
7.1 Local manufacture and sourcing

A railway can be designed for local work without pretending every specialist part will be made locally. The useful question is which knowledge and production steps can be established, inspected and sustained in a particular place. OpenSourceRail's vehicle inventory and civil studies provide a starting bill of materials. The next step is to turn each relevant item into controlled stock, process, assembly and quality information.

Make, buy and qualify

A structural member may begin as an extrusion or folded sheet. Its useful description includes grade, stock size, cut length, joining method, surface treatment, dimensional datums and inspection. A bought bearing needs a qualified specification and an incoming check, even if its internal design is left to the supplier. A door sensor or battery cell adds electrical, environmental and safety requirements. The make-or-buy choice should therefore be recorded item by item, including original material and component origin, instead of applying one local-content percentage to the train.

Candidate processes include concrete precasting, metal cutting and forming, structural fabrication, composite panels, interiors, wiring and final assembly. Proposed local capability does not itself prove a factory's yield, rate, tooling cost or skill base. A workshop may be able to make a prototype while still needing investment and training to supply a whole railway. The cost comparison should include that investment and the value of later repair and renewal work.



Proposed fabrication and inspection workshop

Illustrated workstation. Wheel, bearing and bogie manufacture or inspection would require controlled processes, qualified equipment and recorded checks.

From stock to an assembled train

The current six-car breakdown identifies 605 named leaf-part types in 40 assemblies. It includes the running gear, car structures, seats, doors, glazing, HVAC, service batteries, lighting, wiring and many joints and fasteners. That level of counting makes hidden work visible. It does not replace the next stage: a drawing that establishes mating dimensions and tolerances, an operation sequence, a jig, an inspection plan and a safe removal route.

A plausible vehicle assembly starts with a datum-controlled underframe, then adds body frame, roof and floor, bogies, equipment, panels, doors, glazing, interior and inter-car connections. At each stage the assembly must still be accessible for inspection. A late-discovered interference between a cable tray and a door mechanism is much more expensive than a revised source model before stock is cut. This is why the candidate's geometry, inventory and work instructions need to agree.

For the representative door module, the count of 694 pieces per opening is useful only if the header, carriage, guide, drive, sensor and body interfaces can be made and assembled as described. The 97 proposed locally fabricated pieces are a production opportunity, not a completion certificate. Standard purchases and qualified purchases have distinct incoming requirements.

Repair is a production route in reverse

A module's design should also describe how a technician diagnoses it, isolates power, removes it, fits a replacement, checks it and returns the train to service. The same drawings and part identities should support both factory assembly and depot repair. This helps a local operator keep knowledge after the original builder has left and gives later suppliers a clear interface to compete against.

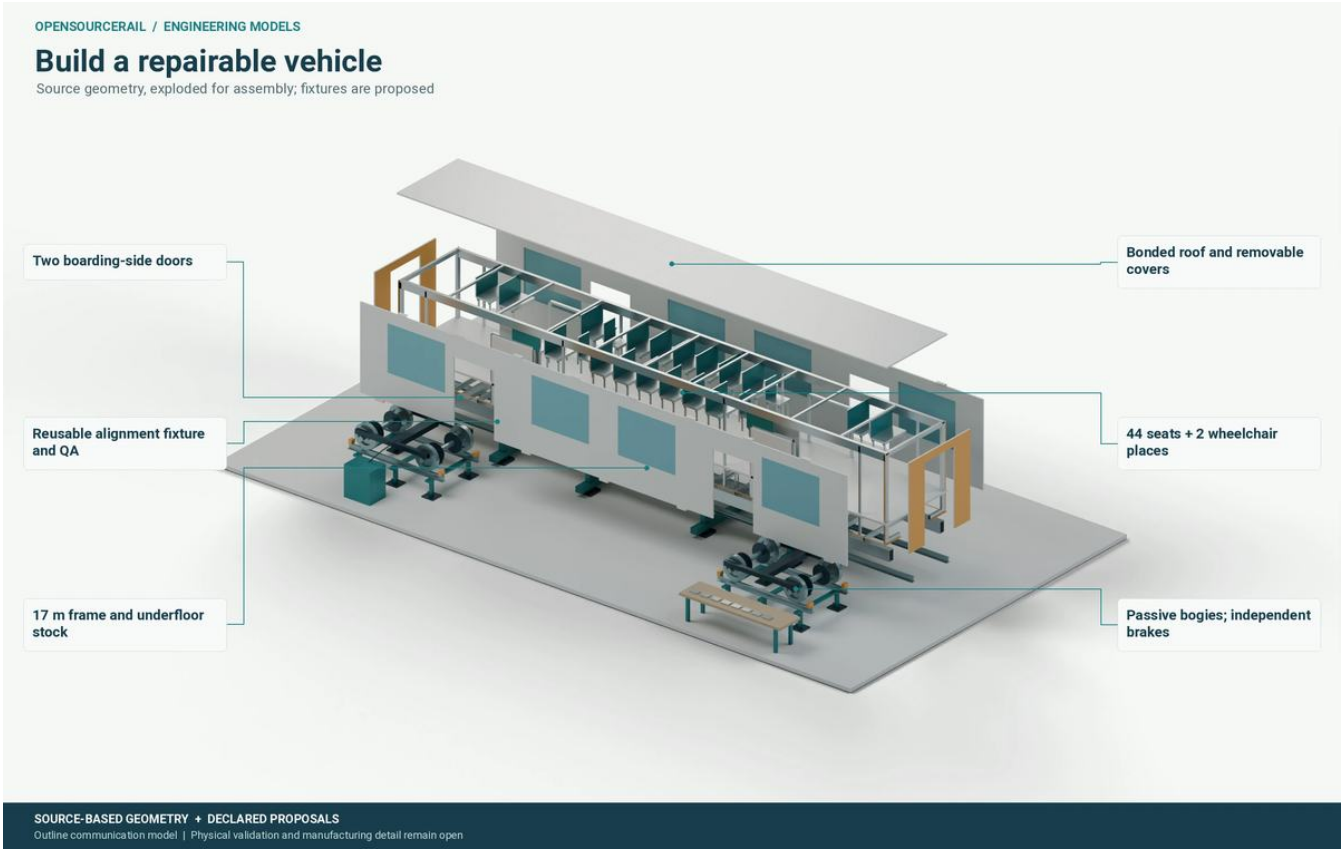
7.2 From factory to finished corridor

An affordable railway is partly a construction problem. A design that uses little material can still be expensive if it demands scarce equipment, specialized long-distance shipments, repeated traffic closures or difficult site adjustment. OpenSourceRail aims to make vehicles, stations and civil parts that local teams can produce, inspect, install and eventually repair. That objective has to be tested against the actual site and available workshops, not inferred from the apparent simplicity of a drawing.

Decide what is made locally

Local manufacture begins with a useful division of work. Some parts may be cut, welded, machined or cast in regional facilities. Others are better purchased as qualified standard items. Safety-relevant bought parts may need supplier testing, traceability and incoming inspection even when they look routine. A train bill of materials must carry these choices for every physical item, including joints, fasteners, seals, wiring, cooling and protective equipment. Omitting the small parts makes both local-content and cost estimates misleading.

The project has a detailed vehicle inventory and proposed process routes. It identifies the door operator as a particularly useful representative module: local fabrication, standard purchase, qualified purchase, electrical integration and future replacement all meet in a compact assembly. The counts are a starting map for work packages. Before manufacture, workshops need released drawings, mating dimensions, material and process specifications, inspection points and a way to record deviations. The fabrication sequence must match the actual vehicle structure, including distortion control, corrosion protection and safe lifting.

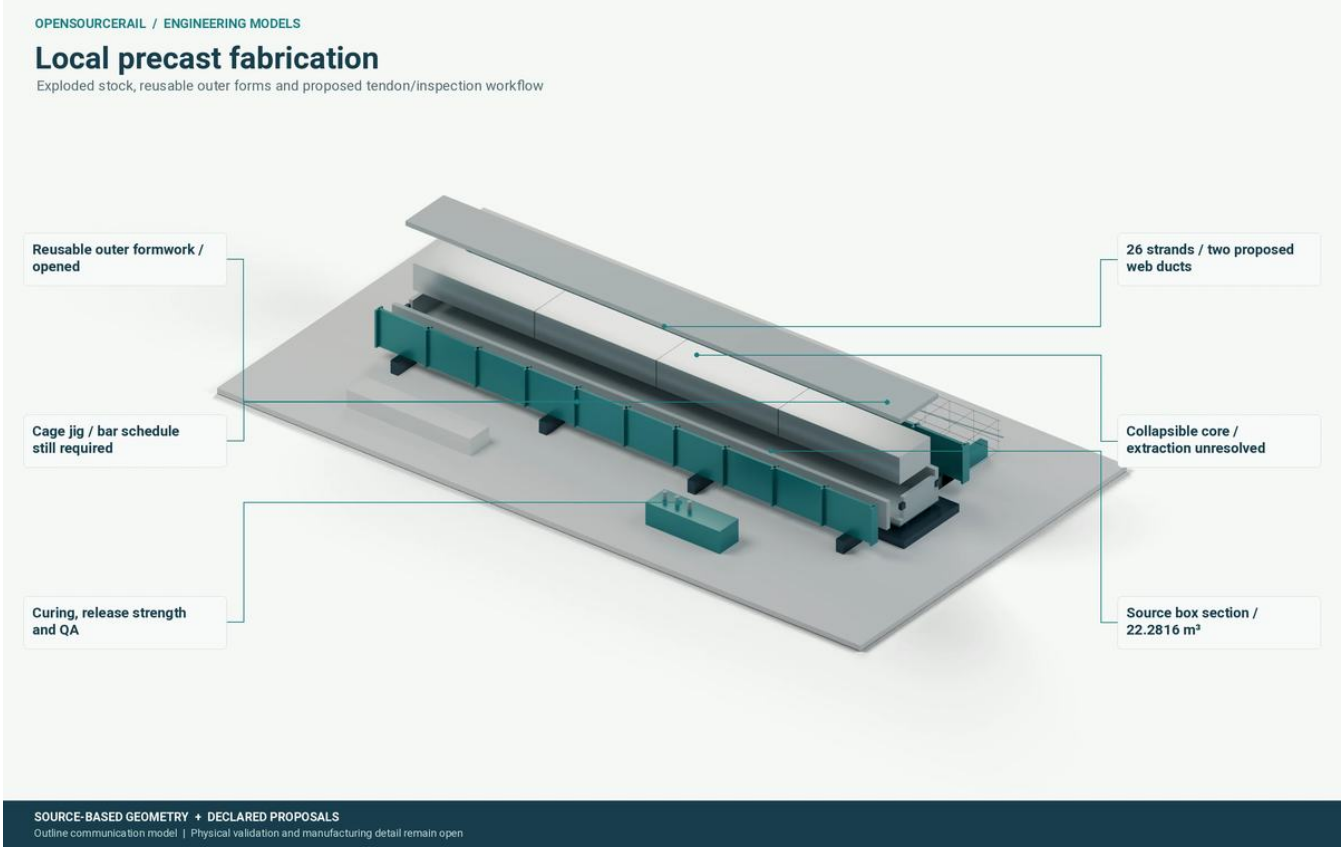


Vehicle fabrication concept

A shop can make and repair substantial portions of the railway, once the design and quality process are proved for its equipment and staff.

Precast work changes the site problem

The civil studies explore repeatable precast guideway components. Casting off site offers controlled dimensions and inspections while potentially reducing work above a busy road. But the finished element still has to be transported, lifted, supported, connected and maintained. Its production size should follow road access, factory moulds, lifting points, transport constraints and the crane that can reach the installation position.



Precast fabrication concept

A repeatable span is only one part of a complete support and erection system.

For a representative span, the construction method should state the temporary support, lifting gear, crane radius and capacity at that radius, traffic or railway closures, bearing installation, post-tensioning or other connections, survey adjustment and safe release of the lifting gear. The span also needs a load path through piers and foundations for normal, crowd, wind, accident and construction conditions. A finite-element result for the finished beam cannot establish the stability of a half-built span.

Delivery is a sequence of approvals and handoffs

The city's outline comes first, followed by survey, geotechnical work, land and utility decisions, a matched service and vehicle candidate, and construction staging. Procurement should identify long-lead items and alternative suppliers while preserving the approved technical interfaces. Factory tests, site inspection, commissioning and staff training then establish whether the built system matches the design. Passenger service requires its own independent safety and operational approvals.

The platform's project and business tools can help keep these handoffs together. A proposed bill of materials can become a production plan, a received component can be inspected, and a fitted module can enter a maintenance record. Existing demonstrations show parts of that digital route, including a synthetic business-system example. The full candidate linked transaction for a real door module remains to be completed. The software should make omissions and conflicting revisions visible before someone orders, builds or fits the wrong part.

Repairability affects the first drawing

A replaceable unit needs reachable fixings, a safe isolation procedure, clearance to remove it, lifting provisions if it is heavy, and enough space to inspect its neighbours. These are geometric facts, not afterthoughts for a manual. A door guide that cannot be extracted through the available opening may force a larger body change. A power cabinet that blocks a crane route may change a station layout. Testing repair sequences early can reduce future service disruption and can inform the cost comparison before a design is selected.

7.3 Cost and whole-life decisions

Affordability is a property of the complete service, not a single vehicle or span price. A city needs to know what it would spend to design, build, equip, operate, maintain, renew and eventually remove the railway. It also needs to know what passenger capacity and journey quality those costs buy. A fair comparison holds the service objective and its safety obligations constant while allowing competing designs to meet them differently.

Build a price from physical work

A useful estimate follows quantities and production routes. A beam needs concrete, reinforcement or tendons, forms, labour, curing space, lifting, transport, bearings, erection and rail fixing. A carbody panel needs stock, cutting, forming, joining, finishing, fasteners, inspection and replacement planning. Bought specialist equipment needs a specification, delivery, integration, testing and eventual renewal. A unit price with no scope definition can hide any of these steps.

The platform keeps factory, delivered, installed and whole-life costs separate. It compares low, base and high design-stage scenarios with explicit rates and allowances. Fourteen cost scopes connect different parts of the railway, including rolling stock, guideway, stations, propulsion and power, construction, operations, maintenance and finance. A rate borrowed from another city is a provisional assumption. A supplier quotation can strengthen one account without making all the other accounts certain.

Cost boundary	Examples of questions
Factory	How much stock, labour, machine time, tooling and QA does the part need?
Delivered	What are packaging, transport, duties, handling and supplier terms?
Installed	What civil works, wiring, lifts, testing and site access are needed?
Whole life	How often are parts renewed, what energy and staff are needed, and what remains at disposal?

The vehicle's detailed parts proposal has a base direct parts-and-process subtotal of about $1.064 million across six cars. It excludes final assembly, shared plant, delivery, qualification, warranty and many whole-life accounts. It is not the purchase price of a finished train. Likewise, the Samawah comparison's base installed capital is about $1.14 billion and its whole-life present cost about $2.26 billion under its own planning assumptions. Neither figure is a construction bid or a city budget.

Price service, not just hardware

Fleet size follows route time, frequency, availability and spare policy. A station change can alter energy or access even if it leaves the number of operational stops and trains unchanged. A different car formation can alter platform length, depot space and civil loads. A powered-wheel train might reduce station equipment but increase vehicle mass and maintenance. These cross-effects are why each economic result must retain its physical and service candidate.

Current scenarios use declared production and civil shift patterns and separate operating-hour cases. They are planning assumptions, not an approved roster or construction schedule. Wages, productivity, supplier prices and ground conditions can all change a ranking. The model should show where a change matters most, so the next survey or quotation is chosen for its decision value.

Affordability is not achieved by erasing a mandatory safety item, calling an unknown mass zero or postponing a required test outside the cost boundary. A more honest early estimate may be wider, but it gives the owner a sounder basis for reducing cost through design, production and service choices.

7.4 A corridor cost comparison

Costs become more useful when tied to a service a city could recognize. OpenSourceRail retains a thirty-kilometre elevated, double-track comparison with twelve stations. It assumes a 72 km/h speed limit, one metre per second squared acceleration, a two-minute peak interval and 276 named places per reference train. The calculation uses 52 six-car trains, including its fleet and spare assumptions. This is a design-stage benchmark, not a priced order or a decision to build that particular corridor.

Two prices can describe different assumptions

One generic-rate Class C central estimate is about US$738 million, or US$24.6 million per route-kilometre. It includes trainsets, stations, civil works, a provisional compact launch and power concept, depot, production and tooling, testing and contingency under its declared rates. A separate project-volume supply comparison reprices the same corridor and fleet using wholesale and manufacturer observations and calculated manufacturing alternatives. Its central installed estimate is about US$548.1 million, or US$18.27 million per route-kilometre. Shared premises, freight, quality work, duties and future replacement setup are counted in that comparison.

The difference is not a verified saving available to every city. The two studies test price and supply assumptions, while the physical design and service control remain provisional. Neither has supplier quotations for a complete qualified railway or a surveyed site. A local owner would need to replace transferred prices, ground assumptions, access, land and utilities, construction method, tax and financing terms with its own evidence.

What is held in the benchmark	What a new city must establish
Thirty-kilometre, twelve-station study scope	Actual route, access, land and ground
Reference train, places and two-minute control	Demand, operating plan and protected headway
Provisional drive and civil families	Candidate-specific equipment and structure
Scenario rates and model quantities	Supplier bids, local labour and site quantities

Use a benchmark to ask better questions

If civil work dominates the price, a city may investigate span form, station placement, construction access and utility diversions. If rolling stock or propulsion dominates, it can compare formations and drive families at equal passenger service. If the fleet is large because round-trip time is long, a station or route change might have more value than a small saving on each car. The point of a model is to direct evidence-gathering and design effort toward the choices that matter most.

The Samawah case illustrates a different scale of question. Its midpoint station edit barely changes the modelled capital total yet increases the whole-life present cost through energy. That is not contradictory. Capital, operating cost and passenger usefulness respond to different parts of the design. A lower first cost does not guarantee a better whole-life choice.

Affordability should be measured alongside journey quality, availability, accessibility and local support. Mandatory safety and passenger protection are constraints on every candidate. The benchmark's figures make the discussion concrete; they become investment evidence only when the selected physical candidate, service, site, suppliers and risks have been reviewed together.

7.5 Control and passenger protection

A safe passenger railway needs more than a train that can move under normal conditions. It needs clear authority for every movement, protection from conflicting movements, reliable detection, a response to faults and means to bring people to safety. OpenSourceRail researches open control software alongside the physical railway, but the current software is a research and simulation platform without live movement authority.

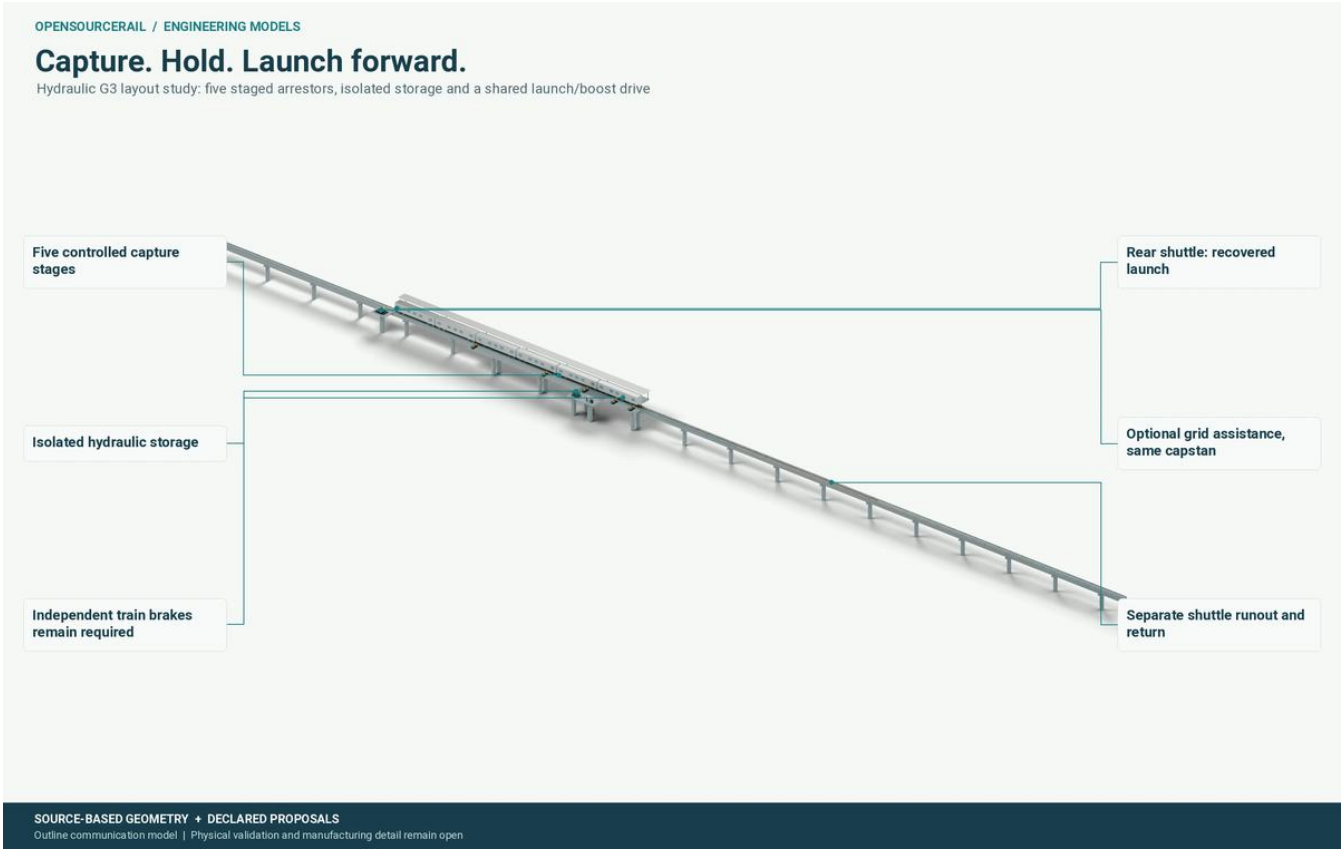
Separate normal service from protection

The service plan describes when a train is intended to depart, stop and arrive. Protection limits where it is permitted to move and how it must stop if the service plan or equipment fails. Station launch equipment may drive a train in the normal journey, but independent onboard brakes must remain able to stop it when station power or communication is unavailable. A protected headway requires validated braking, position, supervision, platform and terminal assumptions, not just dividing an hour by a planned interval.

The reference system must consider train detection, route setting, station dispatch, doors, platform interface, overspeed, emergency stops and recovery. Degraded cases include a missed station handoff, a stopped train, sensor disagreement, loss of power or an occupied section. The response has to fit the actual physical train and route. A correct software test with idealised braking data does not prove the corresponding wet-rail stop.

Open control needs a clear boundary

Rust research software provides a place to define state machines, interfaces and deterministic checks. Simulations can expose contradictory authority, unsafe transitions and unexpected restart behaviour. They can be reviewed independently and rerun as designs change. The resulting evidence is valuable for engineering, but deployment would need qualified sensors, actuators, communication, operating procedures, safety analysis, testing and the approvals applicable to the actual railway.



Station equipment and launch concept

Communication model of a station-side equipment arrangement. The final drive, protection hardware and maintenance access are candidate-specific.

FUXA and the project gateway are explored for equipment observations, alarms, trends and review-linked records. Observation is not the same as commanding a train. A live deployment would need authenticated device bindings, a commissioned network, permissions, fail-safe interfaces and maintenance procedures. The Workbench is a design and review interface; it cannot grant movement authority.

Passenger protection must also shape the physical station and vehicle. A control response is only useful when the doors, platform, evacuation route, communications and rescue method allow people to be cared for. The project therefore keeps control, operations, braking and civil design attached to the same candidate and the same emergency scenarios.

7.6 Business, maintenance and renewal

A railway lasts through many generations of staff and equipment. A component must be received, inspected, installed, maintained, replaced and eventually retired. The information that follows it can make those tasks safer and less expensive. OpenSourceRail connects engineering records with proposed manufacturing, procurement, quality and maintenance workflows so that a physical revision is visible beyond the design office.

One component, one history

Consider a door operator. Its drawing defines its fit; its bill of materials names the parts; a purchasing record identifies the supplier; incoming QA checks the received item; assembly records where it was installed; maintenance records faults, repair and replacement. A technician should be able to see which revision is on the train and which inspection limits apply. If a supplier changes the motor or a body mount moves, affected work instructions and spare parts must change with it.

The current software can prepare candidate-linked drafts and has exercised a bounded synthetic custody example. It can reject a purchased child without a serial, batch, incoming inspection or valid parent consumption record. The complete counted door module has not yet passed a real native ERPNext receipt-to-assembly-to-replacement journey. That is an integration milestone, not a detail to infer from the smaller example.

ERPNext and Frappe HR are relevant to purchasing, stores, work orders, quality, people and training. The project gateway and FUXA can support equipment observations and alarm history. These tools require a real owner, permissions, warehouse and asset masters, authenticated devices and commissioned procedures. A software demonstration does not create legal purchasing authority, accepted equipment or a railway operating organization.

Design for replacement

Maintenance planning begins before a part fails. The door needs isolation points, access, lifting or handling provisions, a removal path, spare stock and a post-replacement check. A service battery needs thermal management, fire isolation and a cassette or other removal route. A guideway bearing needs inspection and an eventual jacking and replacement method. These choices alter mass, civil space, workshop tools and lifecycle cost.

The same principle applies to renewal of software and data. A city owner should retain readable design information, open interfaces, licenses and the ability to move to another supplier. Historical calculation records should remain retrievable so a later engineer can understand why a design was selected or rejected. Current work should be kept compact enough that the evidence remains usable rather than overwhelming the ordinary checkout.

People carry the system

Production and maintenance capability rests on trained people as much as machines. Inspectors need approved criteria; technicians need safe work instructions; planners need to know the assumptions behind costs; operators need fault and rescue procedures. The project can help organize those records, but a real railway would need an accountable employer, accepted competence, independent checks and an operational safety system. The aim of the open platform is to make that work visible and transferable, not to replace it with a database entry.

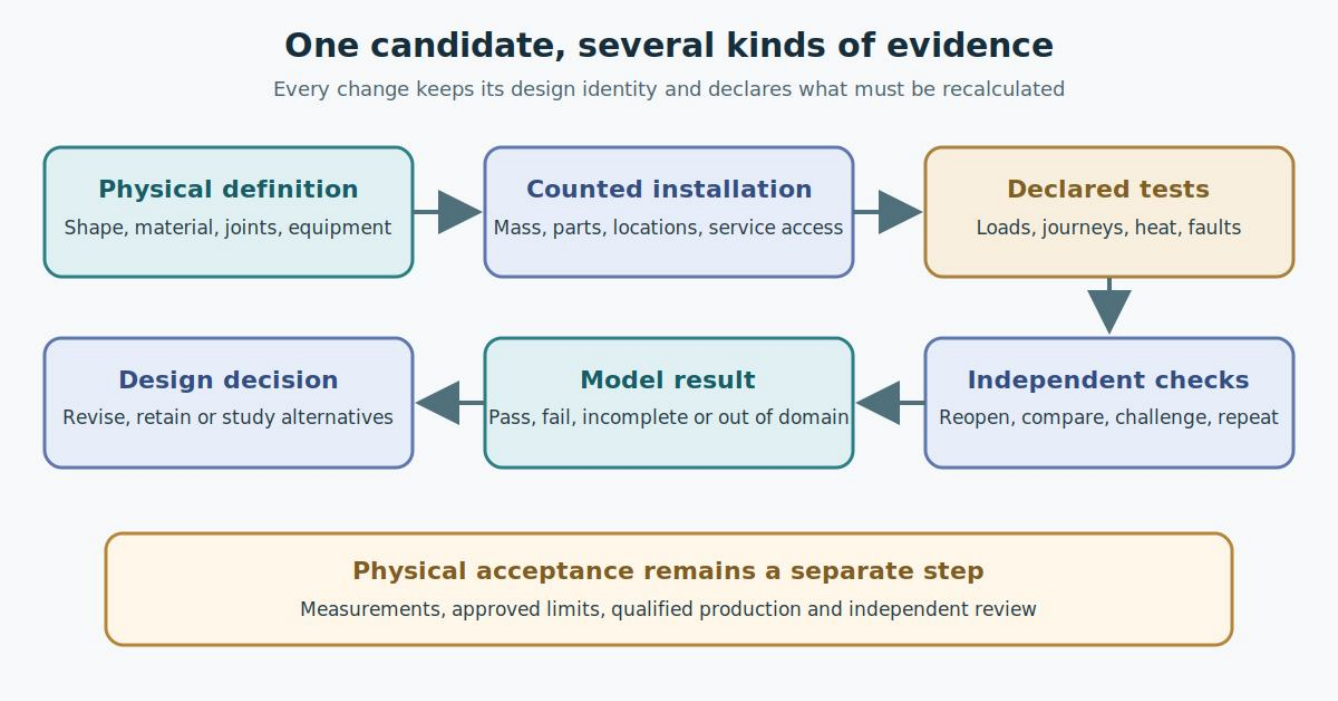
VIII. Digital twins and testing

How models are connected, checked and challenged.

8.1 A digital twin of one candidate

The phrase digital twin can suggest a complete virtual railway. Here it has a more useful, narrower meaning: a connected description of one proposed physical arrangement, with calculations and delivery records that refer back to that arrangement. The value is not that every unanswered question has been solved. It is that the unanswered questions are visible beside the geometry, mass and service claims that depend on them.

OpenSourceRail has generated a research twin for a six-car passive-wheel vehicle, five short wayside accelerator banks, and one directional journey between two berths. Three body families were each tried at two berth spacings, producing six digital research cases. The selected v0.6 development reference remains a separate selection; generating this study does not replace it. The test cell is neither an entire return line nor a city network. It omits the depot, full station buildings, the complete fleet and surveyed foundations.



One candidate through design, calculations and independent checks

The diagram shows how a proposed change moves through the digital work. Physical acceptance remains a separate step requiring measurements and review.

Follow one choice through the system

Suppose a designer changes the body structure. The stock geometry changes, then its counted material and mass may change. The changed mass affects axle loads and the force demanded of propulsion and brakes. The guideway must carry the new train, while the workshop may need different stock and joining operations. Costs should respond to those same changed quantities. A connected twin makes it possible to see that chain and to reject a record that quietly borrows the earlier body's result.

In the research cell, each candidate can produce a geometric scene, located and unlocated inventory, mass account, engineering and manufacturing bills of material, proposed assembly and procurement records, motion demand, fault and load calculations, civil quantities, construction prerequisites, control interfaces and a list of physical tests still required. These are different views of one proposal. A consistent set is useful because a reader can ask whether the same train appears in the beam calculation, the parts list and the cost outline.

The distinction between located and unlocated items is especially important. A body frame may have explicit three-dimensional positions, while equipment such as cooling, wiring or a supplier's internal parts may still have only a quantity or allowance. The twin retains that difference. It must not place an unknown item at the origin or give it zero mass merely because a model needs a number. The account makes retained unknowns visible; it cannot quantify full installed mass until missing equipment is defined.

A journey demand is not an available journey

The cell can calculate the motion needed to travel 1.5 or 2 kilometres and the forces that a proposed accelerator would have to supply. That does not prove that an actual motor can provide those forces at the relevant speed, air gap, temperature and electrical limit. The five modelled banks contain a defined length of primary faces and counted stock, yet finished windings, converters, feeders, cooling and protection remain open. An older motor method with a different active length cannot be substituted as though the hardware matched.

The same boundary applies to the arrival. A candidate may model independent friction receiving and adverse grades, but a physical train still needs validated braking, wheel-slide behaviour, recovery from a stop outside a powered zone and passenger evacuation. A software journey can show a requirements gap. It cannot turn that gap into a safe timetable.

Test the digital package itself

The twin's verification has two layers. First, it checks that the files and calculations agree with the declared candidate: dimensions, materials, positions, volume, mass, bills, costs and downstream results. Second, it tries to reject deliberate inconsistencies, such as a changed part hidden inside an otherwise plausible package or a claim of acceptance without the evidence required for it. This protects the digital record from accidental mixing and easy tampering.

One two-kilometre case also produced native IFC and editable solid geometry, then reopened the exported models independently. That checks whether the geometric information survives a real interchange. The exported shape contains only the stock that was actually located. Reopening a valid solid does not prove that joints, clearances, fatigue, materials, installation or cost are complete.

A useful twin therefore has an honest stopping point. It can be digitally consistent while its physical completion check remains open. The next step is to obtain the missing motor, body, brake, site, supplier and test evidence and attach each result to the exact candidate and evaluation it supports.

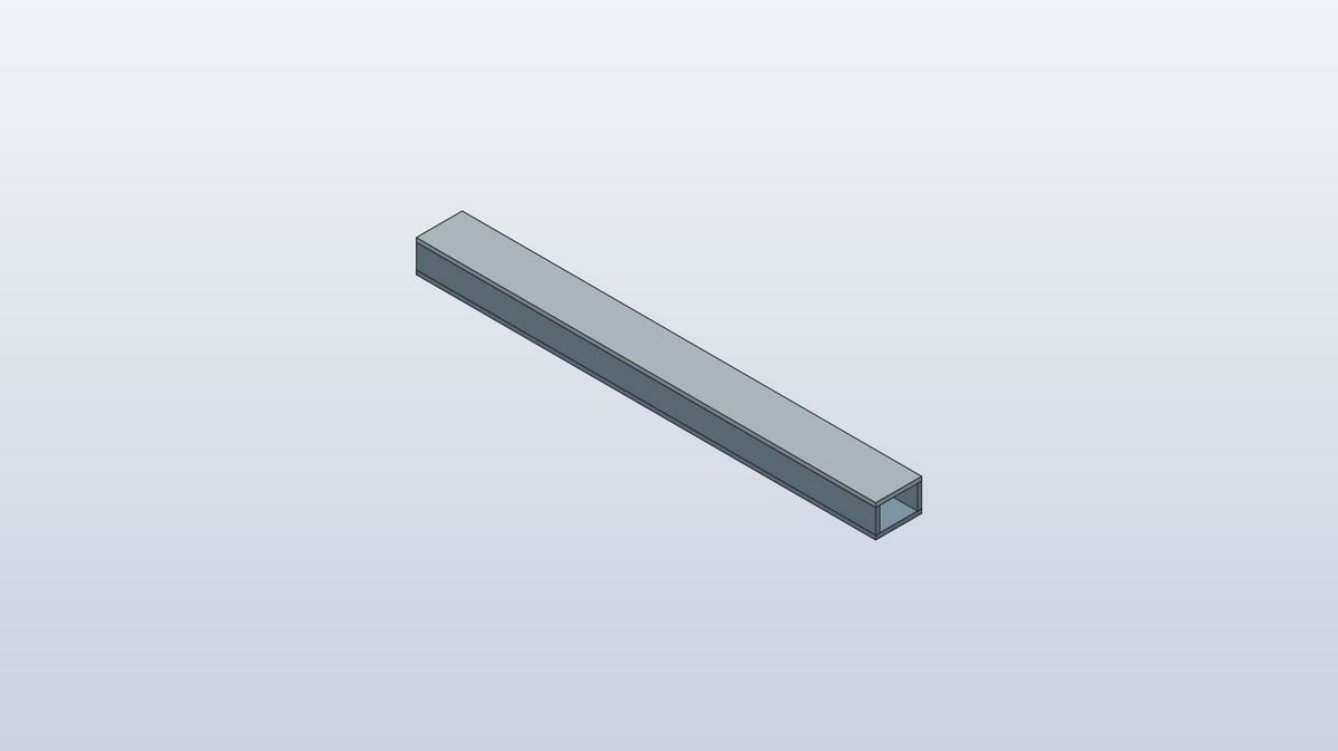
The most useful public view of this work is a readable comparison: what was changed, which conclusions moved, which conclusions were reused, and which questions remain. A twin that can explain those four things helps a city, manufacturer or reviewer take part in the design without treating a large model as a mysterious authority.

8.2 What structural tests show

A structural image is most informative when it shows both the proposed shape and the question asked of that shape. A coloured stress or displacement field can look decisive, but its meaning depends on the material, support, load, mesh and failure modes represented. The OpenSourceRail civil comparisons make that distinction explicit.

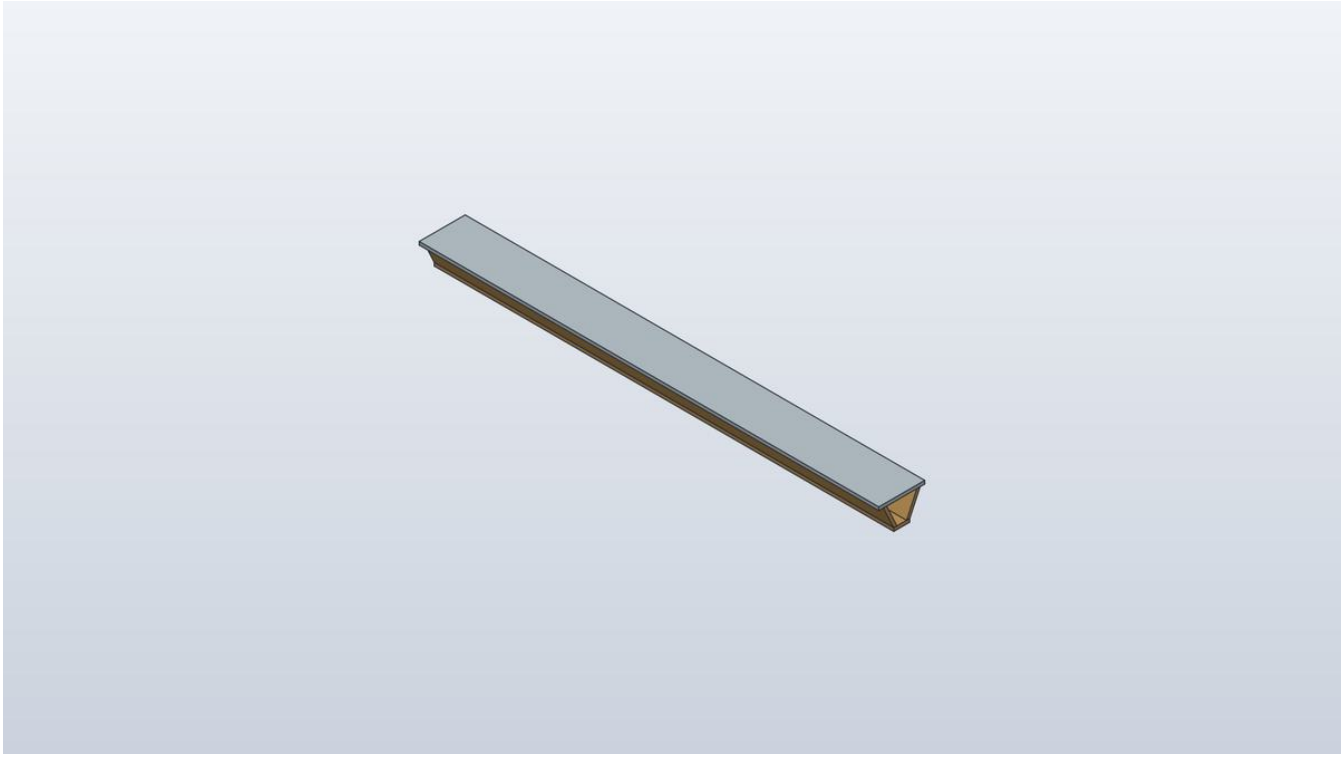
Compare forms under a shared question

The portfolio contains twelve proposed twenty-metre span geometries across six structural families. Their native CAD solids have been reopened and measured in an independent CAD system. The box and folded plate below are two examples. Their forms distribute material differently and would require different mould, reinforcement, lifting and bearing details. A lower stock mass alone cannot rank them as installed structures.



Native box-span geometry

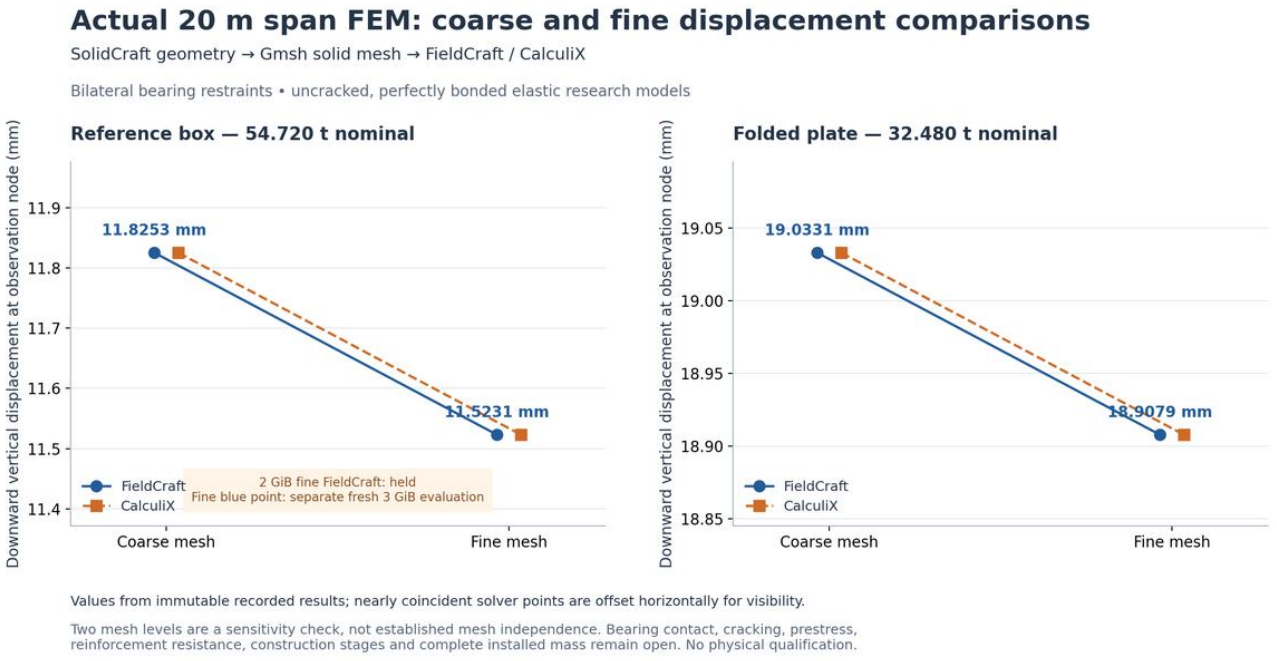
One actual CAD study of a box family. It is a portfolio geometry, separate from the selected beam and not a complete installed span.



Native folded-plate span geometry

A distinct candidate form. Geometry validity does not establish cracking, fatigue, reinforcement or erection capability.

Two matched forms were then solved with both FieldCraft and CalculiX; each form used matched coarse and fine meshes. The displacement changes from the coarse to fine box model were about 2.6%; for the folded plate, about 0.7%. Both met the experiment's declared force-balance, refinement and cross-solver screens. The agreement increases confidence that the two numerical routes represent the chosen model in a comparable way. It does not prove that the chosen model includes the real bearing, rail seat, reinforcement, cracks, prestress and construction stages.



Recorded coarse and fine finite-element comparison

Displacement results for stated idealized models and loads. Two mesh levels are diagnostic; they are not full mesh independence or physical acceptance.

The fine box analysis originally exceeded a smaller memory request. Its successful result belongs to a later, larger resource allocation, not to the failed trial. Keeping both records is useful: the failure estimates the practical cost of refinement and prevents a later report from hiding the experiment's limits. The beam study also uses ideal bilateral support patches, which can carry tensile reactions that a real bearing may not. That is an explicit model limitation, not a discovered property of the construction.

A deformed beam is not a test of every part

The plotted displacement makes the numerical result easier to see, but the physical member may contain joints and local details outside that model. The segmented civil assembly on the preceding pages has its own joints and support questions. A continuous-beam result cannot be used to qualify those segments, their fasteners or their foundations.

The next engineering steps are therefore physical and geometric: model rail and fastener support, contact and bearing compliance, reinforcement and anchorage, moving and accidental loads, fatigue and the temporary lift state. Site ground and utility conditions must enter the chosen corridor. Material certificates, fabrication QA and prototype tests can then be compared with numerical predictions. Until then, the best use of the plots is to choose which uncertainty or candidate deserves attention next, while retaining alternatives that remain credible.

Follow the load path beyond the plot

For a rail beam, the wheel load first reaches the running rail. Fasteners and pads spread that load into a seat or slab, then the main member takes it to bearings, piers and ground. A finite-element picture of only the main member answers only one part of this chain. A rail seat can crack, a fastening can loosen or a bearing can move even if the main beam's ideal elastic stress remains modest. The next model must follow the interface at which the open decision actually lies.

Concrete also changes with time. Prestress can be lost; shrinkage and creep move the structure; temperature changes its length and curvature. Repeated trains can bring fatigue into reinforcement, joints and attachments. Drainage and corrosion can shorten life. These mechanisms need material choices, detailing and inspection plans, not simply a finer version of the first mesh. For an alternative steel or composite form, the decisive issues may instead be welding, local buckling, bonding or fire protection. Different families deserve different models, even when they share the same comparison loads.

Independent numerical agreement is especially useful when the models are genuinely independent and their common assumptions are visible. Two solvers that both hold down an ideal bearing may agree closely while both miss real lift-off. A physical test can then check the most important predictions: deflection under known loads, strain near a critical interface, dynamic response, and how the member recovers after unloading. Measurements should be compared with the predicted range, not chosen afterward to make the model look right.

The purpose of the portfolio is to narrow uncertainty without losing useful alternatives. A heavier form may be easier to manufacture and inspect; a lighter one may need difficult reinforcement or lifting equipment. Structural performance, construction, repair and price need to be considered together for the same corridor and candidate. Only then can a coloured field contribute to a railway decision.

8.3 Journey, control and fault tests

The vehicle, guideway, station drive and control system must work as a journey, including when something goes wrong. A digital model can help identify interactions that are hard to see in a static drawing: a longer dwell changes the next departure; extra mass changes both braking and structure loads; a failed station drive changes rescue and headway. The result is valuable only when the modelled equipment can actually provide the force and protection the calculation assumes.

Start with the demand for motion

For a named corridor, a first calculation can ask how fast a train must accelerate, how much energy it needs, how it coasts over gradients and where it must brake. That produces required force, power and stopping envelopes. It does not provide an available motor map. The next step must connect the proposed device's actual position, speed, current, voltage and temperature limits to those demands. A large idealized force entered into a journey model can make almost any timetable look feasible; the engineering task is to show the source and repeated duty of that force.

The sparse-wayside research twin explicitly stops at this boundary. It counts five short banks and computes the motion they would have to support, but lacks a validated operating-force map for their completed electrical and thermal installation. Its digital package can pass consistency checks while an attempted physical-completion check remains open. This is a productive result: it tells the team which component evidence is needed next instead of inventing an arrival.

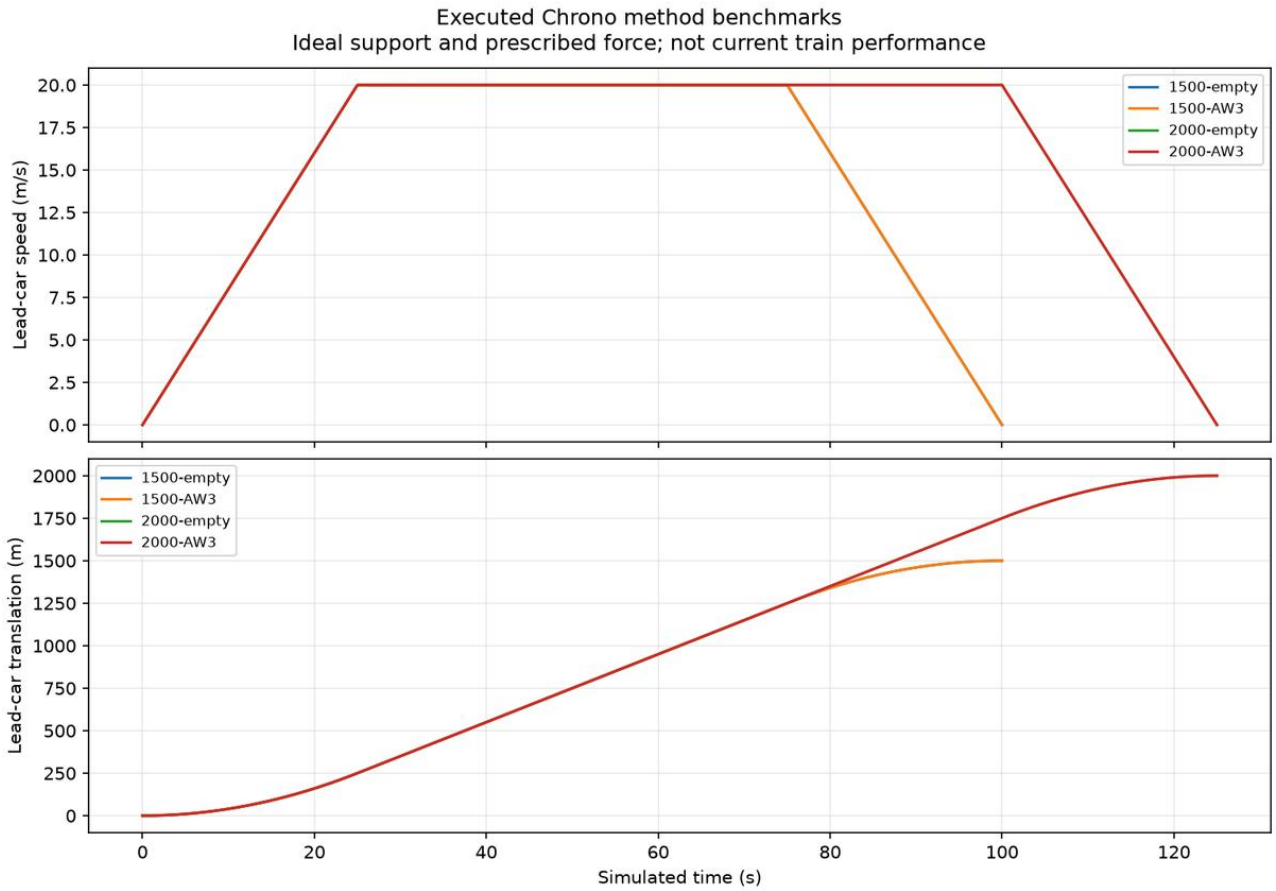
Faults belong beside normal service

Suppose the receiving equipment at a station is unavailable. The train must still stop with independent brakes. If adhesion is poor, stopping distance grows and wheel-slide protection becomes important. If the train stops outside a powered zone, it may need a protected tow or other rescue provision. A fault can affect the following trains as well as the stopped one, so timetable reliability and safe separation cannot be based only on normal dry-rail departures.

The software can test contradictions in the control logic: a train requesting a route that is already occupied, a door state that conflicts with dispatch, or a loss of a necessary signal. Refusal is an important outcome. It shows that the model did not convert missing or inconsistent information into permission to move. A simulator result is still tied to its assumed sensors, actuators and brake behaviour; it is not a commissioned automatic train-control system.

A method plot has a history

The traces below come from a retained earlier magnetic-railway numerical method study. They show how an ideal, prescribed-force benchmark can check a dynamics integration and its time step. They are not a journey of the current wheel-supported train and do not establish a magnetic support or propulsion map. Their inclusion illustrates why plots must be read with the candidate and model named.



Historical ideal-force dynamics benchmark traces

A numerical-method control from a separate historical research configuration. The forces were prescribed; no current train journey or physical performance is proved.

The earlier study completed benchmark trajectories while holding its integrated physical journey for missing equipped mass, flexible-body and force-map evidence. That distinction matters. A close numerical match to a chosen ideal target can establish that a solver follows its equations. It cannot establish that the manufactured train and motor follow the target.

What a complete journey review would join

The selected vehicle would need an installed mass and inertia range, an available propulsion map, actual braking and wheel-rail limits, gradients and curves from the selected alignment, station dwell and access assumptions, thermal endurance, fault response and protected separation. Passenger accessibility, emergency power and evacuation are part of the same review. The output should show normal and adverse cases with their boundaries, rather than a single appealing travel time.

OpenSourceRail's models can already expose many of these dependencies. The next physical work is to measure or justify the missing equipment behaviour, independently review the worst relevant cases and keep the software result linked to that exact candidate. A journey model is a way to ask precise questions; it is not a substitute for the answers.

For a city reader, the useful report is therefore more than a journey time. It should show the station spacing, speed and dwell assumptions, energy and peak demand, the normal and adverse stopping distances, and how a disabled train is cleared. It should say which values come from measured equipment and which remain design requests. Only then can different propulsion ideas be compared at the same passenger service without moving the unresolved risks out of sight.

The comparison should also keep the passenger experience visible. A shorter run time may come from sharper acceleration, but that raises questions about comfort, standing during an exceptional load, wheelchair securement and the repeated electrical duty. A longer dwell may be necessary for accessible boarding and can

alter fleet size. Operations, engineering and accessibility therefore belong in one journey review, with each proposed change carried through the same formation and corridor.

8.4 What twin verification can and cannot prove

A trustworthy digital twin should refuse to make a claim that its own records cannot support. This is different from proving the physical railway safe. The first task is internal consistency: does the beam geometry match the quantity and cost? Does the mass used by braking match the counted vehicle? Does the candidate in a delivery draft match the candidate that was analysed? The second task is physical evidence: are the material, support, motor and brake assumptions true for the item that will be built? The project has made progress on the first task while keeping much of the second open.

Verify the same candidate across views

An engineering model may appear as a geometric scene, CAD solid, bill of materials, factory task, civil quantity, journey model and cost estimate. Each view has a different purpose. Verification checks that they describe the same physical revision, or clearly state when one is a separate research alternative. A body revision that changes mass should reach load, braking and civil calculations. A change to a local price may change the cost estimate without moving any structure. The dependency is part of the evidence.

The sparse-wayside twin generates six research packages and reconstructs their semantic outputs during checking. Deliberately altered geometry, part counts, mass, cost or control claims are rejected rather than trusted because an archive carries a plausible label. A native IFC and solid-model round trip in one case checks that located stock survives export and reimport. The five other cases did not request native geometry, so the one successful round trip is not represented as six executions.

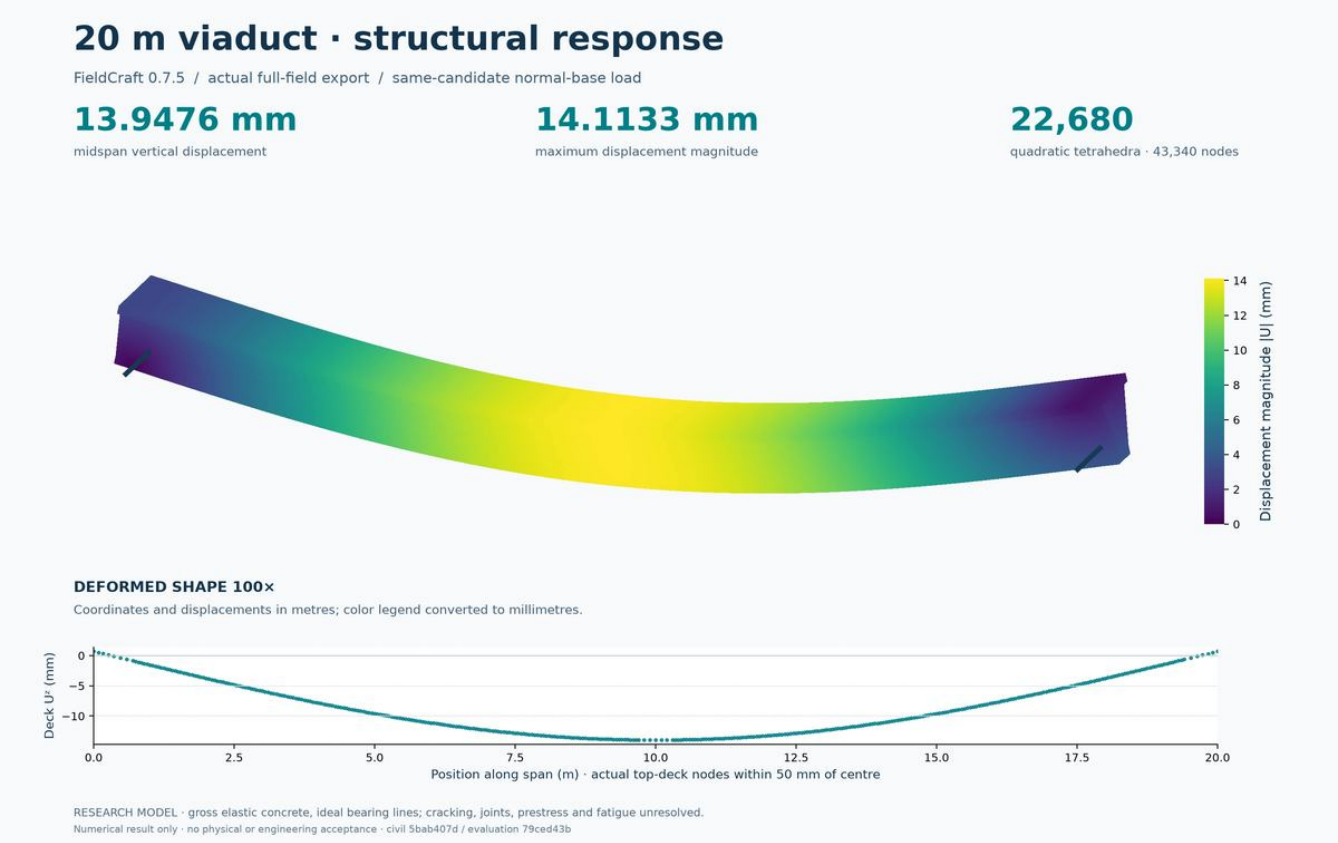
Digital consistency has a limit

Suppose a box beam and two independent solvers produce nearly the same displacement. That may show that the implementations agree under the chosen elastic material, supports and loads. If the real bearing can lift off while the model holds it down, the agreement does not answer the physical question. Suppose a door part appears in every bill and CAD scene. If its guide cannot be removed through the body opening, completeness of the list has not solved maintenance. The next model or test must address the actual interface.

This distinction also protects cost. A complete calculation under assumed labour and material rates may be useful for comparing candidates, but it is not a supplier quotation. An unlocated battery, unknown cooling duty or missing site foundation must not become a zero-cost account. A broad range with explicit gaps is more useful than a precise total that silently drops required work.

Digital question	What it can establish	Physical evidence still needed
Do geometry and inventory agree?	Consistent declared dimensions, positions and counted stock	Fitted interfaces, tolerances, supplier internals and as-built mass
Do two models give similar results?	Agreement within a stated idealization	Real material, load, contact, fatigue and calibration
Does a journey calculation complete?	Behavior under its declared available forces and faults	Measured force, braking, thermal duty and protected operation
Does a business draft connect?	Candidate and document continuity in software	Reviewed transactions, QA and actual installed serial records

The field picture below is a useful example of that boundary. Its deformation is exaggerated so a reader can see the shape of a predicted response. The underlying test is an ideal continuous elastic beam under a defined load. It does not verify the joints, fasteners or foundation of the separately pictured segmented civil proposal. The image becomes trustworthy when its question and its excluded details travel with it.



FieldCraft displacement field for an ideal beam

An ideal continuous-beam field view with exaggerated deformation. It does not qualify the segmented civil assembly shown elsewhere in this book.

Give failure its correct name

A computational job can finish successfully and still fail an engineering screen. A mesh can fail to converge, a candidate can exceed a limit, and a model can be outside its domain. These are different outcomes. Keeping the original inputs and the reason for each outcome lets another team learn from the work without mistaking failure for a missing file or a hidden success.

The twin's completion gate currently remains open for such matters as an available motor map, full installed mass, structural joints, braking, site works and complete prices. That is the honest state of the research. As measurements, supplier information and physical tests arrive, each can be attached to its exact candidate and reviewed independently. A digitally consistent project is a strong foundation for that work; it is not the final permission to manufacture or operate a railway.

Let a reader challenge the claim

The strongest assurance record is one a second person can interrogate. If a cost falls after a design revision, the reader should be able to see whether material quantity changed, a price assumption changed, or a required item disappeared. If a beam receives a passing numerical screen, the reader should be able to see the support and load conditions that made it pass. If a journey is shorter, the reader should find the acceleration and dwell assumptions that changed. This kind of access turns a polished dossier into a reviewable argument.

It also makes honest progress possible. A failed test may eliminate a weak proposal. An unpriced component can point to a supplier task. A nonconverged structural model can show where a better idealization or measurement is needed. None of these outcomes has to be disguised to make the project look complete. The digital twin earns trust by keeping the boundary between proposal, calculation, test and accepted railway clear as the work advances.

IX. Engineering and evidence

What the tools establish and what remains open.

9.1 Engineering tools and evidence

The open engineering platform is meant to keep a railway decision coherent as it passes from city planning to geometry, structural analysis, inventory, cost, manufacture and maintenance. Its value lies in the connections. If a station moves, the dependent service and cost results should update; an unchanged vehicle model should remain available. If a carbody changes, mass, clearance, braking, civil load and factory quantities must be reconsidered.

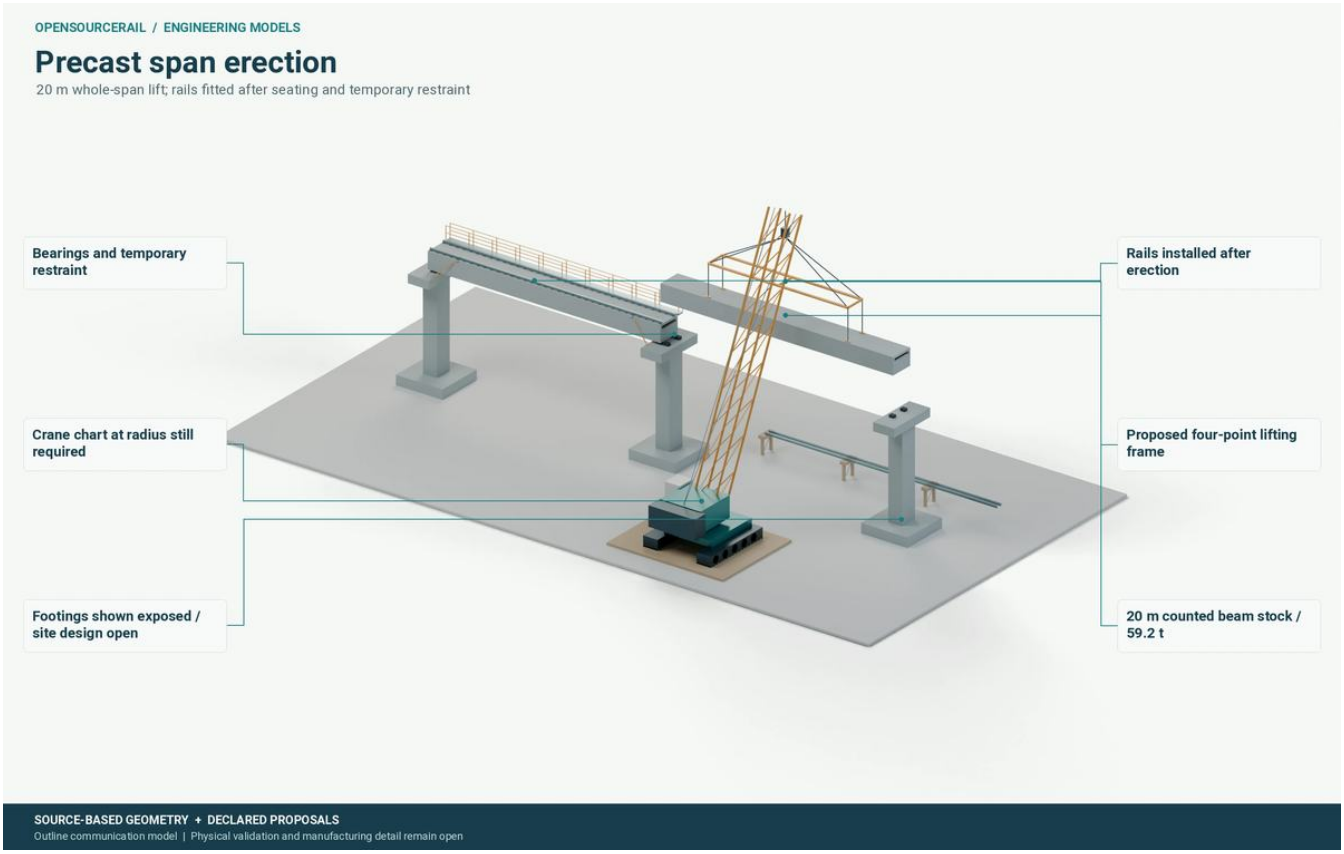
A design has more than one kind of identity

The physical candidate describes the geometry, material, components and interfaces of the proposed railway. A numerical evaluation describes the model, assumptions, loads, tool and conditions used to test it. A selection records which candidate is preferred for a stated purpose and why. Keeping these separate prevents a new calculation from pretending to be a new train, and prevents an old result from being attached to a changed design.

The tools keep both successful and failed evaluations. A body that does not meet an applicable screen can still teach the team about a load path. A nonconverged numerical solve may reveal a modelling problem. An interrupted run should not disappear, because its partial evidence and resource use can guide the next experiment. Reuse is valuable only when the relevant source, candidate, model, load and tool conditions still match.

Models at several levels

Early design uses geometry and analytic checks to discard clear definition failures cheaply. More promising candidates can progress to coarse numerical models, then refined and coupled studies. SolidCraft is being used for candidate geometry and assembly work; FieldCraft for matched mechanics studies. Independent FreeCAD and CalculiX routes provide useful comparisons where their methods apply. RustGIS carries native city edits in the tested domain. A tool's name does not establish that it can represent every joint, material law, electromagnetic transient or railway operating condition.



Engineering model of the proposed construction and vehicle system

A model helps make interfaces visible. The real lift still requires the chosen beam, crane chart, ground, rigging and temporary-works design.

CAD is particularly useful when a part must fit another part or be removed from an installed assembly. FEA can test a declared load path and compare mesh and solver sensitivity. GIS can bring a station edit back into the costed corridor. None of these operations automatically proves manufacture, structural safety or route approval. Each output must state what was represented and what remained outside its model.

A readable result

The Workbench should make the relationship visible to a planner, engineer or reviewer. One project view should show the corridor edit, chosen vehicle, civil option, quantities, costs, drawings and unresolved findings together. The reader should be able to tell whether a result is a proposal, a checked calculation, a verified software workflow or an accepted physical result. A held or stale result is still useful history, but it must be clearly marked and kept out of the supported decision path until renewed.

9.2 Current results and open questions

The v0.6.1 Open Engineering Platform is a published software release. It offers planning and research tools, a Workbench, candidate-linked examples and documented limitations. The Samawah software journey has been installed and demonstrated from the published release. That supports the claim that the software route can be used and reviewed. It does not make a railway physically accepted or a city design ready for construction.

Development since that release has joined the Samawah station edit to service, fleet, fourteen cost scopes and a revised dossier view. Native GIS edits have returned through Samawah and Baghdad calculations. Candidate studies have counted vehicle parts and door components, explored new CAD and mechanics backends, and compared independent numerical results. These are meaningful steps toward a complete engineering workflow. Their findings still belong to the exact candidates and model domains used to produce them.

Area	What exists	What still decides the physical design
Corridor	Comparative route, station, service and cost calculations	Surveyed access, demand, land, ground, utilities and chosen service
Vehicle	Counted components, candidate geometry and analytical controls	Installed fit, complete mass, joints, dynamics, fire and passenger tests
Propulsion and braking	Conditional force, energy and stopping studies	Available installed force, repeated thermal duty and validated braking
Civil works	Span and moving-load models with independent comparisons	Bearings, rail seats, foundations, fatigue and erection at the actual site
Manufacture	Proposed routes, stock, BOM and assembly order	Controlled drawings, qualified processes, supplier evidence and QA limits
Lifecycle	Draft business and maintenance handoffs	Actual candidate-linked transactions, organizations and reviewed records

A numerical result has a boundary

Two solvers can agree closely on a model while its real bearing, fastening or material behaviour remains uncertain. A finite-element mesh can refine toward a stable displacement without proving that the chosen load case is complete. A motor method can reproduce an ideal field while leaving the manufactured primary, current closure and heat transfer absent. The correct response is to keep the useful numerical evidence and name the next measurement or model needed to make a physical decision.

The current installed vehicle still needs resolved door and equipment interfaces, removal paths, mass and inertia. Propulsion needs force-speed and repeated-duty maps linked to the chosen train and corridor. Braking needs independent stopping envelopes. Civil work needs a represented rail/support interface, foundations, moving and accidental loads, and a complete erection method. Cost needs equal-service comparisons, site quantities and stronger supplier evidence. These questions are connected: a heavier fitted vehicle changes the beam and braking problem, while a changed station changes service and energy.

The older integrated reference dossier is retained with a source-binding hold. Its attractive global cards are not treated as current supported results. Historical maglev and other research remain available to study; they do not replace the wheel-supported reference by proximity in a folder or a book. This separation protects both the history and the present design.

9.3 Research choices and independent checks

The wheel-supported railway is a development reference, not the end of exploration. The project keeps earlier maglev work as historical research and allows new candidates to test propulsion distribution, train formation, running gear, speed, station spacing and civil form. The purpose is to find a useful, affordable system under real constraints. An attractive isolated result should lead to a fuller comparison, not silently replace the reference design.

Compare the service that a city receives

A lighter carbody, shorter train or different drive may save capital while changing seats, frequency, power, platform length, repair effort and failure response. The comparison is fair only when each candidate is asked to deliver the same defined transport service and mandatory passenger protection. If a concept offers less capacity, takes longer or needs more vehicles, those differences belong in the result. Local manufacture and repair capability are also part of the objective, because a system that cannot be supported locally may become expensive or unreliable later.

Different questions need different models. A quick analytical calculation can reject an impossible geometry or expose a missing mass account before a large numerical study. A coarse structural model can identify important load paths. Fine models are useful when their mesh, material, supports and loads represent the decision at hand. Physical tests are needed when a model cannot establish a critical interface or material behaviour. Moving up this ladder costs time and storage, so the most valuable question is often what evidence would change the choice between two candidates.

Agreement is useful, with a boundary

Two independent solvers that agree on a span's displacement give confidence that the numerical implementation of that particular model is consistent. They do not prove that the real rail seat, bearing, pier and foundation have the same behaviour. Repeating a mesh at greater resolution can reveal numerical sensitivity. It does not introduce a missing bolt, fatigue case or construction load. A physical measurement, with an approved test method and independently reviewed limit, has a different role again.

The civil work already contains comparisons between newer SolidCraft and FieldCraft methods and independent FreeCAD and CalculiX checks. The new tools can help agents generate geometry and evaluate alternatives, while the older independent route remains valuable where it tests the same physical question through another implementation. A tool should be retired only when its role is either replaced with evidence or clearly preserved for historical reproduction. Convenience alone is a weak reason to lose an independent check.

Keep failures useful

A failed design can teach the project where a load path, interface or model is inadequate. A solver that does not converge, a candidate that exceeds a thermal limit and a mechanism that cannot carry the declared axial load are different findings. Each should retain its inputs, conditions and reason for failure. A rejected pin-jointed mechanism does not reject a bonded shell unless the shell is tested under an appropriate model. An unsupported material law should remain unsupported rather than be replaced quietly by a convenient substitute.

Research alternatives also need ancestry. A change in magnetic-flux hypothesis may be a new evaluation of the same physical geometry. A new bogie or door layout is a physical revision and should propagate through mass, braking, civil loads, manufacturing and cost. A selection records why one candidate is preferred for a stated purpose; it is not a claim that the railway has passed physical acceptance. This discipline lets an open community explore boldly while keeping each conclusion understandable.

X. Building the continuous v0.7 project

A corridor revision linked to design, manufacture and maintenance.

10.1 The v0.7 continuous project

The published v0.6.1 platform lets a reader explore city outlines and review several useful engineering and delivery records. The v0.7 development programme asks for a stronger connection: one change to a city corridor should follow through the service, the selected train, civil work, manufacture, cost and the dossier that explains the result. A reader should be able to tell which parts changed, which remained valid and which are still waiting for engineering evidence.

Imagine a planner moving a station entrance. The location could change the walk that passengers take, a train's dwell time, energy use, the quantities of access structure and the construction allowance. It should not cause an unchanged vehicle door to be analysed again. If a vehicle revision changes the door, on the other hand, its mass, structure, wiring, manufacture and maintenance consequences should travel together. This is the practical meaning of project continuity.

A project has one accountable design

The proposed v0.7 journey keeps a corridor revision, vehicle candidate, civil option, service plan and cost basis attached to each result. The physical vehicle and the question being asked of it are distinct. A door may have the same geometry while receiving a new test under a different load. A research propulsion alternative may require a different bogie, electrical equipment and mass, and therefore becomes a different physical candidate. A selection is yet another decision: why one of the alternatives is being carried forward for a stated purpose.

These distinctions protect a reader from a tempting but false composite. A fleet count from one corridor cannot automatically price another. Structural results for one body cannot validate a modified body. A powered-wheel research car cannot borrow the passive reference's nominal mass or braking result. The Workbench is intended to make such boundaries visible without asking every reader to open a numerical archive.

What already works

The first v0.7 comparison is available in the development Workbench. It recalculates the published Samawah network and a proposed merger of two nearby station markers. The reader can open before-and-after dossiers, inspect all fourteen cost scopes, see a revised delivery draft and download a comparison. The edit keeps the vehicle and unchanged civil quantities attached to their existing evidence. Mixed candidates and stale upstream stages are refused rather than quietly combined.

The city calculation can queue bounded studies and reuse a verified result for an identical request. That speeds up review and reduces needless work, but it is only valid when the relevant inputs and methods are identical. The platform records both fresh calculations and reused work so a warm result is not described as a newly executed one.

What continuity still needs

The corridor comparison does not yet make a complete installed railway. The exact six-car train, its mated doors and equipment, selected drive, brakes, civil support, construction method and candidate-linked production documents still need to be brought into one reviewable project. The current door panel exposes useful counted parts and open fit questions; it is not a released door assembly. The GIS return path has completed automated native-tool round trips for Samawah and Baghdad, but a full supported browser journey through the native editor still needs its own demonstration.

The aim is not to make every box green by assumption. A complete project can be useful while showing that a platform location is unsurveyed, a motor force map is unavailable or an inspection limit has not been approved. It tells the city what decision can be made now and what evidence must come next. That is a more useful meaning of progress than a large collection of disconnected successful calculations.

A useful change report

For a city reader, the most helpful report begins with the parent proposal, the new proposal and a short explanation of the edit. It then separates direct changes from consequences. A moved stop directly changes its location; changed line time or access cost is a calculated consequence. It also explains why apparently related values, such as the fleet or a vehicle part count, stayed fixed. This makes a small edit an opportunity to understand the railway rather than just compare two large totals.

For an engineer, the same report points to the exact load, mass, geometry and material assumptions behind each result. For a workshop, it indicates whether a drawing or part number has changed. For an owner, it shows which prices are local observations and which are transferred estimates. These are different views of the same candidate, not separate versions of the truth. If one view depends on information that has not been supplied, the report should say so and leave that conclusion open.

10.2 Samawah before and after

The Samawah station example is small enough for a reader to follow and large enough to reveal the system's dependencies. Two source markers on one line sit about eighty metres apart. The planning case assumes a 120-metre platform, so treating them as separate full stations would need an explanation. The proposed edit replaces the two markers with one at their midpoint. It does not move the route itself.

Read the starting case

In the development Workbench, the reader opens Samawah in the city atlas and chooses the station-merge comparison. Before looking at the cost delta, it helps to note the scope: the five-line network, the selected vehicle reference, the explicit service assumptions and the fact that the source markers had already been grouped into one operational complex. The network is a planning model. Its stations, ground and walking access have not been fully surveyed and qualified.

The unedited case contains 26 raw markers and 25 operational station complexes. The edit leaves 25 raw markers and still 25 operational complexes. The modelled route remains about 57.87 kilometres and the fleet remains 81 trainsets. Those equalities are meaningful. The edit cleans up a problematic pair of markers but does not, under the current rules, remove a built operational station or one train from the fleet.

Follow the changed accounts

The base installed capital estimate falls by about ten thousand dollars. The base whole-life present cost rises by about two hundred and thirty thousand dollars. Line 1's one-way travel time grows by roughly one and a half seconds; the other four lines do not change. The energy account is the main contributor to the higher whole-life result. An access-repair allowance is replaced by a shorter proposed access length with its own risk provision. The report shows all fourteen cost scopes, including those with no change, so the reader can separate the consequences of this edit from the size of the entire city programme.

Question	What the comparison says	What it cannot decide
Did a station disappear?	One raw marker did; operational complexes stayed at 25	Whether the real station should be at either marker or the midpoint
Did the fleet shrink?	The model still needs 81 trainsets	The final timetable, spare policy or protected headway
Did construction become cheaper?	The provisional capital model falls slightly	Site ground, utilities, access and supplier prices
Is the revised journey better?	One line is a little slower in the model	Actual walking routes and passenger preference

Do not lose the difficult findings

The comparison retains a line-1 coast-reach question and the absence of accepted propulsion and braking force evidence. It continues to display transferred rates, unknown local works, platform fit, catchment and walking-access questions. The existing civil quantity identity and the vehicle/manufacturing reference are reused because this marker edit did not change their defined inputs. That is an engineering statement about the model's scope, not a guarantee that a surveyed site would leave them unchanged.

A retained compact-drive study considered a similar midpoint idea using a different body candidate. Its results remain a research comparison. They are not inserted into this passive-wheel case merely because the city name and station pair match. If a planner wants to compare the drive alternative, they need its own complete vehicle, service, civil and economic result.

The next decision belongs on the ground. A midpoint might shorten the distance between original markers but put an entrance on the wrong side of a road, a utility corridor or an inaccessible parcel. A useful follow-up would map real pedestrian approaches, potential platform edges, land and ground constraints, then run the chosen location through the same comparison. The software makes the trade-off inspectable; it does not choose a station for the people of Samawah by arithmetic alone.

How to use the case in a public discussion

The comparison can support a focused conversation with residents and planners. People can mark the walking routes they actually use, the crossings they avoid, the destinations reached by each original marker and the time of day when access matters most. The city can add land, utility and flood information. A station designer can test platform orientation, lift locations and emergency exits. Each new input can become an explicit proposal for another study rather than an informal annotation left outside the cost calculation.

The result should be presented as a range and a set of trade-offs. The model's ten-thousand-dollar capital difference is small beside the roughly billion-dollar provisional programme total and could be overtaken by one site-specific item. The whole-life change is similarly conditional on energy and service assumptions. The important outcome is the traceable direction of those accounts, not a claim of precision to the last dollar. A city decision may favour accessibility or reduced disruption even when one modelled cost account moves the other way.

10.3 Native city editing with RustGIS

A route or station is easier to understand when it can be edited in a proper map environment. The current development path connects the city atlas to RustGIS, an editable native project, and back to the service and cost study. The intended experience is direct: choose a city, download an editable project with its source tables, make a declared change, return the saved project and review the revised map and results. The original published city stays available for comparison.

What travels with the map

The export includes routes, stations, platforms, original access markers, interchanges, civil segments, access connectors and unlocated requirements. It can also carry available roads, buildings, land use, terrain, water, boundaries and place context, with their source attribution. Tables give the reader service, energy, cost, workforce and delivery context beside the geometry. This makes a project useful offline: the editor can see what a line crosses and which data are genuine source context rather than a new engineering conclusion.

Some layers are proposals that a planner may edit. Others are calculated consequences or original sources and must remain read-only. Current admitted planning edits include route shape, a line's headway or speed cap, station positions and dwell. A derived civil quantity or price is regenerated by OpenSourceRail after the return; editing its displayed table is not a way to approve or import a cost. A change to map colours or camera position does not create a new railway candidate.

The return path

After saving the edited native project, the Workbench presents an import action that asks for an author and a reason. The importer compares the route and station data against the trusted export. A valid change creates a planning revision, then passes through the existing bounded city study. The refreshed map, service, fleet, civil-quantity, price and delivery links belong to that revision. A reader can keep the parent and child side by side instead of replacing the source city silently.

The comparison is intentionally narrow. A new station must belong to a known line and have an explicit dwell. A route or stop with ambiguous geometry needs correction rather than a silent snap to a convenient location. Original source identities are protected. Derived layers remain outputs. A return is refused if it cannot be matched to the exported city and its applicable model conditions.

What has actually been demonstrated

In v0.7 development, Samawah and Baghdad have each completed two consecutive native station edits followed by import, recalculation, refreshed export and reopening. Those exercises used native project files in automated runs, without an observed desktop editing session. They show that the digital round trip can retain source context and produce a new calculation without redrawing unrelated lines. They are not evidence of a complete desktop or browser user-experience test. The published v0.6.1 release did not include a native GIS round trip; its city-atlas export was held by a mismatch between published GIS and outline sources.

Nor does a successful digital edit establish a real alignment. An unsurveyed stop can pass interchange checks while being inaccessible across a road. The map does not reveal all underground utilities, ground behaviour, land rights or construction space. A planning revision is a proposal until the city and engineers provide those inputs. RustGIS is useful because it allows one proposal to be inspected, revised and repriced with its context intact, not because the software can infer missing field evidence.

When to return to the planning team

The map is most helpful when disagreements are visible before a design is frozen. A stop moved to improve access could cross a property boundary, require a longer elevated approach or worsen interchange with a bus route. An editor can prepare several revisions and compare their passenger and cost effects, but the decision should be made with the city's survey and transport evidence. The revised project should preserve the reason for every change so a later reviewer can understand why one alignment was selected.

It is also useful to distinguish a map correction from a design change. Fixing a mistaken source label should not automatically imply a different train or civil structure. Altering the line's actual shape may change length, curves, gradients and foundations, which can invalidate several downstream studies. A trustworthy workflow reports which dependants were recalculated and which were carried forward unchanged. That makes native editing part of the railway's design history instead of an isolated drawing.

10.4 Door operator as a complete module

A passenger door is a fitting test of whether a railway design is truly connected. It must suit the body opening and the station platform, move through an unobstructed path, protect passengers, receive power and signals, survive repeated use, and be removable by a maintainer. The v0.7 programme chose one door operator as a representative module because it joins these mechanical, electrical, production and service questions in a form that can be reviewed end to end.

Begin with the counted proposal

One opening has 74 defined part types and 694 individual pieces in the current proposal. Among them, 97 pieces are proposed for local fabrication, 568 for standard purchase and 29 for purchase subject to qualification. Across the six-car train there are twelve openings. The inventory includes the header, guide, carriage, belt, pulleys, drive, leaves, sensors, seals, wiring and their connections. It is much more informative than a simple door animation, but counting is only the first step of installation.

The stock-and-allowance subtotal for one opening is about fifty kilograms. That number is not an installed mass. Mounts, actual bought equipment, hardware choices, protective enclosures and the way parts are attached can change it. The body structure around the opening must carry the operator and passenger loads. Its motion must remain clear under body deflection, manufacturing tolerance and wear.

Draw the mating relationships

A buildable drawing set needs controlled datums: how the header attaches to the body, where the guide lies, how the carriage runs, how the belt and pulley align, where the motor and reducer sit, and how the leaf meets its stop and seals. Sensors need positions that work in the real motion envelope. Inspection limits must distinguish an acceptable assembled door from one that merely looks plausible in a model.

The current installation study reveals a specific problem. Eleven operator types still occupy a generic centre region of the header rather than resolved installed positions. A two-metre removal envelope does not represent the full 2.9-metre guide stock. These are useful findings: they tell the team what must be redesigned or measured before anyone describes the operator as fitted. Copying the generic coordinates into a drawing would hide, not solve, the conflict.

Make, inspect, replace

A local workshop needs material and process definitions, cutting and machining instructions, assembly order, tooling, tolerances and QA checks. Purchased components need their own supplier and incoming-inspection records. An assembled door needs a serial identity so its exact children, tests and later repairs can be traced. Replacement requires a safe isolation procedure, a path for removing the largest component and a way to verify the fitted replacement against the vehicle revision.

The project already has a source-bound door panel and a typed software route for counted assemblies. These expose part counts, proposed groups, missing inspection definitions and custody requirements. A native ERPNext journey has exercised a simpler synthetic workpiece, and software contracts cover some made and bought child records. The full 694-piece door has not yet passed that actual native lifecycle. Nor has the physical operator completed a mated installation, qualification test or passenger service approval.

The module will be complete for the v0.7 software journey when a user can open the corridor's exact vehicle, examine the controlled operator, follow its bill of materials and inspection record, and see how it is removed and replaced. Physical qualification requires the next, independent step: manufactured parts, measurements, endurance and safety testing under approved limits.

A passenger-facing design review

The operator can also be reviewed from the platform and cabin. Leaf speed, closing force, obstruction detection, visible and audible warnings, emergency release, seal condition and the platform gap affect how safe and confident people feel when boarding. A wheelchair user needs a predictable clear width and enough time to pass; a person with luggage or a child may meet the closing path differently. These behaviours need tests with the actual door and control arrangement, not only a static section drawing.

Maintainers have their own passenger-facing problem: a faulty door can remove a car from service or delay a whole line. Fast diagnosis and a replaceable operator could improve availability, but only if the unit can be isolated and extracted in the real vehicle. The unresolved full guide removal is therefore an operating and cost issue as well as a CAD issue. This is why v0.7 follows one module through the complete system before scaling the method to every component.

10.5 New CAD and mechanics tools

Three-dimensional models make a railway easier to inspect. A reader can see how a door guide sits above an opening, how a bogie carries a body, or how a precast beam reaches its bearing. The newer SolidCraft and FieldCraft tools are being introduced to help generate geometry and run bounded structural studies more directly from a candidate. Their value depends on whether the model represents the physical question and whether another method can challenge the answer.

Geometry before a large solve

SolidCraft has generated a representative door-header stock blank and reopened its exported solid. An independent FreeCAD import reproduced the geometry. This demonstrates a useful interchange: a part definition can leave one tool and be examined in another. It does not turn the blank into a complete door operator. The carriage, belt, motor, sensors, body mounts, tolerances and removal path still have to be mated.

One export format does not preserve every source property, such as material or design provenance. That information must therefore travel with the geometry in a companion record. A viewer should know whether it is seeing a stock blank, an assembled proposal or a physically measured part. The same visible shape can have quite different evidence maturity depending on that context.

What a structural comparison establishes

FieldCraft has been compared with CalculiX on shared meshes for a bounded normal-load civil model. The methods agreed very closely on a roughly ten-millimetre displacement in that model. A mesh refinement comparison also met its declared diagnostic threshold. These are useful checks of the computation, but they do not establish that the real rail seat, bearing, reinforcement, pier and foundation will behave in the modelled way. The fine result still needs convergence and the full physical interfaces to be represented.

The study derives the reference formation's 24 axle positions and keeps normal passenger and crowd structural cases distinct. Moving loads can change the critical position of a beam response. Braking, thermal movement, fatigue and construction loads can change the critical type of response. A model that is excellent for one question may be unsuitable for another; this is why the Workbench should display its load and geometry domain beside its result.

Keep an independent route

The tool migration is selective. SolidCraft is useful for light candidate geometry and FieldCraft for a demonstrated mechanics domain. FreeCAD and CalculiX remain valuable independent checks where they answer the same question with a different implementation. Meshing and specialist tools remain where the current workflow still needs them. Historical evidence is preserved; older tools are retired from default use only when their current role has a supported replacement or a deliberate verification route remains.

For the next candidate, the better sequence is to resolve cheap geometric and load-path questions before running finer models. The door's mating datums, guide removal and installed mass are upstream of a useful detailed door solve. The civil span's rail support, bearing, foundation and lift configuration are upstream of a construction decision. New tools make these studies faster and more accessible when they are asked the right question. They cannot recover a missing physical interface from numerical precision alone.

Choosing a model for a decision

Consider a city asking whether two civil forms can be made in a local precast yard. A coarse geometry and material take-off may be enough to compare handling sizes, moulds and likely transport. A fine stress model cannot answer whether the road can carry a beam to the site. When the question becomes bearing stress or fatigue, the connection and support details must be represented and the numerical model can become more detailed. The amount of computation should follow the question, not the visual sophistication of the model.

The same sequence applies to the door. A stock blank establishes a fabrication envelope and tests whether geometry can be exchanged between tools. A fitted operator study needs its actual joints and moving parts. A qualification test needs manufactured samples and approved limits. Each stage produces evidence that is useful within its boundary. Showing those stages explicitly lets a non-specialist appreciate progress without mistaking an impressive rendering for a finished component.

10.6 Manufacture and maintenance custody

An engineering model becomes useful to a workshop when its parts can be made or received, inspected, assembled and later identified in the field. The v0.7 lifecycle work asks the platform to follow one exact module from its design candidate through those ordinary business steps. The door operator is the test case, because it includes local fabrication, qualified purchases, electrical work and replacement during service.

The bill of materials is a beginning

A counted door proposal names parts and quantities, but production needs more. A maker needs drawings, materials, allowable variation, routing, tooling and inspection points. A buyer needs an approved specification, supplier identity and a way to check what arrives. An assembler needs a sequence and a record of the parts that went into each serialized unit. Without these details, a seemingly complete bill of materials can still allow two workshops to build incompatible doors.

The current delivery tools can prepare reviewable tasks and draft groups from a candidate. They can preserve parent assembly and child relationships without treating an assembly as a featureless single part. They do not authorize a purchase or shop order merely because a draft exists. Cost estimates for assembly operations remain hypotheses until real process routes and measured or justified times are available.

Trace bought and made children

The software custody design distinguishes a made child from a purchased one. A made child should have its own reviewed work record, source parts and accepted quality inspection. A purchased child should be tied to a received item, its serial and batch where applicable, and an incoming inspection. When children are consumed into a parent, the records should account for the exact units, not a convenient count of “one kit.” A replacement parent should point to the specific removed serial and the repair event that made the exchange legitimate.

This is especially important for the door. Of its 694 pieces per opening, 597 are proposed purchases across the standard and qualification routes. Those items need a practical receipt and traceability policy, while the locally fabricated pieces need process and inspection evidence. The current software can reject some mismatched or missing child records and has synthetic examples of made and bought custody. Those tests are useful for the digital contract. They are not an actual receipt and installation of the complete door in ERPNext.

A repair should teach the next design

Once a module is in service, a condition report, alarm, inspection and replacement should remain linked to its candidate and serial. A reader could then ask whether a recurrent fault belongs to one supplier batch, one assembly revision, a station environment or a maintenance method. The answer could lead to a revised component, changed inspection interval or different supplier. The platform's FUXA and gateway integrations can provide observations and maintenance handoffs, but a live device and business deployment need commissioning, roles and review.

The planned v0.7 demonstration should show a real candidate-linked door module moving through draft, receipt or manufacture, QA, assembly, installation, condition event, replacement and business handback. Each step should refuse a missing document or conflicting revision. The existing native fixture exercises a simpler discrete workpiece, not this full chain. Physical installation and operating permission remain separate even after the digital record is complete.

Why the record matters years later

Railway components often outlive the software version that first described them. A replacement door may have a revised drive while sharing a body opening with older cars. A serial record can tell a maintainer which revision is fitted, which inspection limits apply and which spare parts will mate. It can also show whether a supplier change affected only a bearing or required a new assembly test. Without that history, “like for like” can become an unsafe assumption.

Local repair capacity grows when these records remain accessible to the owner and its workshops. Staff can learn from failure rates, rework, actual labour hours and incoming defects rather than relying on a catalogue estimate forever. That feedback belongs in the next design revision and cost model. The goal is an accountable lifecycle that a city can operate and improve, while keeping purchasing, engineering and safety approvals in the hands of the responsible people.

10.7 Efficient repeated studies

An open design platform should make revision inexpensive enough that engineers and cities can explore alternatives. If changing one station requires rebuilding every train, remeshing every structure and republishing every city, the workflow becomes slow and wastes storage. v0.7 therefore works on a simple principle: recalculate what the changed input actually affects, retain the rest with its verified identity, and refuse reuse when the physical or numerical question has changed.

Follow dependencies, not folders

In the Samawah midpoint example, the route geometry, selected vehicle and most civil quantities remain unchanged. The station and access inputs change, so the dependent service, energy, cost and dossier results must be re-evaluated. A later change in a labour rate could alter factory or construction cost without changing the map. A new bogie would have a much wider effect: mass, dynamics, braking, beam loads, manufacturing and whole-life cost would all need review.

The Workbench's project envelope now rejects mixed candidates and stale upstream stages. It can show that a result is genuinely reused when the relevant request is unchanged. This saves more time than accelerating an unnecessary calculation. Reuse is not a shortcut around evidence. It requires the applicable inputs, model domain and tool version to agree.

Keep work within the computer's resources

City studies use a bounded queue with at most two analytic workers by default and shared reservations for processor, memory and disk use. A local comparison showed a pair of fresh studies completed sooner in parallel than in a serial run, with matching numerical outputs. That is a measured improvement for that case, not a promise that more workers always help. Parallel jobs can use more total processor time and contend for memory or storage. Heavy native analyses therefore need their own admission budgets.

The project has also kept successful compressed results under a budget and uses an explicit disk reserve before starting work. Failed or interrupted studies retain useful evidence rather than disappearing into an automatic cleanup. This distinction matters to researchers: a failed solve can show an invalid model, a resource limit or a genuinely poor candidate. Such records should not be silently recast as a green result or deleted as mere waste.

Use Rust where it wins on the whole task

Rust is already part of the control and city tooling, and a bounded delivery scheduling kernel has a verified Rust implementation. In a same-input comparison, the complete delivery compilation became modestly faster with that optional backend, while the physical and policy outputs matched. Python remains the default path. A separate experiment in canonical encoding found only a small overall benefit, so it was not promoted broadly. These findings support targeted migration rather than a wholesale rewrite of already working engineering models.

A reader benefits when a warm city study responds quickly and when a changed input receives a distinct, trustworthy answer. They do not need to know which language ran a particular stage. The most useful performance report therefore includes waiting time, resource use, verified reuse and the scope of the result. For v0.7, speed is a means of making a more complete and reviewable project, not a substitute for missing engineering.

What should be measured next

The most useful performance measurements follow a representative task from a user's click to a sealed result. They include the time spent waiting in the queue, the calculation itself, verification, writing and opening the report. A native structural run needs memory and temporary disk peaks in addition to wall time. A city study needs to distinguish new service work from reused geometry or unchanged price inputs. These measurements reveal where an optimisation would actually help the reader.

One bottleneck can move after another is solved. Faster cost arithmetic will not help much if the worker spends its time loading a large archive. Parallelism may lower wait time for two small jobs but harm a large finite-element solve on the same machine. A compact summary can make the Workbench responsive while full numerical evidence remains available on demand. The platform should report these trade-offs alongside numerical agreement, so speed never hides a changed answer.

10.8 The Workbench project journey

The intended v0.7 Workbench journey begins with a place, not a solver. A user chooses Samawah, sees its proposed lines and stations, opens the close-marker question, and compares a before and after case. From there, the project should lead naturally to the train, civil works, costs and delivery records that would be affected by the decision. The interface has to make both the answers and their missing inputs understandable.

What a reader can do now

The development Workbench offers the Samawah station-merge comparison. It calculates the two cases under the same explicit service and cost policies, shows the raw and operational station counts, fleet and whole-life differences, and makes the two dossiers and revised delivery draft available. A source-bound door panel gives the reader another view into the physical project: part definitions, proposed assembly groups and open inspection and removal questions. These are connected software views, not a complete physical build package.

An identical city request can be retrieved from a verified result without rerunning the same calculation. If the user changes a substantive input, the result must have its own evaluation and updated dependants. The Workbench should show an unchanged result explicitly. A zero difference in fleet, for example, tells the reader that the chosen station edit did not move the model across a fleet threshold. It should not vanish from the report simply because it is not dramatic.

Guide the reader through decisions

A good project view starts with a question: What changed, why, and for whom? It then gives a plain-language result before the detailed tables. In Samawah, the raw marker count falls by one, but the operating stop count and trainsets stay the same. The model's capital total falls a little while whole-life cost rises. A reader can open the cost scopes to see the energy and access contributions, then return to the city map to consider whether the proposed station position helps people.

The next layer explains evidence. The route is a planning outline; the walking path is not surveyed; the propulsion and braking evidence does not yet support an actual operating timetable. The Workbench should present those limits beside the figures that depend on them. A result that cannot be supported should be held visibly, with the reason and the next evidence needed, rather than replaced by an attractive global number from another revision.

Move from the corridor to an asset

The v0.7 target is to open the exact vehicle and choose its representative door operator from the same project. The reader should see its fitted geometry, material and make-or-buy route, controlled inspection criteria, production and removal sequence, and maintenance record. A changed door that alters installed mass or clearance should trigger the appropriate vehicle and service reviews. A station location change should not regenerate an unchanged door. Both kinds of edit should produce a new comprehensible dossier.

This complete journey remains development work. The existing Samawah browser comparison and door evidence panel are demonstrated software steps. A native GIS round trip has been exercised through automated native-tool editing, and the full door lifecycle and installed railway integration remain open. The Workbench will be most valuable when a new reader can follow these boundaries without needing to interpret a collection of specialist files.

A useful set of screens

The city overview can present the map and passenger question first. A comparison screen can then place the parent and edited station side by side, with a concise account of why the change was made. A service screen can show offered trains, journey time, fleet and energy. The cost screen can group factory, installed and whole-life figures without hiding the assumptions underneath. The engineering screen can show the vehicle, door and civil option alongside their outstanding fit and test findings.

The last screen should be a readable dossier: what was selected, what changed, what evidence was used and what remains unresolved. Different readers may stop at different depths, but the deeper record should always be available. A planner should not need to interpret a mesh file to see that a bearing is unverified. An engineer should be able to find that bearing question and its exact model when ready to work on it. This layered presentation is the Workbench's central promise.

10.9 Generalising to Baghdad

Samawah is a good first case because its close station markers make a specific, understandable decision. A platform that works only for this one edit, however, is not yet a city planning method. Baghdad provides a second setting with a larger published network and different route and right-of-way questions. The v0.7 programme uses it to test whether the same project continuity can survive a different city, not to declare Baghdad's outline ready for construction.

Start from the right maturity level

The published v0.6.1 release provides a read-only Baghdad atlas comparison. Its retained study covers fourteen lines, route and service alternatives, and provisional costs. The older release did not offer a supported native GIS export for that atlas, because the GIS and outline publications did not agree at that source stage. A reader should not infer that a download button in later development was part of that published feature set.

In current v0.7 development, Baghdad has completed an automated native GIS return path with two sequential station edits, repricing and a refreshed project. The result demonstrates digital exchange, preservation of unrelated line inputs and recalculation under a bounded city workflow. It does not mean that a Baghdad editor has completed a fully observed desktop-to-Workbench user journey or that any route has been surveyed.

What a second city can reveal

The Baghdad case should test conventions that seemed harmless in Samawah: how a station is tied to its line, how chainage and route shape are interpreted, how different service policies affect fleet, and how land or utility allowances are presented. A local price from one city may not be appropriate in another. A right-of-way estimate can depend on legal and physical conditions absent from a general map. The second city should therefore show which assumptions are shared, which are transferred for planning only and which require new local evidence.

The same candidate discipline applies. The vehicle and propulsion architecture must be named for each Baghdad comparison. If a different formation or drive is proposed, its mass, service, civil and cost effects need their own evaluation. A Baghdad fleet count from a retained atlas cannot be attached to a Samawah construction estimate merely because both use six-car trains.

The reader's comparison

A useful Baghdad journey would let a planner choose one corridor and make a controlled station or alignment edit. The resulting report would show the parent and revised maps, passenger access questions, line service, fleet, civil quantities, energy and all fourteen cost scopes. Unchanged lines would remain visible as unchanged. The project view would explain any refused edit, ambiguous station placement or missing site cost. It would also show that the cost and delivery drafts belong to the edited Baghdad case rather than to a generic city default.

The purpose of the second city is confidence in method. It can expose assumptions that a first demonstration overlooked, while retaining the important boundary between a functioning planning workflow and a buildable railway. A surveyed alignment, demand study, local rates, ground and utilities, station access and candidate-specific operation will still be needed before a Baghdad investment or engineering choice.

Compare without transferring false certainty

Suppose the Baghdad edit changes a station location but leaves the line length almost constant. The fleet could remain unchanged while walking access, land and station costs move substantially. Another edit could alter a line's geometry and trigger civil quantities and travel-time changes even with the same stations. The reports should make these different dependency patterns easy to see. A shared vehicle design may be reusable, while its city-specific power, structure and service calculations may not be.

The second city should also help improve the questions asked of the first. If Baghdad reveals that a transferred utility rate is especially uncertain, Samawah's estimate may deserve the same scrutiny. If the native editor exposes an ambiguity at an interchange, the city model should record it rather than forcing an arbitrary resolution.

Reuse and generalisation are valuable when they uncover such differences, not when they make two unlike places appear identical.

10.10 What completes v0.7

v0.7 is a development programme, not a released railway. Its software goal is a complete and repeatable journey: edit one corridor, compare service and costs, inspect the same candidate's vehicle and civil work, follow a representative module into production and maintenance planning, and receive a dossier that states both results and limits. A good release would let a new user repeat that journey from a clean installation and understand where each conclusion came from.

Complete the installed candidate

The current Samawah comparison keeps the same vehicle reference on both sides. The next step is to bind one detailed six-car candidate with its door, service equipment, mass and inertia, running gear, propulsion, brakes and civil interfaces. Mating dimensions and removal paths must be resolved rather than filled with placeholder positions. A complete installed mass needs actual equipment, joints, fasteners, wiring, cooling and protection, with unknowns explicitly bounded. If the manufacturing child differs physically from the selected reference, it must receive its own service, structural and economic evaluations.

Propulsion needs a force and thermal duty that can support the proposed journey. Braking and rescue must work independently under the relevant faults and poor-adhesion cases. The civil choice needs rail support, bearings, piers, foundations, moving loads and construction stability. The crane must be capable at the actual installation radius. These are not small finishing touches: they can change train mass, station power, cost and the passenger timetable.

Complete one controlled module

The door operator should have a mated geometry and a practical removal route. Controlled drawings, tolerances, materials, process steps and inspection limits should follow the same revision. The bill of materials should identify bought and made parts and their origin. A full candidate linked module should then pass through actual receipt or manufacture, quality review, assembly, installation, fault or condition event, replacement and handback in the business and supervision systems. Current synthetic software custody tests are foundations for this route; they are not its completion.

Complete the city journey

The native GIS exchange needs a supported reader workflow from edit to recalculation and refreshed project, with Samawah first and Baghdad as the generalisation check. The dossier should give before-and-after maps, service and fleet, quantities, energy and all fourteen cost scopes, including unchanged results and explicit unknowns. Transferred rates and unsurveyed conditions should be easy to see. A reader should be able to tell whether a held result comes from an incompatible source, an unsupported model or absent field evidence.

Verify the software and name the physical work

The clean-install acceptance case should reproduce the guided journey, check the downloaded report and source identities, and show that an unchanged study is reused appropriately while a changed one receives a new result. Independent review should challenge the candidate links, numerical models and the completeness of the published claims. The v0.7 release notes should describe available functions, previews and holds in ordinary language.

Physical prototypes, supplier qualification, real measurements and the approvals for manufacture and passenger operation remain their own programme. A complete software journey will make those needs clearer and more traceable. It will not transform a calculated railway into a safe operating one by changing the colour of a dashboard.

A reader's test of completeness

A new reader should be able to choose the supported Samawah example, understand the proposed edit, inspect the before and after results and find the source of a cost or fleet change. They should be able to see which result is held because of absent site evidence and which component still needs a drawing or physical test. They should then open the door module and follow its proposed making and replacement route. If an unrelated result is reused, the interface should explain why. If a physical candidate changes, the interface should refuse a stale result.

The same sequence, adapted to one Baghdad corridor, is a stronger test than a list of isolated features. It exercises planning, service, engineering, production and review as one product. It also gives the team a clear stopping condition for a software release: a supported journey that another user can repeat, with open physical work stated honestly. The subsequent prototype programme can then use that traceable project as a starting point for measurements and approved decisions.

10.11 One complete candidate for v0.7

The immediate v0.7 target is a journey through one project. A user changes a Samawah station or alignment and sees the resulting service, fleet, civil, energy, cost and dossier changes. From that same project, they can open one vehicle module and follow its geometry, manufacture, inspection and maintenance. Every result should refer to the same candidate or explain why it is a separately evaluated alternative.

The corridor and its train

The Samawah midpoint comparison is the first working step. The next work should bind the corridor to one selected six-car installed candidate, one representative civil and station arrangement, and a propulsion/braking evaluation that can support the proposed journey. A station edit should recalculate only its dependants. A vehicle change should trigger the mass, force, structure, manufacturing and cost work it actually affects. The Workbench should show those changes and the open site inputs without asking a reader to inspect a directory of numerical archives.

Baghdad provides the second city test. Repeating the same method on a different corridor checks that the project is not tuned only to one set of Samawah assumptions. It also exposes which rates, route conventions and station interfaces are genuinely shared and which must be replaced with local information.

One module through its life

The door operator is the representative module. Its inventory already reveals much of the required work; v0.7 needs a mated, installed design with controlled header, rail, carriage, drive and sensor dimensions. Door motion and removal must fit the vehicle. The same module should then have a manufacturing drawing set, tolerances, tools, assembly operations, incoming inspection and service replacement instructions. Its actual candidate-linked bill of materials can be carried through business and maintenance records.

This is a deliberately narrow demonstration. A full train has many other critical systems, and physical passenger safety still requires more than a working software journey. One complete, reviewable path through the tools is more valuable than thousands of superficially filled records.

Engineering work that follows

The vehicle team must close installed mass, equipment access, joints and running gear. The propulsion team must establish available force, electrical and thermal duty, and independent stopping. The civil team must model the actual rail and support interface, bearings, foundation and erection. Cost and service work must compare alternatives at equal passenger service with the same candidate quantities. Physical tests, supplier information and independent review have their own milestones.

The platform's tool choices should follow that work. A measured slow stage may justify more Rust or vectorisation, while a dependency error should be fixed before making the wrong calculation faster. SolidCraft, FieldCraft and RustGIS can become preferred where they have demonstrated the required geometry, mechanics or editing domain. Other tools remain valuable for independent checks until an equivalent route has been shown. Historical research stays retrievable even when an obsolete default installation is retired.

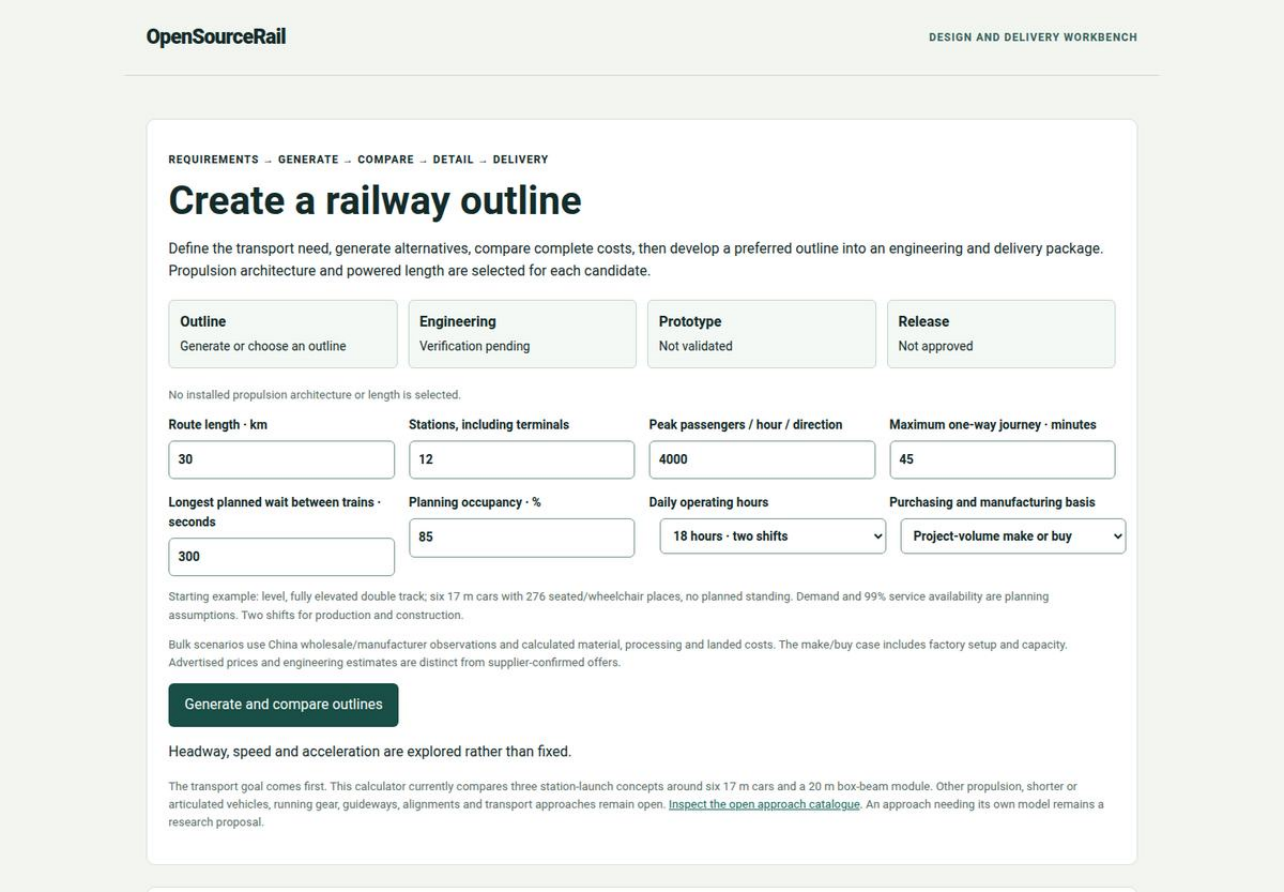
The outcome sought for v0.7 is a comprehensible comparison and a genuine production handoff, with its unresolved evidence plainly stated. That is the right foundation for a physical prototype and, later, a railway that a city can understand, build and maintain.

XI. The interfaces in use

A visual tour of the Workbench, planning and parts views.

11.1 The current Workbench

The Workbench is the project's common reading room. It brings city plans, candidate engineering, status and delivery results into a browser so a person can follow a design question without opening every research archive. Its strongest role is to show the relationship among a proposed route, the train used in its calculation, the resulting service and the evidence still needed. The screen is an engineering review interface, not a railway operator's desk.



Current development Workbench landing view

A real browser view of the current development Workbench. Visible cards and holds describe software and engineering status; they grant no physical or operating approval.

Begin with the object being studied

A reader should first locate the selected development reference and the city or study under review. The passive-wheel reference is a defined train arrangement, while propulsion alternatives are separate research candidates. A city plan has a route and station proposal of its own. The Workbench should make those scopes distinguishable before displaying a price or travel time. A number without a clear object is easy to carry into the wrong discussion.

The published v0.6.1 software gives an accessible first journey in Samawah. It can calculate a bounded city study and show the input assumptions, fleet, energy, civil quantities and fourteen cost scopes. Other panels expose retained research and links to engineering material. Some attractive global views remain held where their source binding is stale. Keeping a card visibly unavailable is a useful behavior: it tells the reader which conclusion the software cannot support at this revision.

Later v0.7 development adds a project comparison that follows a specific Samawah station-marker edit through updated service, cost and delivery. This is a stronger continuity test than simply displaying two unrelated reports. The browser shows what changed, what was reused, and what cannot yet be assembled into a complete installed-candidate dossier. The development view should not be described as a feature of the fixed v0.6.1

release.

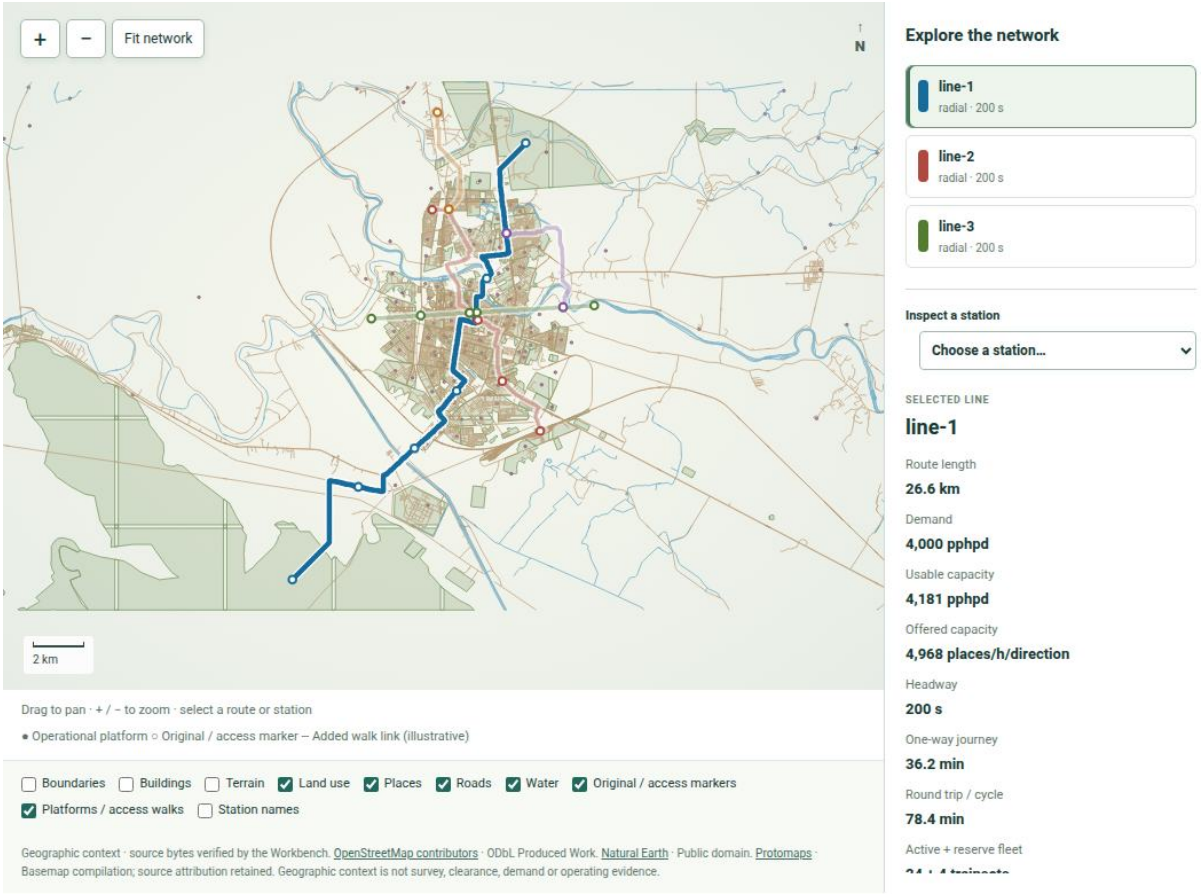
Read a panel in three passes

First ask what is selected: city, route, vehicle and study revision. Next ask what was calculated: a map, a fleet, a cost scenario, a structural result or a delivery draft. Finally ask what limits the claim: transferred rates, unpriced site work, a missing force map, an unmated assembly or a withheld global card. This order is more useful than looking for a single green or red symbol. A result can be a successfully completed calculation while its engineering screen remains open.

The Workbench links to downloadable records for readers who need deeper evidence. A city owner may use the summary to decide what to survey next; an engineer may inspect the exact load and candidate; a workshop may follow a module into its parts and inspection proposal. These different uses should lead back to the same declared design. The interface makes the handoff visible, but the people responsible for a physical decision must still review its evidence.

11.2 Planning and maps on screen

A planning screen helps a city team discuss geography together. A coloured line shows where a proposed service might go; circles and markers show operational station groups, original access points and interchanges. Those symbols are useful only when the reader understands the distinction between a source location, a planning edit and a calculated result. The map should invite questions about walking, land, utilities, flood risk and transfer, not persuade an owner that a corridor has already been surveyed.



Samawah city atlas in the current Workbench

The current city atlas displays the mapped planning case and its source context. The image is a browser screenshot, not a measured alignment or station design.

From an idea to a study

The ordinary planning workspace lets a user propose route, stop, service and origin–destination changes. The saved proposal records the declared change and its source context. It does not by itself contain a recalculated train, guideway or budget. A calculated study is the next step. It applies an identified vehicle, service and cost basis, then reports fleet, energy, civil quantity classes, capital and whole-life scenarios. Keeping those two stages separate prevents an edited marker from looking like an engineered station.

The city atlas is broad, with 266 planning cases and many different line patterns. Readers can compare a compact five-line outline in Samawah with the more extensive Baghdad network, or look at the maps later in this book for other regions. The scale of the drawing is not a measure of readiness. A large, polished map can still rely on common demand and price assumptions; a small corridor with good local survey may support a more confident decision.

What a map click should lead to

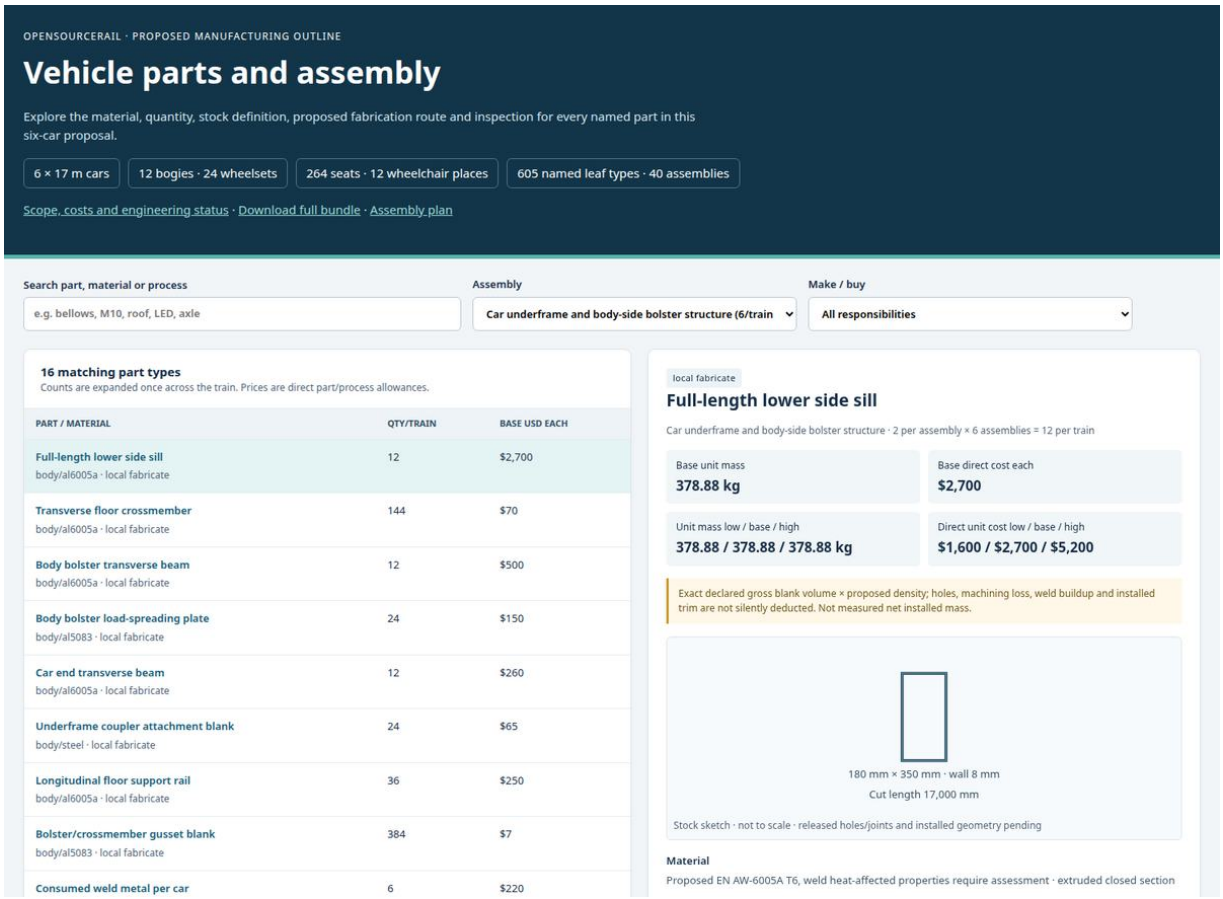
A useful station view should answer where the proposed stop sits, what line serves it, how it connects to other lines and which source marker or edit produced it. It should also show what remains unknown: actual accessible walking routes, platform footprint, land, ground, utilities and the safe construction method. For a line, the reader needs route length, stop pattern, headway assumption and the train used in the service calculation.

The current GIS return path adds a stronger editing loop in development. A planner can request an editable native project, revise permitted route or station information and submit the saved project with an author and reason. The returning study retains the original city for comparison and recalculates affected service and cost outputs. Automated native cycles have been exercised on Samawah and Baghdad; the book does not treat them as proof of a fully tested desktop user journey or surveyed geometry.

The safest way to discuss the screen is to pair every coloured route with one sentence about its maturity. These are calculated planning outlines. They help choose surveys and compare alternatives. They are not construction drawings or operating timetables.

11.3 Parts and lifecycle on screen

A product image tells a workshop what a module might look like. A useful parts interface goes further: it lets a reader identify each item, its proposed material and source, the assembly that uses it, and the work still needed before it can be made or purchased. The vehicle parts catalogue is one of the project's practical bridges from a railway concept to local manufacturing and repair.



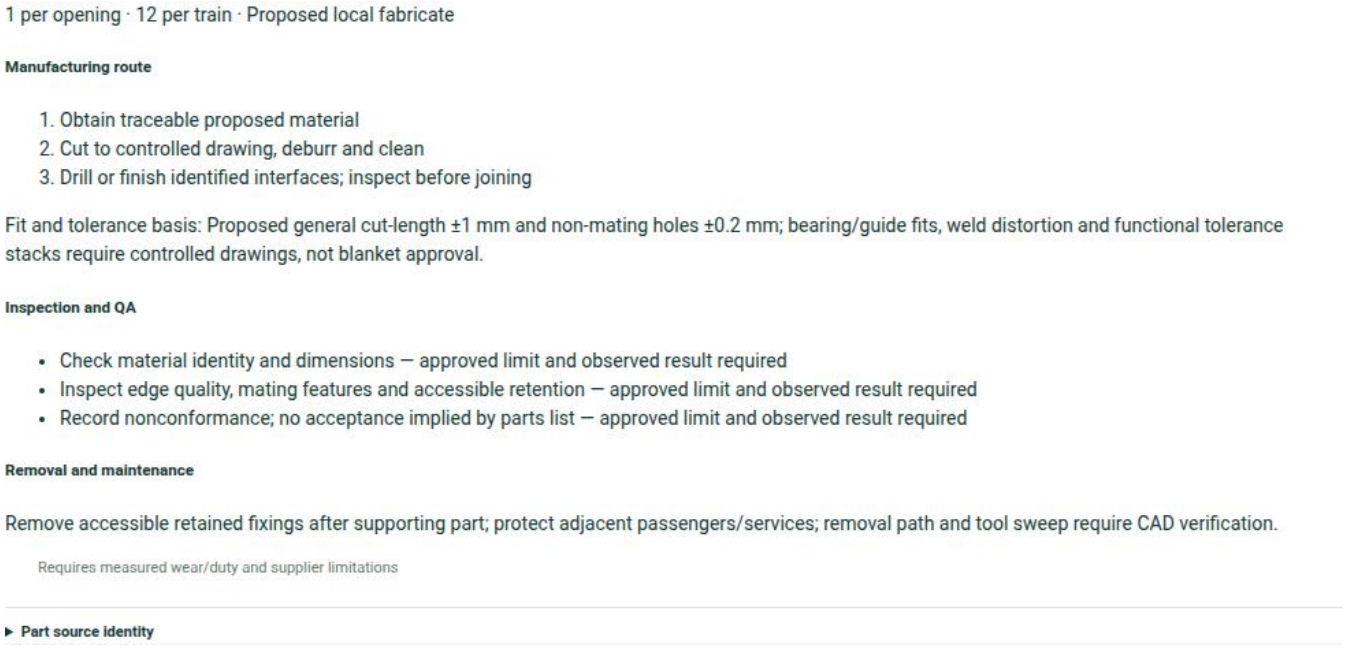
Searchable vehicle parts catalogue

The searchable parts interface exposes counted proposed items and their assembly relationships. It is a genuine product view, while its estimates and drawings remain subject to the stated engineering holds.

Search by an assembly, then follow its children

The door operator is a good example. One opening contains a counted proposal of 74 part types and 694 individual pieces. The inventory distinguishes parts proposed for local fabrication, standard purchase and purchase subject to qualification. A reader can move from the familiar word “door” to a header, guides, carriages, belt, motor, sensors, leaves, seals and many fasteners. That detail matters because every child part needs a fitting position, source, inspection and eventual replacement route.

The view must not make a counted list look like an installed door. Several operator items still occupy a generic header zone, and a shortened removal proxy does not represent the full guide. A controlled module needs mating dimensions, tolerances, wiring and safe access. If the user asks whether a part can be removed, the answer must come from the actual geometry and procedure, not from its presence in a bill of materials.



Current development part-detail panel

An actual Workbench part-detail view shows proposed manufacturing, inspection and removal questions. The panel is clipped below its source-identity line; none of its displayed suggestions is an approved drawing or test limit.

Follow the same item beyond design

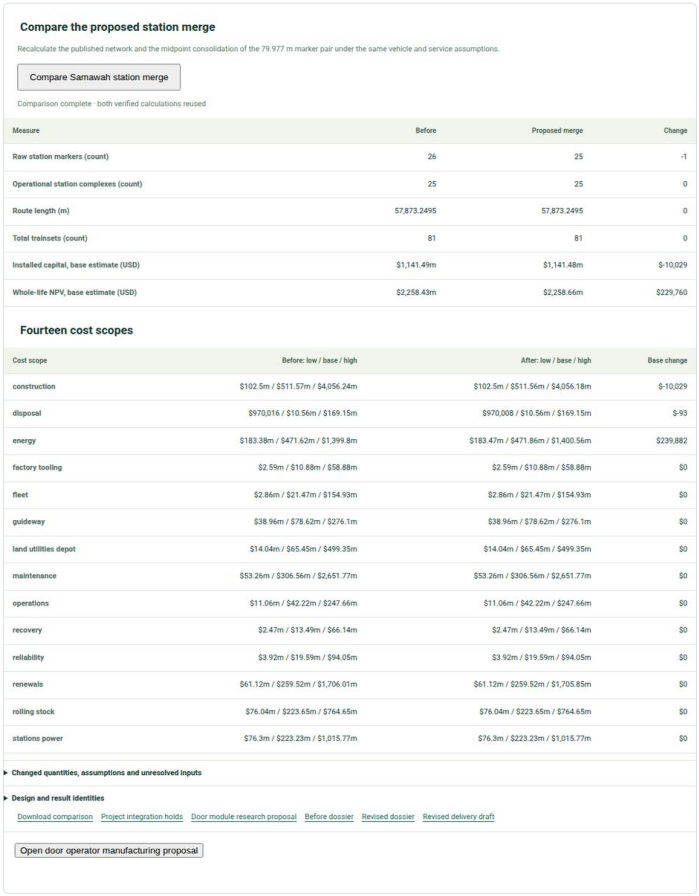
Once a module is engineered, the owner needs to know how it moves through receipt, incoming inspection, assembly, installation, condition monitoring, repair and replacement. The optional business integrations can draft the corresponding records. In v0.7 development, software contracts have begun to check serialized child custody and reviewed inspection for a counted assembly. The complete door has not yet been carried through a native business-system transaction, so this is a development pathway rather than a finished operational feature.

The supervision side raises a related question. An alarm is meaningful only when it identifies the actual installed item and its position. The optional equipment gateway and FUXA integration can help represent observations and maintenance handoffs in a configured environment. They do not grant live train control. A repair record should lead back to the source design and supplier or production lot so engineers can decide whether the fault is local wear, installation, environment or a broader design issue.

For a workshop, the practical test of the screen is simple: can it identify the part that must be made or replaced, the exact design it belongs to, the evidence for its dimensions and quality, and the next responsible reviewer? Where an answer is missing, the interface should say so. That honesty is what makes an open parts catalogue useful rather than merely impressive.

11.4 Reading status and holds

An engineering Workbench can display a great many results. The reader needs a small vocabulary for judging each one. Proposed means a design idea or edit has been recorded. Calculated means a declared model has run. Verified digital evidence means the result and its source relationship have passed the checks that apply to that software method. Physically accepted would require appropriate hardware, measurements, approved limits and independent review. The first three states can be useful while the fourth remains open.



Compare the proposed station merge

Recalculate the published network and the midpoint consolidation of the 79.977 m marker pair under the same vehicle and service assumptions.

Compare Samawah station merge

Comparison complete · both verified calculations reused

Measure	Before	Proposed merge	Change
Raw station markers (count)	26	25	-1
Operational station complexes (count)	25	25	0
Route length (m)	57,873.2495	57,873.2495	0
Total trainsets (count)	81	81	0
Installed capital, base estimate (USD)	$1,141.49m	$1,141.48m	$-10,029
Whole-life NPV, base estimate (USD)	$2,258.43m	$2,258.66m	$229,760

Fourteen cost scopes

Cost scope	Before: low / base / high	After: low / base / high	Base change
construction	$102.5m / $511.57m / $4,056.24m	$102.5m / $511.56m / $4,056.18m	$-10,029
disposal	$970,016 / $10.56m / $169.15m	$970,008 / $10.56m / $169.15m	$-93
energy	$183.38m / $471.62m / $1,399.8m	$183.47m / $471.86m / $1,400.56m	$239,882
factory tooling	$2.59m / $10.88m / $58.88m	$2.59m / $10.88m / $58.88m	$0
fleet	$2.86m / $21.47m / $154.93m	$2.86m / $21.47m / $154.93m	$0
guideway	$38.96m / $78.62m / $276.1m	$38.96m / $78.62m / $276.1m	$0
land utilities depot	$14.04m / $65.45m / $499.35m	$14.04m / $65.45m / $499.35m	$0
maintenance	$53.26m / $306.56m / $2,651.77m	$53.26m / $306.56m / $2,651.77m	$0
operations	$11.06m / $42.22m / $247.66m	$11.06m / $42.22m / $247.66m	$0
recovery	$2.47m / $13.49m / $66.14m	$2.47m / $13.49m / $66.14m	$0
reliability	$3.92m / $19.59m / $94.05m	$3.92m / $19.59m / $94.05m	$0
renewals	$61.12m / $259.52m / $1,706.01m	$61.12m / $259.52m / $1,705.85m	$0
rolling stock	$76.04m / $223.65m / $764.65m	$76.04m / $223.65m / $764.65m	$0
stations power	$76.3m / $223.23m / $1,015.77m	$76.3m / $223.23m / $1,015.77m	$0

▸ Changed quantities, assumptions and unresolved inputs

▸ Design and result identities

Download comparison · Project integration holds · Door module research proposal · Before dossier · Revised dossier · Revised delivery draft

Open door operator manufacturing proposal

Current Samawah station comparison in the Workbench

A current development comparison gives both sides of a station edit and its outstanding questions. Its numerical deltas are planning results, not site measurements or construction approval.

A visible hold is information

The v0.6.1 Workbench keeps an older integrated dossier and its portfolio selector held because their source binding does not match the release. That is a specific software evidence limit. It does not invalidate every city input or the separately checked Samawah journey. Conversely, a successful Samawah journey does not repair the held global dossier. The interface should let a reader see both statements at once.

The same discipline applies to a physical finding. A door may have a complete count but unresolved fit. A beam may pass an idealized finite element comparison while its bearing and erection remain open. A city cost may be complete within a declared model while its ground, land and supplier prices remain provisional. Treating those as separate statuses helps the next experiment target a real gap instead of repeating a calculation that already succeeded.

Compare before and after with the same scope

In the Samawah development comparison, the raw marker count falls from 26 to 25, while operational station groups and trainsets remain fixed. Capital and whole-life costs move in different directions. A reader should ask which assumptions and cost scopes actually changed, whether the vehicle and civil quantities stayed the same, and which local facts could overturn the provisional ranking. The report's zero changes are part of the evidence: they show that unaffected work was preserved.

A different candidate demands a separate comparison. A powered-wheel research result cannot be used to fill a station-drive force gap in the passive-wheel reference. A detailed manufacturing child with a different inter-car gap cannot inherit its parent's platform, fleet and cost results. The Workbench's candidate and revision identity protects against those errors when a project moves from one screen to another.

Take the right next step

For a city planner, a hold may call for a survey, demand count or access review. For a vehicle designer it may call for a mated drawing or a removal mock-up. For a civil engineer it may call for a bearing, foundation or construction-stage model. For a workshop it may call for a supplier qualification and inspection definition. For an operator it may call for measured braking and fault-response evidence. The interface is most useful when it names that next piece of work instead of presenting a vague “not ready” label.

The final approval remains with the organizations responsible for a real railway. The Workbench can make the design history clearer and the missing evidence harder to overlook; it cannot approve manufacture or movement because a screen looks complete.

XII. Adoption and open knowledge

Using the platform, contributing and reading deeper.

12.1 Who can use the platform now

There are several useful starting points, depending on what a reader knows and what decision they need to make. None requires pretending that a railway is already approved. The most productive first use is a bounded question: Which corridor deserves survey? Which service assumption drives fleet size? What part of a proposed train still lacks an installed interface? Where would a workshop need a drawing or a qualified supplier before it could make a module?

A city or public owner

A city team can start with the mapped catalogue and the planning workspace. It can inspect a proposed line, stop locations and nearby context, then add local knowledge about walking access, demand, land, utilities or flooding. The planning handoff records route, stop, service and origin–destination proposals. That handoff is a proposal, not a costed or validated railway. A calculated city study is a different object: it also carries modelled service, fleet, energy, quantities and costs under stated assumptions.

This distinction helps an owner commission the right next work. A new stop may demand a catchment survey before its price is worth refining. A new headway may require a protected-service and fleet study. A guideway crossing may require ground and utility evidence before choosing a span. Samawah shows how even a small marker edit can reveal these dependencies. Baghdad provides a different large-city planning context, with a retained read-only comparison in v0.6.1. Neither example is a substitute for local transport engagement or engineering surveys.

A builder or local workshop

A fabrication team can use the vehicle and civil studies to ask whether a proposed part suits its available materials, machines, quality methods and transport. The assembly views, parts catalogue and make-or-buy records are designed to expose the number of items and operations hidden by a concept picture. They can support requests for information and a prototype plan. Before making a safety-relevant part, the team still needs controlled dimensions and tolerances, process qualifications, inspection limits and responsibility for changes.

The door operator is a useful example. Its counted inventory is detailed, yet several mating positions and the full removal route remain unresolved. A workshop contribution that establishes those physical interfaces, or shows why a proposed guide cannot be removed, may be more valuable than a larger set of attractive renderings. The same principle applies to precast spans: the ability to cast a beam is only one part of transporting, lifting, connecting and inspecting it.

A university, reviewer or engineer

Researchers can reproduce a declared calculation, compare candidate families, challenge an assumption and identify a useful measurement. The project retains failed and historical work because it can show where an idea or a model stopped being credible. A close match between two solvers is useful evidence for the exact model they share. It does not prove that an unmodelled bearing or joint is safe. A strong review states which claim has been tested, which evidence supports it and what remains uncertain.

The software's source-bound records make disagreements more productive. Instead of arguing over a single cost or displacement value, collaborators can identify the candidate, service, load, rate or material assumption that produced it. They can then propose a new evaluation without overwriting the earlier result. Physical measurements and independent checks should be added as their own evidence, rather than retroactively changing what an old simulation meant.

A future deployment partner

An organization considering the business and supervision components can test them in a disposable environment using its own roles, item masters, warehouses, inspection steps and equipment model. Draft records should be reviewed before any real purchase, work order or asset installation. A gateway should be bound to known devices and kept outside movement authority unless an independently approved control system permits it.

The path from demonstration to operation is a sequence of local decisions: select a bounded project, name its owners, establish site and supplier evidence, develop the installed candidate, review production and safety records, build and test, and commission under the applicable authority. The platform can organize those decisions. The organizations responsible for a real railway must make and approve them.

12.2 Keeping the work open and useful

Open design knowledge is valuable when another team can understand what was assumed, what was tested and what changed. A drawing with no dimensions is hard to manufacture. A cost with no scope is hard to compare. A numerical result with no candidate or load case is easy to misuse. OpenSourceRail therefore treats explanation, evidence and revision history as parts of the engineering product, not as paperwork added after the technical work.

A contribution can be small and decisive

A useful contribution may correct a unit, document an overlooked doorway clearance, test a control fault, check a supplier's bearing specification, or improve a city's walking-access information. Practitioners in operations, maintenance, quality, civil works and vehicle assembly can identify failures that are difficult to see in a software model. Local knowledge can overturn an apparently convenient station location or production assumption.

The project welcomes focused changes, reproducible observations and independent reviews. A proposal should explain the condition under which it applies and what it would change. If a new vehicle part changes mass or clearance, its effects should be followed into braking, civil load, manufacturing and cost. If a city edit changes only a labour rate, the geometry need not be redrawn. This keeps open collaboration fast enough to be useful while preserving the meaning of earlier work.

Distinguish review from approval

An independent engineer can confirm that a calculation was reproduced or challenge its model domain. A production reviewer can approve a drawing for a stated prototype. A railway operator and its assessors must address the larger deployment safety case. These are related but different acts. The repository's software tests and research archives provide inputs to review; they do not appoint an operator, qualify hardware or grant movement authority.

Open work also means preserving negative results. A beam that failed a load case, a motor calculation outside its tested speed, or a door removal path that collides with the body should remain visible with the condition that caused the finding. Later work may repair it or choose another family. Deleting the failure would erase useful design knowledge and make a later selection harder to trust.

Licensing and stewardship

The project uses different open licences for different kinds of work: Apache 2.0 for software, CERN-OHL-S v2 for open physical designs and CC-BY-SA 4.0 for documentation, subject to the file-specific notices and upstream rights recorded in the project. That split is intended to let software, drawings and explanations be shared and improved while keeping their origins clear. A contributor should check the terms of any supplier drawing, data source, image or code before adding it.

The project is currently a pre-1.0, founder-led effort. Maintainers review changes to the shared repository; domain reviewers examine the subjects they know; a deployment partner supplies local site, equipment and operational evidence. Larger changes to architecture, safety claims or city economics deserve a written rationale and review. A promising open design can become a reliable public asset only when the people who will build, maintain and operate it can also understand and govern it.

The next chapters and releases should continue to serve two audiences at once: the general reader who needs a clear account of the railway, and the specialist who needs exact evidence. This ebook provides the first path. Maintained engineering records and their separate verification receipts provide the second. Both are necessary for a project that aims to remain locally useful long after its first prototype.

12.3 Sources, terms and licensing

This is a reader's edition of the current platform and engineering work. The separately published source receipt records the exact files and illustrations used to build the PDF. The maintained project guides provide the detailed calculations, software behavior and historical records behind the narrative. The following titles are useful starting points in the project's documentation. Begin with the published release guide if you want to use the software; use the engineering guides when you want to examine a calculation or proposed physical design. Each retains its own evidence limits.

Further reading	Main subject
OpenSourceRail platform guide	Three products, passenger railway concept, delivery method and local capability
v0.6.1 installation and quickstart	Published downloads, local setup and a first Samawah session
v0.6.1 release and known limitations	Published software scope, installation and visible holds
v0.6.1 feature maturity and demonstrations	Available, preview and held features, with three example journeys
First adoption path	Ways a city, workshop or reviewer can use the current tools
City design catalogue and route atlas	Maps, service assumptions and costed planning outlines for the city examples
v0.7 Samawah corridor continuity	Before-and-after station comparison and its planning boundaries
v0.7 execution and evidence	Current development results and unresolved interfaces
City planning workspace and RustGIS guide	Planning proposals, native edits and revised city studies
Three-dimensional engineering gallery	Vehicle, bogie, launch and civil views with their distinct research origins
Vehicle parts, manufacture and assembly	Counted stock, proposed processes and assembly breakdown
One coordinated railway experiment	Same-candidate engineering propagation and research alternatives
End-to-end research digital twin	Connected geometry, mass, analysis, cost and verification boundaries
SolidCraft and FieldCraft backend pilot	CAD and mechanics comparisons and their model domains
Viaduct alternative studies	Span geometries, structural comparisons and remaining civil interfaces
Assembly lifecycle integration	What a counted module must prove before business handback
v0.7 corridor-to-lifecycle roadmap	The integrated work still needed

Terms used in this book

Candidate means one physical arrangement of the railway and its relevant interfaces. Evaluation means a calculation or test under declared conditions. Selection means a decision to use one candidate for a stated purpose. Source-bound means that a result identifies the inputs and method to which it applies. Whole-life cost includes construction, operation, maintenance, renewal and end-of-life work under an explicit period and discounting assumption. Installed mass includes the equipment, mounts, wiring, cooling, protection and other items actually needed on the vehicle, beyond a convenient stock subtotal. Physical acceptance would require appropriate measurements, approved limits and independent review; it is not conferred by a software demonstration.

Images marked concept communicate intended form or use. Engineering model views illustrate declared geometry and methods, but are only evidence for their exact model and checks. No image in this book is an approved manufacturing drawing, surveyed railway route or operating permission.

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